

Mathematics atlas

Investigation plans

90 ways into substantial mathematics

A facilitator library extending from objects, drawings and games to algebra, calculus and named advanced prerequisites.

24 families Algebra and discrete mathematics

36 families Geometry and analysis

30 families Probability and applications

Start with the question. Keep the real prerequisites.

Each family preserves its exact opening rules, learner choices, access gates, worked example, boundary case, hints, source evidence and proposed hour. These are planning cards for facilitators; they contain answers and are not student worksheets.

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Scope and review

The companion [research map](#) surveys 63 MSC fields and records 231 problem seeds. This book develops 90 selected families. Neither count means every subject or interesting problem has been exhausted. Independent mathematical/editorial review checks the stated instances; no family is classroom-piloted.

MSC attribution

MSC2020 classification codes are from Mathematical Reviews / zbMATH, downloaded 25 September 2026.

Classification-derived content is used under [CC BY-NC-SA 4.0](#). The activity designs and annotations are original atlas work. This is an educational planning edition.

Choose one investigation

Read the launch and try the smallest instance before reading the solution. Choose a family whose question interests you, then check its entry, explanation and proof gates separately.

A workable hour

The typical arc is 0-10 minutes to handle the materials, 10-15 to launch and check legal moves, 15-35 to investigate, 35-40 to move/reset, 40-55 for a second attempt or extension, and 55-60 to share. Each card customizes that menu. It is not a requirement to finish every question.

Say the launch and provide the objects or sketch. Keep the solution beside you. Offer hints one at a time and preserve time for the learner to choose, conjecture and explain. The satisfying stop is useful even when a general proof remains beyond reach.

What a connection means

exact-special-case: The activity is a specified genuine instance of the mathematical object or theorem.

faithful-representation: Objects and legal moves encode the stated mathematical structure.

shared-mechanism: The reasoning has a common mechanism, but the activity does not establish the full advanced claim.

motivation-only: The comparison invites interest; it is not itself an instance or proof of the advanced theory.

Evidence is local to each claim

Source entries state what was inspected: a full passage, an excerpt, a located topic, or a continuation. A citation is not a substitute for matching theorem hypotheses. Checked finite examples are distinct from general research continuations; a simulation is not a proof for all inputs.

The independent reviews, exact-check scripts and limitations are preserved in [plans/atlas/reviews/](#). Preparation times and proposed hour plans are editorial estimates, not classroom measurements. Advanced cards retain hard prerequisites when removing them would remove the mathematics.

Returning learners

Before scheduling a family, consult its prior-use note and the collection use log. Record the exact board, starting numbers, rule set, function and explanation encountered. A familiar object can support a new question; changing only the story or the amount of arithmetic does not make a new investigation.

The six-year goal is approximately 120 varied sessions along a child's path. This book is a broader library, not a fixed age sequence. Select, trial and adapt families; develop detailed student pages after choosing a route.

Capabilities, not age labels

A learner may have some of these tools and not others. O does not mean cognitively easy: spatial tracking, working memory, language and abstraction are specified separately in each card.

O Follow a short spoken rule; sort, match, move or compare objects. Reading optional.

C Count small sets and track cases; add or subtract whole numbers as required.

M Multiply/divide and reason about factors or remainders.

F Fractions, ratios and signed numbers; the card specifies which are needed.

V Coordinates, vectors or geometric measurement in the stated representation.

A Variables, equations and functions; the card specifies the kind of algebra.

P Systematic proof: cases, induction, contradiction or quantified claims.

L Linear algebra: vectors, matrices, systems or eigenvectors as specified.

D Differential calculus: limits and derivatives as specified.

I Integral calculus: integrals and their meaning as specified.

X A named advanced prerequisite, such as groups, metric spaces or measure theory.

Four distinct gates

Enter: Join the activity and make a meaningful first move.

Explore: Track enough cases or calculations to investigate a pattern.

Explain: Justify the stated concrete result in an appropriate representation.

Prove: Meet the prerequisites for the more general theorem or formal argument.

The hard stop names the point where a prerequisite becomes essential. Reading, arithmetic and reasoning demands supplement these codes. A proof can be an exhaustive arrangement, a drawing or an oral argument; formal notation is not required unless it carries the actual mathematics.

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Algebra and discrete mathematics

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Probability and applications

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AD-01 / Participation and launch

Build a world that defeats a rule

MSC primary 03 - Mathematical logic and foundations | Secondary 06

The opening challenge

Put any cards in your world. Every red object must be round. Can you make that rule true while 'every round object is red' is false? Keep both rules visible. Empty trays are allowed later, but begin with at least one object.

Learner choices: Choose the world's inhabitants, remove unnecessary objects, and invent a new proposed inference for another group to challenge.

Prerequisite gates

Enter: O: distinguish two colors and two shapes, or equivalent tactile labels; spoken rules suffice.

Explore: C: compare several objects while holding one rule fixed.

Explain: P: identify the exact premise and conclusion and point to a witness.

Prove: P: reason over the four possible combinations of two predicates, including an empty predicate class.

Reading: None with adult reading; rule cards can use pictures.

Arithmetic: Only small counts, if any.

Reasoning: Language and quantifier scope are the real demands; avoid adding memory pressure through hidden objects.

Hard stop: Stop at explicit finite models before discussing arbitrary structures or metatheorems.

Materials and preparation

10 minutes estimated prep. A tray and twelve red/blue circle/square cards, with spare duplicates. Draw a persistent four-cell property table. Prepare three rule cards, not a long worksheet.

A flexible hour

0-10 freely sort and invent descriptions; 10-15 test one sample world together; 15-35 build smallest counterworlds; 35-40 move to room corners representing four property types; 40-55 compare converses and empty classes; 55-60 display one world with an explanation. The first countermodel is a complete stopping point.

Questions to explore

- Which single object would break the first rule?
- Can your world contain a blue circle?
- Does adding a red square change truth or merely make the world larger?
- What happens if there are no red objects?

Satisfying stop: A learner builds a smallest counterworld and explains which object defeats the conclusion while the premise survives.

AD-01 / Facilitator reasoning

Build a world that defeats a rule

The mathematical question

How do models distinguish a valid inference from a statement that happens to hold in a few examples?

For predicates R and S on a specified finite universe, 'every R is S ' means no object satisfies R and not S . Its converse is a different sentence. A countermodel satisfies every premise while falsifying the conclusion.

Worked instance

Problem: Make every red object round, but make not every round object red. Then test whether every round object being blue follows.

Solution: A world containing one blue circle satisfies the first sentence because it has no red counterexample, and falsifies the converse. If a red object is required, use one red circle plus one blue circle. This second world also falsifies 'every round object is blue'. A red square is forbidden by the premise; blue squares are irrelevant to either implication.

Boundary: An empty tray satisfies both universal sentences, so it cannot defeat the converse. This is not the same as saying a red object exists. Explicitly separate universal and existence claims.

Hints to reveal in order

1. Mark forbidden card types rather than guessing entire worlds.
2. To defeat 'every round object is red', look for a round object that is not red.

Extensions

- Introduce an arrow relation 'visits'. Compare 'everyone visits someone' with 'someone is visited by everyone'; two vertices with only self-arrows separate them.
- Ask which two-predicate inferences are valid by considering all four object types; general finite-model theory is a later continuation, not established here.

Connection and its limits

Bridge type: exact-special-case

Preserves: A tray is a finite domain; color and shape are unary predicates. Building or altering a tray is constructing a model, and checking each object is evaluating the quantified sentence.

Limits: These finite unary examples do not establish completeness, compactness, or a decision procedure for general first-order logic.

Sources and inspection scope

[Open Logic Project](#), [Sets](#), [Logic](#), [Computation](#)

Chapter 7 semantics and satisfaction; chapter 12 completeness/compactness inspected in survey

Inspected: excerpt only; the finite countermodels below are established directly

Prior use / what changes: Year-1 truth/logic material is related; exact prior instances require the use log. Week 5 tests consistency and uniqueness, whereas this activity makes learners construct countermodels to proposed implications.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-02 / Participation and launch

The list that cannot catch your pattern

MSC primary 03 - Mathematical logic and foundations | Secondary 05

The opening challenge

Your partner lists four patterns, each four counters long. Build a pattern missing from their list. Then make a method that works against any list they choose. You may inspect the whole list; rearranging their patterns is not required.

Learner choices: The list maker chooses patterns, including duplicates. The challenger chooses an initial strategy and later decides which coordinate will guarantee disagreement with each row.

Prerequisite gates

Enter: O: match two colors and locate a marked square in each row.

Explore: C: track row and column indices with fingers or paired labels.

Explain: P: assign a guaranteed disagreement to each opponent row.

Prove: P and X: for the infinite extension, understand functions on positive integers and universal quantification over possible enumerations.

Reading: None for colored cards; a short symbolic rule for the infinite stage.

Arithmetic: Counting to four or five; no calculation.

Reasoning: Coordinating two indices is harder than the arithmetic. Use one row at a time and retain comparison markers.

Hard stop: A participant may finish with the finite guarantee. Do not infer understanding of infinity from success at four rows.

Materials and preparation

5 minutes estimated prep. Two-color counters, a four-by-four grid, and a four-place answer strip. Label matching rows/columns with four pictures if numeral reading is a barrier.

A flexible hour

0-10 make and compare patterns; 10-15 play one challenge without prescribing a method; 15-35 trade adversarial lists and look for a guarantee; 35-40 stand in four numbered places and change one gesture each; 40-55 explain certificates or discuss a precisely defined infinite list; 55-60 show one disagreement per row. Infinity is optional.

Questions to explore

- How can you certify four differences without comparing every square?
- Must your answer be the only missing pattern?
- Can five rows of four bits always be defeated by the same diagonal recipe?
- What statement would a truly complete infinite list make?

Satisfying stop: A four-counter pattern with four visible mismatch certificates and an explanation that the procedure survives any changed list.

AD-02 / Facilitator reasoning

The list that cannot catch your pattern

The mathematical question

How can diagonalization construct an object different from every member of a proposed list, and what changes when the list is infinite?

For any n by n binary array, choose the i th bit of a new length- n word opposite the entry in row i , column i . The new word differs from each listed row. For an infinite sequence of infinite binary sequences, the same coordinate rule constructs a sequence absent from the enumeration.

Worked instance

Problem: Defeat rows 0000, 0110, 1011, 1101 with a guaranteed rule.

Solution: The diagonal is 0,1,1,1, so choose 1000. It differs from row 1 in position 1, row 2 in position 2, row 3 in position 3, and row 4 in position 4. These four witness positions prove absence from the list, regardless of other matches. Other missing words may also work.

Boundary: All sixteen four-bit words can be listed, leaving no missing word. Therefore the claim is not 'every finite list misses a finite word'; the relation between number of rows and available coordinates matters.

Hints to reveal in order

1. Reserve a different answer position for each opponent row.
2. At the reserved position, deliberately choose the other color.

Extensions

- Use three symbols: any fixed rule choosing a different symbol works, so binary is convenient rather than essential.
- For infinite binary sequences, define $b(i)=1-a_i(i)$. Explain why no row index can equal the resulting sequence; avoid decimal expansions and their nonunique representations.

Connection and its limits

Bridge type: exact-special-case

Preserves: Rows are binary words; changing diagonal entries gives the exact finite construction used by the infinite argument.

Limits: The finite game proves a finite avoidance result. Uncountability requires explaining an arbitrary infinite list and a rule defined at every positive integer, not drawing many rows.

Sources and inspection scope

[Open Logic Project, Sets, Logic, Computation](#)

Chapter 4, printed pp. 58-60, binary-sequence diagonalization and Theorem 4.18

Inspected: full relevant text

Prior use / what changes: Week 6 concerns identifying hidden finite codes, not diagonal construction or uncountability. This is a distinct question even though it reuses binary counters.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-03 / Participation and launch

The largest collection with no nested sets

MSC primary 05 - Combinatorics | Secondary 06

The opening challenge

Choose as many cards as possible. You may not choose two when every picture on one is also on the other. Sharing some pictures is allowed. Build a large legal collection, then convince another team that a larger one cannot exist.

Learner choices: Choose a collection, exchange small sets for large ones, choose how to organize the complete deck, and invent a certificate rather than chase the adult's target arrangement.

Prerequisite gates

Enter: O: recognize which pictured objects belong to each card; no arithmetic beyond small counts.

Explore: C: distinguish containment from overlap and organize all sixteen subsets.

Explain: P: argue that a legal collection can take at most one card from each nested chain.

Prove: P: construct or justify the six-chain partition; general n needs a separate inductive chain construction or counting proof.

Reading: None with picture sets; names may be replaced by four symbols.

Arithmetic: Count up to sixteen and compare collection sizes.

Reasoning: Track a universal pairwise condition; overlapping cards are legal unless one contains the other.

Hard stop: Stop at the exact four-object optimum if the general combinatorial proof is not accessible.

Materials and preparation

15 minutes estimated prep. Sixteen cards, one for every subset of four pictures; counters to mark selected cards. Prepare six blank chain strips and optionally the certificate on a facilitator sheet.

A flexible hour

0-10 sort and compare set cards; 10-15 demonstrate containment and a legal overlapping pair; 15-35 compete cooperatively for large collections; 35-40 form nested groups of participants; 40-55 build chain certificates; 55-60 show a matching construction and bound. Finding six is a worthwhile early stop.

Questions to explore

- Do all chosen cards need the same number of pictures?
- Why is the empty card difficult to combine with others?
- Can one long chain help limit your selection?
- How many chains would prove six is best?

Satisfying stop: A six-card construction together with a complete six-chain certificate that explains optimality.

AD-03 / Facilitator reasoning

The largest collection with no nested sets

The mathematical question

How large can a family of subsets be when no member contains another, and how can a chain decomposition certify optimality?

An antichain in the Boolean lattice of subsets of an n -element set has size at most $\binom{n}{\lfloor n/2 \rfloor}$. The exact four-element problem admits a six-chain partition certificate, with six two-element subsets attaining the bound.

Worked instance

Problem: Determine the maximum for subsets of $\{1,2,3,4\}$.

Solution: All six pairs form a legal collection. Partition the deck into chains: $\text{empty} < 1 < 12 < 123 < 1234$; $2 < 23 < 234$; $3 < 13 < 134$; $4 < 14 < 124$; 24 ; 34 . Each of the sixteen subsets appears once and every chain is nested. A legal collection uses at most one member of each of six chains, proving the maximum is six.

Boundary: For three objects the three singleton cards and the three pair cards each give a maximum, but choosing both full layers is illegal: for example $\{1\}$ is contained in $\{1,2\}$. Equal layer sizes do not permit merging layers.

Hints to reveal in order

1. Arrange cards by number of pictures, then inspect all two-picture cards.
2. Group the entire deck into nested chains. One selected card per chain is the most possible.

Extensions

- Find the exact answer for five objects; the middle layer has ten sets, and a ten-chain certificate is a concrete next proof challenge.
- Investigate antichains in a divisibility poset. The largest same-size layer need not solve arbitrary partially ordered sets; the Boolean-lattice hypothesis matters.

Connection and its limits

Bridge type: faithful-representation

Preserves: A card displays a subset of four named objects; legal collections are precisely antichains under inclusion. Partitioning cards into nested chains proves an upper bound.

Limits: The supplied four-object certificate proves only that case. It does not establish a chain decomposition or Sperner's theorem for every n .

Sources and inspection scope

[Richard P. Stanley, Topics in Algebraic Combinatorics, 2013 author draft](#)

Chapter 4, Sperner property and Corollary 4.8; chain discussion

Inspected: full relevant theorem discussion; the particular six-chain certificate is original and checked below

Prior use / what changes: No matching current Week 1-10 mechanism. Week 5 also distinguishes a good construction from a proof of optimality, but its clue constraints and reasoning are different.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-04 / Participation and launch

Send a whole tree in two numbers

MSC primary 05 - Combinatorics | Secondary 68

The opening challenge

Build a connected network on 1,2,3,4 with no loops around a cycle. Remove the smallest leaf and write its neighbor. Repeat once. Can your partner reconstruct exactly your edges using only the two-number message? Labels stay attached to their vertices.

Learner choices: Invent the tree, choose challenging messages, look for ambiguity, and decide how to prove the decoder cannot choose differently.

Prerequisite gates

Enter: O and C: connect four labeled vertices, recognize a leaf as a vertex with one incident edge.

Explore: C: follow the smallest-label tie-breaking rule and keep track of deleted vertices.

Explain: P: explain why decoding is unique rather than merely successful in a few trials.

Prove: P: induct on successive leaf deletions, using the relation between remaining code occurrences and degrees.

Reading: Short algorithm cards; partner can read and record.

Arithmetic: Order four small integers; count occurrences; powers only in the later counting interpretation.

Reasoning: Dynamic state tracking is substantial; physically remove edges and retain a vertex ledger.

Hard stop: Complete the four-vertex correspondence before general coding proofs; avoid treating a large unverified catalog as proof.

Materials and preparation

8 minutes estimated prep. Four numbered cards and reusable strings, with a two-place message slip. A second color can mark removed vertices; prepare a visible algorithm card.

A flexible hour

0-10 build connected networks and remove redundant edges; 10-15 demonstrate one deletion; 15-35 exchange messages and debug decoding; 35-40 enact leaves leaving a human network; 40-55 organize all sixteen messages and argue uniqueness; 55-60 explain why two numbers suffice. General n is optional.

Questions to explore

- Which label cannot yet be removed if it still appears in the message?
- What does a repeated number tell you about the tree?
- Why must the same smallest-leaf rule be used every time?
- Can any two-number message fail?

Satisfying stop: Two partners exchange one tree with a message and explain why the reconstruction is forced.

AD-04 / Facilitator reasoning

Send a whole tree in two numbers

The mathematical question

Why do labeled trees on n vertices correspond exactly to words of length $n-2$ on those labels?

Prüfer encoding repeatedly removes the smallest labeled leaf and records its neighbor until two vertices remain. For labels 1 through n , $n \geq 2$, this gives a bijection with length- $(n-2)$ words, proving Cayley's count n^{n-2} .

Worked instance

Problem: Decode message (2,4), then check by encoding. Count all trees on four labeled vertices.

Solution: Initially labels 1 and 3 are absent, so remove the smaller 1 and add edge 1-2. With message (4), surviving label 2 is smallest absent; add 2-4 and remove 2. Join survivors 3 and 4. The edges are {1,2}, {2,4}, {3,4}; encoding removes leaf 1 then leaf 2, recovering (2,4). Every two-entry word from four labels decodes by the same rule, and the inverse recovers it, giving $4 \times 4 = 16$ trees.

Boundary: If labels are ignored, there are only two shapes on four vertices: a path and a three-leaf star. The sixteen count must not be reported as sixteen unlabeled shapes.

Hints to reveal in order

1. At decoding, choose the smallest surviving label absent from the remaining message.
2. Join that label to the first message entry, remove the label and entry, then connect the last two survivors.

Extensions

- For five labels, choose a three-entry code first and predict which vertices will be leaves; degree equals one plus its number of code occurrences.
- Compare Prüfer's bijection with determinant-based spanning-tree counting in AD-05; the latter handles restricted edge sets but is less direct as a communication code.

Connection and its limits

Bridge type: exact-special-case

Preserves: Labeled dots and string edges are actual trees; deleting leaves and reconstructing them are the encoding and inverse algorithm.

Limits: The count is for labeled simple trees. Rotating a drawing does not produce a new tree, but changing which named vertices connect can.

Sources and inspection scope

[Richard P. Stanley, Topics in Algebraic Combinatorics](#)

Chapter 9 appendix, three combinatorial proofs; Prüfer proof and Cayley count

Inspected: relevant text inspected; exact $n=4$ encoding and decoding proved below

Prior use / what changes: Week 10 uses routes in networks, not labeled-tree encoding. Network materials can be reused without treating the old route puzzle as this new investigation.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-05 / Participation and launch

How many minimal networks survive?

MSC primary 05 - Combinatorics | Secondary 15, 90

The opening challenge

Keep every place connected while using no unnecessary link: removing any chosen link must disconnect the network. Find every different network you can make using only these five available links. Two drawings count the same when they use the same links.

Learner choices: Choose which links survive, decide a recording scheme, invent an organizing question, and test another team's completeness argument.

Prerequisite gates

Enter: O: join four places with removable strips and trace connections.

Explore: C: distinguish connectedness, a cycle, and minimal connectedness; compare edge sets rather than drawing positions.

Explain: P: split all possibilities by whether the diagonal is selected.

Prove: L and P for the general Matrix-Tree Theorem: incidence matrices, determinants, and a determinant expansion such as Cauchy-Binet.

Reading: None for the graph; symbolic matrix reading for the optional upper stage.

Arithmetic: Count to eight; determinant stage uses signed products and sums.

Reasoning: Completeness and duplicate control matter more than arithmetic. Give each edge a distinct label.

Hard stop: A correct combinatorial count is complete mathematics; stop there when determinant prerequisites are absent.

Materials and preparation

10 minutes estimated prep. A square with vertices A,B,C,D in cyclic order, diagonal AC, and five removable edge strips. Eight small recording frames or photographs can replace repeated drawing.

A flexible hour

0-10 build and break networks; 10-15 agree on necessary links and duplicates; 15-35 enumerate minimal connected networks; 35-40 form a human cycle then remove one link; 40-55 prove the count or compute a cofactor; 55-60 compare the two descriptions. A classified eight-tree list is enough.

Questions to explore

- How many links must a minimal connected four-place network have?
- What changes when the diagonal is included?
- Can a three-link selection contain a cycle?
- Why should deleting a different matrix row and column give the same answer?

Satisfying stop: Eight drawings organized into two four-case groups, with a reason no other edge set works.

AD-05 / Facilitator reasoning

How many minimal networks survive?

The mathematical question

How can a determinant count spanning trees of a graph, and what combinatorial structure explains the answer?

For a finite connected loopless graph, any principal cofactor of its Laplacian $D-A$ equals its number of spanning trees. The core is a four-cycle with one diagonal; direct classification proves eight trees without requiring determinants.

Worked instance

Problem: Count spanning trees of edges AB, BC, CD, DA, AC and verify one Laplacian cofactor.

Solution: Without AC , deleting one of the four cycle edges gives four trees. With AC , choose AB or BC for B and CD or DA for D , giving four more. Other diagonal-containing three-edge choices leave B or D isolated. Deleting D 's row/column from the Laplacian gives $\begin{bmatrix} 3 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 3 \end{bmatrix}$, whose determinant is $15 - 4 - 3 = 8$.

Boundary: A disconnected graph has no spanning tree on all its vertices. If a second diagonal is added, its crossing is not a vertex unless declared; declaring a junction changes the graph and the count.

Hints to reveal in order

1. First handle networks without AC : delete exactly one edge of the square.
2. With AC present, choose one connection for B and one for D ; explain why those choices suffice.

Extensions

- Add the second diagonal BD , treating its crossing as not a vertex. The graph becomes K_4 , which has sixteen spanning trees by AD-04.
- Assign positive edge weights: products of tree-edge weights sum to a weighted Laplacian cofactor. This continuation requires a separately checked weighted statement before use.

Connection and its limits

Bridge type: exact-special-case

Preserves: Choosing edge subsets that connect every vertex without a cycle is exactly choosing spanning trees. The later Laplacian calculation describes this same graph.

Limits: The direct eight-case count does not prove the general determinant formula. No claim is made that all networks or optimization problems reduce to this small example.

Sources and inspection scope

[Richard P. Stanley, Topics in Algebraic Combinatorics](#)

Theorem 9.8 Matrix-Tree Theorem; chapter 9 incidence matrices

Inspected: full relevant theorem and proof discussion

Prior use / what changes: Week 10 minimizes repeated travel, whereas this family counts minimal connecting subgraphs. AD-04 shares trees but studies an encoding of all labeled trees rather than restricted-edge counts.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-06 / Participation and launch

When the usual mixing rule breaks

MSC primary 06 - Order, lattices, ordered algebraic structures | Secondary 08

The opening challenge

For two tiles, find the lowest tile at or above both (their join) and the highest tile at or below both (their meet). A tile counts as at or above/below itself. Start with picture-set cards ordered by containment, then try the five-tile diagram. For three chosen tiles a, b, c , compare a meet $(b$ join $c)$ with $(a$ meet $b)$ join $(a$ meet $c)$. Do these two recipes always agree?

Learner choices: Choose pairs, invent an order diagram, challenge whether a proposed common bound is the least or greatest one, and search for a shortcut failure.

Prerequisite gates

Enter: O: follow upward comparisons and point to common places above two tiles.

Explore: C: coordinate two partial-order searches and keep incomparability visible.

Explain: P: evaluate two parenthesized expressions one operation at a time.

Prove: P: check all pair types in M3 have a meet and join, then present one distributivity failure.

Reading: Spoken operation names suffice; symbolic expression strips are optional.

Arithmetic: None; 0 and 1 are names for bounds, not ordinary arithmetic values.

Reasoning: Abstract comparison and nested operations are substantial. Use movable intermediate-result markers.

Hard stop: Do not introduce general representation or duality theorems without set-theoretic proof prerequisites.

Materials and preparation

10 minutes estimated prep. Five labeled tiles, six arrows for $0 < a, b, c < 1$, and a separate deck of the eight subsets of three pictures. Use 'floor' and 'ceiling' markers.

A flexible hour

0-10 arrange picture sets by containment; 10-15 name common floor and ceiling; 15-35 investigate the diamond and compare operations; 35-40 participants stand at diagram vertices; 40-55 evaluate two competing expression routes; 55-60 show a counterexample. Do not require every participant to manipulate notation.

Questions to explore

- Are a and b comparable even though both sit above 0?
- Can there be several common upper bounds but only one least one?
- What is the ceiling of b and c ?
- Do both expression routes finish at the same tile?

Satisfying stop: Two legal calculation routes end at different tiles, with each intermediate result justified from the diagram.

AD-06 / Facilitator reasoning

When the usual mixing rule breaks

The mathematical question

Which order structures admit greatest common lower bounds and least common upper bounds, and why can distributivity fail?

A lattice is a partial order in which every pair has a meet and join. The five-element diamond M_3 has bottom 0, top 1, and three pairwise incomparable middle elements a, b, c . It is a lattice but is not distributive.

Worked instance

Problem: Compare a meet $(b$ join $c)$ with $(a$ meet $b)$ join $(a$ meet $c)$ in M_3 .

Solution: Distinct middle elements have only 1 as a common upper bound and only 0 as a common lower bound. Thus b join $c = 1$, so the first result is a meet $1 = a$. Each of a meet b and a meet c is 0, so the second result is 0. Since a differs from 0, distributivity fails. Equal middle elements and pairs involving 0 or 1 also have obvious unique bounds, verifying this is still a lattice.

Boundary: A poset with two incomparable upper bounds and no least common upper bound for some pair is not a lattice. 'There is something above both' is insufficient.

Hints to reveal in order

1. List all common upper bounds before choosing the least; do the dual search below.
2. For three distinct middle tiles, compute b join c first on one route, and a meet b plus a meet c first on the other.

Extensions

- For divisors of 12 ordered by divisibility, compute meet as gcd and join as lcm; compare the diagram with subsets.
- Investigate the lattice of subspaces of a two-dimensional vector space: distinct lines supply the same distributivity failure. This exact connection requires a real linear-algebra gate.

Connection and its limits

Bridge type: faithful-representation

Preserves: Tiles are elements; upward arrows encode order, including transitive comparisons. Finding the lowest common ceiling and highest common floor is computing join and meet.

Limits: The order diagram is not a road map: traversing downward then upward does not establish comparison. One counterexample distinguishes lattice from distributive lattice, not all varieties of lattices.

Sources and inspection scope

[Burris and Sankappanavar, A Course in Universal Algebra](#)

Chapter I, §§1 and 3, lattice definitions and distributive/modular lattices

Inspected: relevant definitions and examples inspected; the M_3 calculation is explicit below

Prior use / what changes: No exact current-week match. Tiling flip orders may eventually produce lattices, but that additional structure must be proved separately rather than assumed from Week 1.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-07 / Participation and launch

Everything your starting clues force

MSC primary 06 - Order, lattices, ordered algebraic structures | Secondary 03, 08

The opening challenge

Choose any starting tokens. You may add a token whenever a displayed rule allows it, and you never remove tokens. Keep going until no new token can be added. Can two different legal orders end with different collections? Which starting collections already need no changes?

Learner choices: Choose starting sets, choose which eligible rule fires first, challenge a claimed stopping point, and invent two closed collections whose union triggers something new.

Prerequisite gates

Enter: O: place named tokens and follow one visible if-then rule.

Explore: C: track several rules without losing previously obtained tokens.

Explain: P: distinguish a set already closed from a set still requiring additions.

Prove: P: prove termination by finite growth and minimality by induction on additions.

Reading: Three short rule cards, orally supported and drawn with arrows.

Arithmetic: Count at most four tokens; no calculation.

Reasoning: Conditional language and tracking prerequisites matter; put every rule and obtained token in view.

Hard stop: Keep rules positive and finite until the fixed-point argument is secure; arbitrary logical consequence is a separate advanced subject.

Materials and preparation

8 minutes estimated prep. Tokens a,b,c,d and rule cards: a gives b; b together with c gives d; d gives c. Prepare a recording strip for each starting set and a closed/not-yet-closed marker.

A flexible hour

0-10 explore tokens and make informal recipes; 10-15 demonstrate one implication and one two-token prerequisite; 15-35 compare orders and catalog closed sets; 35-40 pass prerequisite cards around a circle; 40-55 investigate intersections and unions; 55-60 present one surprising forced addition. The full catalog is optional.

Questions to explore

- What does starting with only a force?
- Can a rule be used before all its inputs are available?
- Can an extra starting token ever make you lose an eventual token?
- Why can two individually finished collections have an unfinished union?

Satisfying stop: Two finished collections combine to force a new token, and the learner can explain why order does not change the final closure.

AD-07 / Facilitator reasoning

Everything your starting clues force

The mathematical question

Why do closure operations generate lattices of closed sets, and why does joining two closed sets sometimes create new consequences?

On a finite set, repeatedly apply implications X implies y by adding y whenever every member of X is present. The least fixed point containing the starting set is extensive, monotone, and idempotent. Intersections of closed sets are closed; their join is the closure of their union.

Worked instance

Problem: Find closures of $\{a\}$, $\{c\}$, and $\{a,c\}$; decide whether closed sets stay closed under union.

Solution: Starting at $\{a\}$, the first rule adds b and nothing else fires, so closure is $\{a,b\}$. Starting at $\{c\}$ adds nothing. Their union $\{a,b,c\}$ activates b -and- c , adding d , so its closure is $\{a,b,c,d\}$. Thus union need not be closed. Any closed set containing the starting set must contain each forced addition, by induction; hence every completed run produces the same least such set.

Boundary: If a rule deletes a prerequisite, order can matter: from $\{a\}$, 'replace a by b ' and 'replace a by c ' give different terminal sets. The addition-only hypothesis is essential.

Hints to reveal in order

1. Underline prerequisites already present and mark each newly obtained token.
2. For a union surprise, try one closed set containing b and another containing c .

Extensions

- Enumerate all closed sets: empty, $\{b\}$, $\{c\}$, $\{c,d\}$, $\{a,b\}$, $\{b,c,d\}$, $\{a,b,c,d\}$. Verify intersections remain on the list.
- Connect to subalgebra generation or linear span only after defining their operations; they are further exact closure systems, not identical problem stories.

Connection and its limits

Bridge type: exact-special-case

Preserves: Chosen token sets and forced additions are an actual finite closure operator. Order of processing eligible rules does not change the eventual closed set if all eligible rules are eventually applied.

Limits: These are monotone addition-only implications, not arbitrary instructions with deletion or negation. The result does not promise order independence for every rule game.

Sources and inspection scope

[Burris and Sankappanavar, A Course in Universal Algebra](#)

Chapter I §5, closure operators and closed sets

Inspected: relevant definition inspected; finite rule-system claims proved by the explicit procedure

Prior use / what changes: Week 5 concerns clue uniqueness; this family studies closure under explicit implication rules. No exact reuse is claimed; actual prior encounter remains governed by the use log.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-08 / Participation and launch

Which numbers may share a name?

MSC primary 08 - General algebraic systems | Secondary 11, 20

The opening challenge

Rename some numbers so they share a bag. To add two bags, choose any number from each bag, add around the four-ring, and report the answer's bag. Your renaming is legal only if the answer bag never depends on the representatives chosen. Invent a legal renaming and an illegal one.

Learner choices: Choose partitions, choose adversarial representative pairs, and decide what extra identifications repair a failing renaming.

Prerequisite gates

Enter: O and C: use a supplied four-by-four addition table or move on a four-position ring.

Explore: C: choose two representatives independently and compare result bags.

Explain: P: exhibit a representative-dependent result as a counterexample, or check parity for all cases.

Prove: P and M: propagate one identified pair by adding the same residue to both; use cyclic translation.

Reading: Bag labels and a small table; oral instructions suffice.

Arithmetic: Addition modulo four, supplied concretely; odd/even reasoning for explanation.

Reasoning: Distinguish the object's name from the whole class and test all legal choices, not just one convenient representative.

Hard stop: Universal-algebra theorems about all algebras remain beyond this finite quotient investigation.

Materials and preparation

8 minutes estimated prep. Cards 0,1,2,3, transparent colored bags or drawn class circles, and an addition table modulo four. Duplicate cards let pairs of inputs be shown simultaneously.

A flexible hour

0-10 play with ring addition; 10-15 demonstrate bag-level ambiguity without giving the valid partition; 15-35 test and repair partitions; 35-40 participants become bags and choose representatives; 40-55 classify all possible legal renamings; 55-60 show either a quotient table or a decisive failure witness.

Questions to explore

- What happens if only 0 and 1 share a bag?
- Does putting 0 with 2 force 1 to share with 3?
- Could every number share one name?
- Which information survives the parity renaming?

Satisfying stop: A two-bag addition rule with a reason every representative choice agrees, plus a concrete failing partition.

AD-08 / Facilitator reasoning

Which numbers may share a name?

The mathematical question

When does identifying elements preserve an operation, so that a quotient algebra is well-defined?

For addition modulo four, an equivalence relation is a congruence when equivalent inputs always have equivalent sums. Its congruences are equality, parity, and the indiscrete relation. The specific parity quotient has two classes with addition modulo two.

Worked instance

Problem: Test bags $\{0,2\}$ and $\{1,3\}$; compare with bags $\{0,1\}, \{2\}, \{3\}$.

Solution: Even+even and odd+odd land in the even bag; even+odd lands in the odd bag, independently of representatives. The second partition fails: choosing $0+0$ gives the first bag, while $1+1$ gives $\{2\}$, even though both input bags were the same. For classification, any identified adjacent residues force all four together by translation. Identifying opposite residues forces exactly the parity partition unless further identifications are added. If no distinct residues are identified, equality remains.

Boundary: The singleton partition and the one-bag partition are both valid, although the latter discards all distinctions. Validity does not mean the quotient retains useful information.

Hints to reveal in order

1. Hold one input fixed while changing the other representative.
2. If two numbers share a bag, adding 1 to both must give two numbers that also share a bag.

Extensions

- Repeat for addition modulo six and investigate partitions corresponding to remainders modulo two or three; require new witnesses and a proof before generalizing.
- Add the operation that maps 0 to 0, 1 to 0, 2 to 1, 3 to 1. The parity partition now fails because 0 and 2 have differently classified outputs.

Connection and its limits

Bridge type: faithful-representation

Preserves: Colored bags are equivalence classes. Asking whether a bag-level sum depends on chosen representatives is exactly the congruence condition.

Limits: The experiment concerns a specified operation. A partition valid for addition may fail for an additional operation; checking one algebraic structure does not check all signatures.

Sources and inspection scope

[Burris and Sankappanavar, A Course in Universal Algebra](#)

Chapter II §5, congruences and quotient algebras

Inspected: relevant definition inspected; classification for the four-element example proved here

Prior use / what changes: Week 4 uses cyclic arithmetic; the new central question is well-defined identification and information loss. The same ring alone is a representation reuse, not a new mechanism.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-09 / Participation and launch

Two clocks, one hidden position

MSC primary 11 - Number theory | Secondary 08, 20

The opening challenge

A hidden time is one of 0 through 11. Both clocks start at 0 and move one position per step. The three-clock reports 2 and the four-clock reports 1. Find the time, then design a pair of reports that no one can solve-or explain why you cannot.

Learner choices: Choose clues, choose a search method, organize the full pair inventory, and choose changed clock sizes to challenge the conjecture.

Prerequisite gates

Enter: O and C: move two counters once for each shared step and recognize paired positions.

Explore: C: record ordered pairs without swapping clock roles.

Explain: M and P: reason about multiples, remainders, and the first joint return.

Prove: M and P: use coprimality and a modular inverse or Bezout identity for the general theorem.

Reading: Clock symbols suffice; an adult can read remainder clues.

Arithmetic: Core uses counting to twelve; later stage needs multiplication, division, gcd and lcm.

Reasoning: Track synchronized movement and distinguish uniqueness within one period from uniqueness among all integers.

Hard stop: Do not require a general Bezout proof for the complete small-cycle result; supply exact restrictions on the hidden time.

Materials and preparation

10 minutes estimated prep. Clock faces labeled 0,1,2 and 0,1,2,3; two counters; a strip numbered 0 through 11. A second pair of four- and six-position clocks is optional.

A flexible hour

0-10 explore synchronized clocks; 10-15 solve one clue collaboratively; 15-35 invent reports and build a complete table; 35-40 walk two floor rings in synchrony; 40-55 try clocks with a shared factor and explain failures; 55-60 state exactly what can and cannot be guaranteed.

Questions to explore

- When do both counters first return to 0?
- Can a paired report repeat before then?
- Does the answer remain unique if time is unrestricted?
- What must a four-clock and six-clock agree about?

Satisfying stop: A hidden time with a uniqueness argument and one impossible pair explained by parity.

AD-09 / Facilitator reasoning

Two clocks, one hidden position

The mathematical question

When can independent remainder reports be combined into one residue, and how does a common divisor obstruct compatibility?

For positive coprime moduli m, n , every residue pair has a unique solution modulo mn . For general positive moduli, a pair is solvable exactly when the residues agree modulo $\gcd(m, n)$, and any solution is unique modulo $\text{lcm}(m, n)$. The core proves the 3-and-4 case by a complete cycle.

Worked instance

Problem: Solve $t \equiv 2 \pmod{3}$ and $t \equiv 1 \pmod{4}$ for $0 \leq t < 12$; then test residues $1 \pmod{4}$ and $2 \pmod{6}$.

Solution: The times matching the first clue are 2, 5, 8, 11; only 5 is $1 \pmod{4}$. Two times sharing both residues differ by a multiple of both 3 and 4, hence by 12, so the chosen interval gives uniqueness. For the changed clocks, $t \equiv 1 \pmod{4}$ makes t odd, whereas $t \equiv 2 \pmod{6}$ makes t even; no solution exists.

Boundary: Without a bounded interval or residue-class interpretation, 5, 17, 29 and infinitely many other times satisfy the first pair. State uniqueness modulo 12, not a unique integer.

Hints to reveal in order

1. List the times matching just one report, then inspect the other clock.
2. For four and six positions, track whether both reported residues are odd or even.

Extensions

- For moduli four and six, classify all twenty-four residue pairs: exactly twelve compatible same-parity pairs occur, with period twelve.
- For three pairwise coprime clocks, combine two first and then the third; prove the construction's uniqueness rather than assuming pairwise agreement is enough in unrelated constraint problems.

Connection and its limits

Bridge type: exact-special-case

Preserves: Two counters recording one step at a time are the map from integer time to a pair of residues. Returning to both origins means a common multiple.

Limits: A twelve-state picture proves its own case, not the theorem for arbitrary moduli. Independent clocks cannot produce every pair when their sizes share a factor.

Sources and inspection scope

[Keith Conrad, The Chinese Remainder Theorem](#)

Theorem 1.1 and §2 existence/uniqueness proof

Inspected: full relevant coprime text; general compatibility is a separately explained continuation

Prior use / what changes: Week 4 follows one modular orbit and Week 9 uses common multiples. This family combines simultaneous reports and exposes compatibility; prior ring movement can reduce setup time.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-10 / Participation and launch

Square dots that also make a triangle

MSC primary 11 - Number theory | Secondary 12

The opening challenge

Use the same counters to make both a square and a triangle with one more counter in each successive row. One counter works. Find a larger collection that works both ways. Later, decide whether a proposed recipe makes a new example every time.

Learner choices: Choose which totals to test, invent a search table, explain failed candidates, and decide whether the recurrence should be trusted from examples or from an identity.

Prerequisite gates

Enter: O and C: build rows of lengths 1,2,3,... and square arrays.

Explore: M: compare triangular totals with square totals, using a supplied table if desired.

Explain: A: expand squares and translate between m,n and x,y ; a finite dot explanation remains possible earlier.

Prove: A and P: verify a polynomial identity and show the recurrence preserves positivity and strictly increases size. Completeness needs a separate descent argument.

Reading: Spoken launch; optional symbolic recurrence card.

Arithmetic: Core uses thirty-six counters; extension uses multiplication of two-digit integers and polynomial expansion.

Reasoning: Coordinate two different side-length parameters; label triangle height m and square side n separately.

Hard stop: Stop after a verified common arrangement when symbolic equations remove the learner's agency; the Pell proof is an explicit upper gate.

Materials and preparation

10 minutes estimated prep. Thirty-six counters per pair, square grid and triangular-row mat, plus a calculator or prepared arithmetic strip for large extensions. Do not prepare 1,225 physical counters.

A flexible hour

0-10 freely arrange dots; 10-15 define the exact triangle convention; 15-35 search and verify thirty-six; 35-40 physically form increasing rows; 40-55 choose a drawing-based explanation or symbolic recipe; 55-60 distinguish a verified new example from a claimed complete list. The symbolic phase is optional.

Questions to explore

- Why does a triangle of eight rows use thirty-six counters?
- Can a formula replace moving thousands of counters?
- Which quantities must remain whole numbers?
- Does a recipe that works forever necessarily find every solution?

Satisfying stop: Thirty-six dots visibly form both arrangements, with a count explaining why the match is exact.

AD-10 / Facilitator reasoning

Square dots that also make a triangle

The mathematical question

How can a geometric counting question become a Pell equation, and how does multiplication of quadratic expressions generate infinitely many solutions?

For nonnegative integers m, n , $m(m+1)/2 = n^2$ is equivalent to $(2m+1)^2 - 2(2n)^2 = 1$. Multiplication of $x+y\sqrt{2}$ by $3+2\sqrt{2}$ preserves $x^2-2y^2=1$ and sends (x,y) to $(3x+4y, 2x+3y)$.

Worked instance

Problem: Verify $m=8, n=6$ and use the recurrence to generate another triangular-square total.

Solution: Eight rows use $8 \times 9/2 = 36 = 6^2$. Set $x=17, y=12$; $x^2-2y^2=289-288=1$. The recurrence gives $x'=99, y'=70$, hence $m'=49, n'=35$. Indeed $49 \times 50/2 = 1225 = 35^2$. Algebraically $(3x+4y)^2 - 2(2x+3y)^2 = x^2 - 2y^2$, so the equation is preserved. Odd x and even y remain odd/even, allowing the conversion back.

Boundary: The identity $x^2-2y^2=1$ also has negative solutions; blindly converting them may give negative row lengths. Restrict to positive x, y for the displayed growth recipe, and do not claim the generated next value is the next smallest without a completeness proof.

Hints to reveal in order

1. Compare a list of triangular totals with 1, 4, 9, 16, 25, 36.
2. If $m(m+1)=2n^2$, multiply by four and add one to make $(2m+1)^2$.

Extensions

- In m, n coordinates the recipe is $m'=3m+4n+1$ and $n'=2m+3n+1$. Start at $m=n=1$ and verify that it reaches 8, 6 then 49, 35.
- An advanced continuation proves all positive solutions arise from powers of $3+2\sqrt{2}$. Inspect and prove the descent/completeness argument separately before reporting an exhaustive classification.

Connection and its limits

Bridge type: exact-special-case

Preserves: The same counter total admits triangular and square arrangements. The algebraic re-encoding and recurrence preserve exactly the required equality.

Limits: Producing successive larger examples does not by itself prove that every triangular-square number has been found, or that no omitted one lies between generated examples.

Sources and inspection scope

[Keith Conrad, Pell's Equation, I](#)

§3 Theorem 3.1, triangular-square numbers; §4 quadratic-unit multiplication

Inspected: full relevant text

Prior use / what changes: Year-1 material includes figurate numbers; whether this exact triangular-square investigation was used is unknown. Current Weeks 1-10 do not contain this Pell mechanism.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-11 / Participation and launch

Four symbols where every nonzero number divides

MSC primary 12 - Field theory and polynomials | Secondary 13, 94

The opening challenge

In this arithmetic, two copies of the same coefficient cancel, and t times t becomes $1+t$. Find a partner for each nonzero symbol so their product is 1. Can ordinary arithmetic modulo four do the same? You may derive products with counters or consult a checked table.

Learner choices: Choose a product to derive, choose how to record inverses, test a proposed relabeling to modulo-four arithmetic, and challenge whether multiplication can lose information.

Prerequisite gates

Enter: O and C: manipulate two-position cards and cancel matching pairs; read a supplied multiplication table.

Explore: M or A: apply distributivity and reduce repeated terms with $1+1=0$.

Explain: P: identify an inverse for each of the three nonzero elements.

Prove: A and P: justify arithmetic as a polynomial quotient, or verify all finite field axioms from tables without omitting associativity.

Reading: Four symbol labels and short rules; adult can mediate table use.

Arithmetic: Addition of bits; later polynomial multiplication and cancellation, with no fractions required.

Reasoning: An unfamiliar operation is not ordinary integer arithmetic. Keep the two models, F4 and mod four, visually separate.

Hard stop: Do not call the system a field solely because a few products work; either supply quotient justification or explicitly keep the investigation at the inverse-table result.

Materials and preparation

15 minutes estimated prep. Cards 00,10,01,11 labeled 0,1,t,u where $u=1+t$; two-color coefficient counters; multiplication workspace. Prepare a separate modulo-four table for comparison.

A flexible hour

0-10 explore two-bit addition; 10-15 demonstrate distributivity and the t -square rule; 15-35 complete nonzero products and inverse pairs; 35-40 walk the multiplication-by- t cycle; 40-55 compare with modulo four and discuss irreducibility; 55-60 show the precise difference between the systems. Quotient proof is optional.

Questions to explore

- What is $t(1+t)$?
- Can multiplying by a nonzero symbol send two different inputs to the same output?
- How many times must you multiply by t to return to 1?
- Why cannot simply renaming 0,1,2,3 repair modulo-four multiplication?

Satisfying stop: A complete nonzero multiplication triangle and a clear reason every nonzero symbol can be undone, unlike modulo four.

AD-11 / Facilitator reasoning

Four symbols where every nonzero number divides

The mathematical question

How can a field have four elements when arithmetic modulo four has zero divisors?

The polynomial t^2+t+1 has no root over the two-element field and is irreducible. Its quotient $F_2[t]/(t^2+t+1)$ is a field with elements $0, 1, t, 1+t$. Arithmetic uses $1+1=0$ and $t^2=t+1$.

Worked instance

Problem: Find all nonzero products and inverses in $\{1, t, u\}$, $u=1+t$.

Solution: The identity leaves every symbol fixed. $t^2=u$. Also $tu=t+t^2=t+(t+1)=1$. Finally $u^2=(1+t)^2=1+t^2=t$ because the two cross terms cancel. Thus 1 is self-inverse and t, u are mutual inverses; multiplication by t cycles $1, t, u$. In contrast, modulo four, $2 \times 2 = 0$, so 2 has no multiplicative inverse.

Boundary: Using the reducible modulus t^2 instead would make nonzero t square to zero. The same four coefficient pairs would then form a ring but not a field.

Hints to reveal in order

1. Expand first, then replace t^2 with $t+1$ and cancel duplicate terms.
2. Once t times u equals 1, use it to test whether either multiplication can collapse distinct inputs.

Extensions

- Prove the Frobenius map $z \rightarrow z^2$ fixes 0, 1 and exchanges t, u ; applying it twice restores every element.
- Construct a field of eight elements only after choosing and checking an irreducible cubic. Having eight labels by itself specifies neither the operations nor a field.

Connection and its limits

Bridge type: faithful-representation

Preserves: Two-bit cards represent polynomial coefficient pairs. XOR addition and the stated multiplication reduction are exactly finite-field operations, not numerals interpreted modulo four.

Limits: Table play alone does not prove the existence or uniqueness of every prime-power field. The reason this quotient is a field depends on irreducibility.

Sources and inspection scope

[Keith Conrad, Finite Fields](#)

Theorem 1.1 and warning on p.1 about $Z/(p^n)$

Inspected: full relevant text; the specific four-element arithmetic is checked directly

Prior use / what changes: Week 2 binary addition and Week 4 modular rings are preparatory representations. This family introduces irreducible-polynomial field arithmetic, a distinct mechanism.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-12 / Participation and launch

A cube that ruler and compass cannot double

MSC primary 12 - Field theory and polynomials | Secondary 51

The opening challenge

A unit cube has volume one. Construct a segment that would be the edge of a cube with volume two, using only unmarked straightedge and compass. First construct $\sqrt{2}$ exactly and compare that problem. If your best drawing is close, decide what would certify exactness or prove impossibility.

Learner choices: Choose constructions to attempt, choose coordinate descriptions, compare permitted-tool variations, and locate the precise step at which a proposed impossibility argument needs a theorem.

Prerequisite gates

Enter: V: use compass and unmarked straightedge to construct a square diagonal; participation need not begin with fields.

Explore: A: solve linear and quadratic equations and distinguish exact expressions from decimals.

Explain: A and P: show the doubled-volume edge must satisfy $x^3=2$; prove the cubic has no rational root.

Prove: X and P: field extensions, minimal-polynomial degree, and the tower law are essential for the impossibility proof.

Reading: A symbolic proof sheet, read together if necessary.

Arithmetic: Squares, cubes, rational arithmetic, polynomial degree; calculator only for comparison with approximations.

Reasoning: Distinguish failure to find a construction from a theorem ruling all legal constructions out.

Hard stop: The complete obstruction stays at the field-extension gate. Younger participants may finish with exact square-root constructions, without being told they have proved cube-duplication impossibility.

Materials and preparation

10 minutes estimated prep. Compasses, unmarked straightedges, unit segment, graph paper, and one page showing equations of lines/circles. A prepared field-tower diagram reduces copying.

A flexible hour

0-10 construct and compare lengths; 10-15 fix exact rules and the volume target; 15-35 derive line/circle equations and square-root steps; 35-40 reset with a physical cube-volume comparison; 40-55 follow the field-degree obstruction at the appropriate gate; 55-60 state the conclusion with its tool hypotheses. Exact $\sqrt{2}$ is an earlier stop.

Questions to explore

- Does doubling the edge double the volume?
- What new equation appears when a line meets a circle?
- Why can two circle equations be reduced by subtraction?
- Could a degree-three number fit inside a tower whose total degree is a power of two?

Satisfying stop: Either an exact square-root construction with justification, or the complete degree-three contradiction with its tool restrictions stated.

AD-12 / Facilitator reasoning

A cube that ruler and compass cannot double

The mathematical question

How can algebra certify an impossible geometric construction even when approximation is easy?

Starting from a unit segment with unmarked straightedge and compass, constructible coordinates lie in a tower of extensions of degree at most two over $\mathbb{Q}$. Therefore any constructible real algebraic number has degree a power of two over $\mathbb{Q}$. The irreducible cubic x^3-2 excludes the exact doubled-cube edge.

Worked instance

Problem: Compare exact construction of $\sqrt{2}$ with exact construction of the positive root of $x^3=2$.

Solution: The unit-square diagonal has length $\sqrt{2}$ by Pythagoras. A doubled cube needs edge $\alpha=\sqrt[3]{2}$. A rational root of x^3-2 would be an integer dividing 2, and none of $\pm 1, \pm 2$ is a root; a cubic without rational roots is irreducible over $\mathbb{Q}$, so α has degree three. Each permitted intersection adds at most a quadratic extension. By the tower law, the degree of any constructible coordinate divides a power of two, contradicting degree three.

Boundary: The power-of-two degree condition is necessary, not sufficient for an arbitrary algebraic number to be constructible. Also, using a marked ruler changes the permitted operations, so this theorem no longer settles the task.

Hints to reveal in order

1. Draw a unit square: its diagonal solves a quadratic equation.
2. For the obstruction, track coordinates after each operation, not the visual sophistication of the whole drawing.

Extensions

- Construct $\sqrt{3}$ from a right triangle with legs $\sqrt{2}$ and 1, then describe its quadratic tower.
- Investigate trisection of a 60-degree angle through its cubic cosine equation, preserving the distinction between some trisectable angles and a universal trisection method.

Connection and its limits

Bridge type: exact-special-case

Preserves: The permitted lines and circles are the classical construction operations; coordinate equations faithfully track the algebraic extensions they create.

Limits: A rough physical drawing is an approximation. The impossibility concerns exact finite constructions with the specified tools, not marked rulers, origami, or numerical methods.

Sources and inspection scope

[Thomas Judson, Abstract Algebra: Theory and Applications](#)

§21.3, Lemma 21.3.6, Theorem 21.3.7, doubling-the-cube subsection

Inspected: full relevant text; correct power-of-two statement includes degree one, unlike the page's $k>0$ wording

Prior use / what changes: Week 1 shape manipulation is unrelated to field-degree obstructions. This deliberately keeps an advanced proof gate rather than presenting an elementary approximation exercise as field theory.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-13 / Participation and launch

The monomials that survive the forbidden corners

MSC primary 13 - Commutative algebra | Secondary 05, 15

The opening challenge

A marked forbidden point spoils itself and every point reachable by moving right or up from it. Mark the forbidden starts $(2,0)$, $(1,1)$, $(0,3)$. Which places survive? Convince a partner that drawing a larger grid will not uncover more survivors. Later, name a right step x and an up step y .

Learner choices: Choose how to shade, how to certify the infinite region, how to multiply survivor positions, and which generator to move for a new family member.

Prerequisite gates

Enter: O and V: move right/up on a grid and recognize a forbidden region.

Explore: C: classify all surviving positions and investigate how the boundary changes.

Explain: P: prove every point outside the displayed survivors is forbidden, including points beyond the drawn window.

Prove: A and L: understand polynomial linear combinations and basis independence for the formal quotient interpretation.

Reading: Picture directions suffice initially; polynomial labels optional until the upper stage.

Arithmetic: Small nonnegative coordinates; exponent addition for algebraic extension.

Reasoning: Reason about an unbounded grid from a finite boundary; distinguish geometric squares from algebraic coefficient values.

Hard stop: The low entrance establishes an exact monomial-membership puzzle; abstract quotient dimension requires linear combinations and polynomial independence.

Materials and preparation

10 minutes estimated prep. A grid showing coordinates 0 through 5 in each direction, movable markers at $(2,0)$, $(1,1)$, $(0,3)$, transparent northeast shading, and four survivor tokens. Axes begin at zero.

A flexible hour

0-10 explore right/up paths; 10-15 set the three forbidden starts; 15-35 find survivors and test multiplication moves; 35-40 enact coordinate steps; 40-55 compare changed corners or translate to polynomial quotients; 55-60 give a finite certificate for the whole infinite grid.

Questions to explore

- Can a point above a forbidden point ever recover?
- Which survivor dies when multiplied by x ?
- Is any forbidden start redundant?
- How would the answer change if xy were allowed?

Satisfying stop: Four surviving positions and a complete reason there are no hidden survivors beyond the paper.

AD-13 / Facilitator reasoning

The monomials that survive the forbidden corners

The mathematical question

How does a monomial ideal turn algebraic quotient structure into a staircase of lattice points, and what does that staircase measure?

In $k[x,y]/(x^2,xy,y^3)$, for a field k , the residue classes of $1,x,y,y^2$ form a basis. A monomial belongs to the ideal exactly when it is divisible by at least one displayed generator. Multiplication by x or y moves one step on the exponent grid, possibly to zero.

Worked instance

Problem: Find the survivors and explain why they form a basis of the stated quotient.

Solution: At $i=0$, heights $0,1,2$ survive; at $i=1$, only height 0 survives; $i \geq 2$ never survives. The monomials are $1,y,y^2,x$. Every polynomial reduces to their linear combination by removing divisible terms. Every polynomial in the ideal contains only monomials divisible by x^2,xy , or y^3 , so no nonzero linear combination of the four survivors belongs to the ideal. This proves spanning and independence, hence dimension four.

Boundary: If the sole relation is $x-y=0$, neither x nor y should be erased: their classes are equal, not zero. The forbidden-region rule applies specifically to monomial generators.

Hints to reveal in order

1. Separate the vertical axis, the column $i=1$, and all columns $i \geq 2$.
2. Any column $i \geq 2$ is killed by $(2,0)$; in column $i=1$, every positive height is killed by $(1,1)$.

Extensions

- Remove generator xy . The quotient $k[x,y]/(x^2,y^3)$ has six survivor monomials in a two-by-three rectangle; compare the multiplication maps.
- Move the corners and count survivors by rows. Gröbner bases extend this normal-form approach to more general ideals, but require a separately defined term order and reduction theory.

Connection and its limits

Bridge type: faithful-representation

Preserves: Grid point (i,j) represents $x^i y^j$. Forbidden northeast regions are exactly the multiples of ideal generators; surviving points encode a quotient basis.

Limits: Grid play describes a monomial ideal, not arbitrary polynomial ideals. For relations involving sums, erasing every involved monomial would destroy the quotient's actual information.

Sources and inspection scope

[J. S. Milne, Algebraic Geometry](#)

Chapter 1 rings/ideals; chapter 2 Hilbert basis theorem and quotient-ring framework

Inspected: relevant framework inspected; this specific monomial basis has a self-contained proof below

Prior use / what changes: Week 1 also uses boards, but these positions encode monomial divisibility rather than tilings. The family adds commutative-algebra content rather than relabeling a packing puzzle.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-14 / Participation and launch

A nonzero symbol whose square vanishes

MSC primary 13 - Commutative algebra | Secondary 14, 26

The opening challenge

Invent numbers $a+b\varepsilon$ under the exact rule $\varepsilon^2=0$. Addition works by columns; multiplication distributes before ε^2 terms disappear. Find nonzero numbers whose product is zero, and find surprising numbers that still have multiplicative inverses. You may choose your own coefficients.

Learner choices: Choose trial elements, design a failed cancellation argument, classify invertibility, and decide what is lost when every ε coefficient is discarded.

Prerequisite gates

Enter: A and F: distribute two linear expressions and add signed rational coefficients.

Explore: A: reduce products using $\varepsilon^2=0$ and distinguish formal equality from substitution $\varepsilon=0$.

Explain: P: show ε cannot vanish as a residue class and solve for an inverse.

Prove: A and X: quotient rings and ideals; polynomial independence justifies the two-coordinate normal form.

Reading: Symbolic rule card; paired oral explanation is useful.

Arithmetic: Signed addition, multiplication, and rational reciprocals.

Reasoning: Hold an unfamiliar but consistent algebra separate from ordinary real arithmetic; errors in ε^2 usually expose this conflation.

Hard stop: This is an algebra-gated investigation. Do not replace the relation with ' ε is almost zero' to lower the prerequisite.

Materials and preparation

8 minutes estimated prep. Two-column coefficient mats labeled ordinary and ε , signed counters for integer trials, and paper for rational examples. Prepare comparison cards for $\mathbb{Q}[t]/(t)$ and $\mathbb{Q}[t]/(t^2)$.

A flexible hour

0-10 explore two-column addition and products; 10-15 agree that ε is a formal basis symbol; 15-35 hunt zero divisors and inverse pairs; 35-40 reset with pairs of ordinary/ ε roles; 40-55 derive a general inverse rule or polynomial first-order expansion; 55-60 present one statement distinguishing this ring from $\mathbb{Q}$.

Questions to explore

- Why does $\varepsilon^2=0$ not force $\varepsilon=0$ here?
- Can $1+\varepsilon$ be invertible despite containing a nilpotent part?
- Which coefficient of a product decides whether it could equal 1?
- Do equations $t=0$ and $t^2=0$ have the same rational points but different quotient rings?

Satisfying stop: One checked inverse pair and one nonzero zero divisor, with an explanation of why ordinary cancellation is unavailable.

AD-14 / Facilitator reasoning

A nonzero symbol whose square vanishes

The mathematical question

What algebraic information is retained by nilpotents even when ordinary point sets look identical?

The dual-number ring $\mathbb{Q}[\varepsilon]/(\varepsilon^2)$ consists uniquely of $a+b\varepsilon$ with rational a, b and product $(a+b\varepsilon)(c+d\varepsilon)=ac+(ad+bc)\varepsilon$. Its element ε is nonzero but $\varepsilon^2=0$; $a+b\varepsilon$ is invertible exactly when a is nonzero.

Worked instance

Problem: Invert $2+3\varepsilon$ and compare its behavior with ε .

Solution: For an inverse $c+d\varepsilon$, the equations are $2c=1$ and $2d+3c=0$. Thus $c=1/2, d=-3/4$, and direct multiplication gives 1. For a general nonzero a , the inverse is $1/a-(b/a^2)\varepsilon$. If $a=0$, the ordinary coefficient of every product is zero, so no inverse exists. The element ε is nonzero because the polynomial t is not a multiple of t^2 , although its square is in that ideal.

Boundary: Evaluating every dual number at $\varepsilon=0$ collapses ε to zero. That map preserves arithmetic but forgets information; it is not an injective identification of dual numbers with rational numbers.

Hints to reveal in order

1. Compare ordinary coefficients before trying to solve the ε coefficient equation.
2. Multiply $(a+b\varepsilon)(c+d\varepsilon)$ and set its two coefficients equal to those of 1.

Extensions

- Expand $(1+\varepsilon)^n$. For nonnegative integer n all terms of degree two or higher vanish, leaving $1+n\varepsilon$; prove this by induction.
- For a polynomial f , expansion gives $f(a+b\varepsilon)=f(a)+b f'(a)\varepsilon$. This identity is exact in the ring; interpreting it as differential geometry requires a stated calculus/tangent-space gate.

Connection and its limits

Bridge type: exact-special-case

Preserves: Two coefficient boxes encode an actual quotient ring. The ε^2 rule is an exact algebraic relation, not a numerical approximation.

Limits: ε is not a tiny nonzero rational or real number. The activity does not establish schemes, tangent functors, or general infinitesimal geometry; those need additional definitions.

Sources and inspection scope

[J. S. Milne, Algebraic Geometry](#)

Chapter 2, discussion of nilpotents/reduced quotients preceding Theorem 2.16 and Aside 2.24

Inspected: full relevant nilpotent/radical discussion; dual-number computations proved directly

Prior use / what changes: No current Week 1-10 equivalent. AD-13 supplies quotient normal forms, but this family investigates invertibility and nilpotent information rather than monomial counting.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-15 / Participation and launch

Find every rational point by aiming a line

MSC primary 14 - Algebraic geometry | Secondary 11, 51

The opening challenge

Aim a line from $(-1,0)$ across the unit circle. Choose a rational slope and find the second intersection exactly. Can this method find every point on the circle whose coordinates are fractions? Find any point it misses and explain why.

Learner choices: Choose slopes, search for small denominators, invent a way to recover the slope from a point, and decide how to represent the exceptional case.

Prerequisite gates

Enter: V: draw a circle and lines through one marked point; independent arithmetic can initially be deferred.

Explore: F and A: understand rational slopes and solve a quadratic after substituting a line equation.

Explain: A and P: factor out the already known intersection and account for the exceptional point.

Prove: A and P: derive both formulas and show each rational point in the claimed domain is recovered.

Reading: A coordinate diagram and equations; orally supported derivation is possible.

Arithmetic: Fractions, squares, polynomial factoring, and signed coordinates.

Reasoning: Switch between geometric point and parameter without forgetting which domain excludes a point.

Hard stop: A complete rational parametrization needs symbolic algebra. Projective or scheme language is a later continuation, not required decoration.

Materials and preparation

10 minutes estimated prep. Unit-circle template with $(-1,0)$ marked, ruler, coordinate grid, and a slope table. A calculator can check arithmetic but not replace exact fractions.

A flexible hour

0-10 draw and compare secants; 10-15 state the rational-coordinate question; 15-35 derive or use the formula and exchange slope challenges; 35-40 move along a large circle while tracing lines from the fixed point; 40-55 prove the converse and connect to triples; 55-60 show an exact point and its recovery rule.

Questions to explore

- Why is one quadratic root already known before calculating?
- What happens at slope zero?
- How do you recover t from a given point?
- Which line through $(-1,0)$ has no distinct second intersection?

Satisfying stop: One exact fractional point and a reversible slope construction, including the missing-point explanation.

AD-15 / Facilitator reasoning

Find every rational point by aiming a line

The mathematical question

When does a geometric projection give a rational parametrization, and how must exceptional points be handled?

For the circle $x^2+y^2=1$, a line of rational slope t through $(-1,0)$ meets it again at $((1-t^2)/(1+t^2), 2t/(1+t^2))$.

Conversely every rational point except $(-1,0)$ yields rational $t=y/(x+1)$. This is an exact parametrization with one explicitly handled exception.

Worked instance

Problem: Use slope $t=1/2$ and prove the general formula covers every rational point other than $(-1,0)$.

Solution: Substitution gives $(x+1)((1+t^2)x+t^2-1)=0$. The second root is $x=(1-t^2)/(1+t^2)$; $y=t(x+1)=2t/(1+t^2)$. At $t=1/2$ this gives $(3/5, 4/5)$, and $9/25+16/25=1$. Conversely, for a rational circle point with $x \neq -1$, $t=y/(x+1)$ is rational and its line recovers the point. The sole circle point with $x=-1$ is $(-1,0)$, excluded from the finite-slope formula.

Boundary: A line tangent at $(-1,0)$ is vertical and has no finite slope; its intersection is the known point counted twice. Omitting this exception would make the 'every point' claim false.

Hints to reveal in order

1. Substitute $y=t(x+1)$ into $x^2+y^2=1$.
2. The result has factor $x+1$ because $(-1,0)$ lies on both curves.

Extensions

- Take $t=p/q$ with integers p,q and obtain the triple $(q^2-p^2, 2pq, q^2+p^2)$. State the further coprimality/parity conditions before claiming primitive triples.
- Try a different conic with a known rational point. A general cubic does not inherit this parametrization merely because a line can be drawn through it.

Connection and its limits

Bridge type: exact-special-case

Preserves: The drawn circle, rational-slope line, and coordinate calculation are an actual rational map and its inverse on the specified punctured circle.

Limits: A drawing alone cannot certify rational coordinates. Not every algebraic curve admits such a parametrization, and the missing point must not be concealed.

Sources and inspection scope

[Ravi Vakil, The Rising Sea, January 29, 2015 draft](#)

§6.5, printed pp.191-193, rational circle parametrization

Inspected: full relevant passage; source uses $(1,0)$, while this reflected version is rederived below

Prior use / what changes: Week 9 uses rational slopes for billiards; this family uses them to parametrize solutions of an algebraic equation. The geometric language is related, the mathematical question is new.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-16 / Participation and launch

A curve made of seven finite-field points

MSC primary 14 - Algebraic geometry | Secondary 11, 12

The opening challenge

Place a counter at (x,y) exactly when y times y and x cubed minus x have the same remainder after division by five. Find every permitted point without testing all twenty-five pairs separately. What symmetry can shorten the search?

Learner choices: Choose whether to search by columns or square residues, predict partners of known points, organize a completeness certificate, and compare a changed modulus or polynomial.

Prerequisite gates

Enter: M and V: use a five-by-five coordinate grid and supplied residue/square tables.

Explore: M: cube, subtract, and compare residues, with a calculator or lookup table if needed.

Explain: P: exhaust all five x -values while using the complete square-residue table.

Prove: A and X: projective coordinates for the extra point; algebraic nonsingularity or elliptic group-law proof is a separate gate.

Reading: Coordinates and one equation; partner can record values.

Arithmetic: Modulo-five addition/multiplication; supplied tables lower calculation demand without changing the mathematical question.

Reasoning: Distinguish the number of y -solutions from the number of square residues, and affine from projective counting.

Hard stop: Finish with the seven affine points when projective coordinates are unavailable; do not quietly add an unexplained eighth point.

Materials and preparation

10 minutes estimated prep. Five-by-five grids labeled 0 through 4, counters, a square table, and a separate ordinary real-coordinate sketch for comparison. Prepare an infinity card only for the projective stage.

A flexible hour

0-10 explore residue multiplication; 10-15 place two sample points and reject one; 15-35 complete the curve and explain missing columns; 35-40 stand in residue positions paired by negation; 40-55 change the equation or introduce projective infinity at its real gate; 55-60 report exactly which point set was counted.

Questions to explore

- Which remainders can a square have?
- Why does a nonzero y normally have a partner?
- When do y and $-y$ coincide?
- Does a smooth real-looking drawing determine the number of points modulo five?

Satisfying stop: A seven-point picture with a five-column table proving completeness and a visible $y \mapsto -y$ symmetry.

AD-16 / Facilitator reasoning

A curve made of seven finite-field points

The mathematical question

How does the choice of ground field change the solutions of a polynomial equation, and what structure remains when a continuous-looking curve becomes a finite set?

Over F_5 , the affine equation $y^2 = x^3 - x$ has seven solutions. Its projective completion $Y^2Z = X^3 - XZ^2$ has one point with $Z=0$, giving eight projective F_5 -points. The short Weierstrass discriminant is nonzero modulo five, so this is a nonsingular elliptic curve with its marked point at infinity.

Worked instance

Problem: Find every affine F_5 -point of $y^2 = x^3 - x$ and, if projective coordinates are understood, the additional point at infinity.

Solution: For $x=0,1,2,3,4$ the right sides are $0,0,1,4,0$. Square roots give $(0,0), (1,0), (2,1), (2,4), (3,2), (3,3), (4,0)$, seven points. At $Z=0$ the homogeneous equation forces $X=0$; a projective point cannot have all coordinates zero, so $Y \neq 0$ and all such triples represent the single point $[0:1:0]$. The projective total is eight. The discriminant is $-16(4(-1)^3) = 64$, which is 4 modulo 5 and nonzero, as required for nonsingularity.

Boundary: Using modulus four would not give a field and changes square-root behavior; formulas requiring division by any nonzero residue would fail. Counting affine points and reporting eight without explaining infinity is also incorrect.

Hints to reveal in order

1. Compute the squares of $0,1,2,3,4$ once and reuse that table.
2. For each x , compare $x^3 - x$ with $0,1,4$, the possible square residues.

Extensions

- Over F_7 the same curve has seven affine points and one point at infinity; verify independently rather than assume the total follows the modulus.
- With field division and the actual group-law formulas, investigate addition of points. Associativity is a theorem, not a consequence of the finite picture; keep it a separately sourced continuation.

Connection and its limits

Bridge type: exact-special-case

Preserves: Coordinates are residues with arithmetic modulo five; placed tokens are exactly the rational points of the displayed affine curve over that field.

Limits: The grid is not a sampled real curve and adjacent dots should not be joined as if continuity were present. Point enumeration alone does not establish the elliptic-curve group law or cryptographic security.

Sources and inspection scope

[J. S. Milne, Elliptic Curves, second edition](#)

Introduction, printed p.1: projective Weierstrass equation, nonsingularity condition, and distinguished point; I §1 Example 1.15 for the point at infinity

Inspected: full relevant introductory statement and point-at-infinity example; finite enumeration proved below; broader finite-field theory is continuation

Prior use / what changes: Week 4 supplies modular arithmetic, but polynomial curves over fields are a new mathematical object. AD-15 contrasts rational parametrization with changing the ground field.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-17 / Participation and launch

The average stays, the disagreement shrinks

MSC primary 15 - Linear and multilinear algebra; matrix theory | Secondary 37, 39

The opening challenge

Put markers at two chosen numbers. In one round, each moves one quarter of the old gap toward the other. Both move using the old positions. Predict where they will settle and how far apart they will be after several rounds without doing every round.

Learner choices: Choose starting pairs, compare pairs with equal midpoint or equal gap, invent a faster prediction, and decide whether exact meeting ever occurs.

Prerequisite gates

Enter: O and V: move two markers on a number line toward one another by a demonstrated quarter of the gap.

Explore: F: use halves and quarters to record exact positions.

Explain: A and P: calculate sum/difference before and after one move, or use a symmetric geometric argument.

Prove: A and P: induct on iteration number; L for the eigenvector and diagonalization interpretation.

Reading: Number line and one move rule; full matrix notation optional.

Arithmetic: Fractions and signed difference; start with multiples of four to delay awkward denominators.

Reasoning: Both new positions must be computed from the old pair simultaneously, not sequentially.

Hard stop: Convergence and its error estimate are meaningful without general spectral theory; arbitrary networks require new hypotheses.

Materials and preparation

5 minutes estimated prep. Large number line, two differently colored markers, a midpoint marker, and old/new recording columns. Prepare a quarter-gap strip or folding paper for the geometric move.

A flexible hour

0-10 explore marker moves; 10-15 establish simultaneous updating; 15-35 record pairs and seek invariants; 35-40 two participants act out shrinking gaps; 40-55 derive an iteration formula or change the step fraction; 55-60 explain the fixed midpoint and shrinking disagreement. Eigenvector notation is optional.

Questions to explore

- What never changes?
- How much gap is removed in one round?
- Can the markers meet after finitely many rounds?
- What would change if each moved three quarters of the gap?

Satisfying stop: A prediction for any round from the midpoint and one halving rule, with a reason exact meeting need not occur.

AD-17 / Facilitator reasoning

The average stays, the disagreement shrinks

The mathematical question

How does decomposing a linear transformation into invariant directions explain every iterate without repeated calculation?

The real linear map $T(x,y)=((3x+y)/4,(x+3y)/4)$ preserves $s=x+y$ and sends $d=x-y$ to $d/2$. Thus $(1,1)$ and $(1,-1)$ are eigenvectors with eigenvalues 1 and $1/2$, and T^n is explicitly determined by these two coordinates.

Worked instance

Problem: Start at $(0,8)$. Find the pair after three rounds and describe round n exactly.

Solution: The midpoint is 4. The gaps are 8,4,2,1, so the pairs are $(0,8),(2,6),(3,5),(3.5,4.5)$. After n rounds the pair is $(4-4/2^n, 4+4/2^n)$. Algebra gives $x'+y'=x+y$ and $x'-y'=(x-y)/2$, proving the formula by induction. For unequal starting positions the gap remains nonzero at every finite n , although it tends to zero.

Boundary: If each marker moves the entire old gap, the two positions merely swap: the signed difference changes sign and never shrinks. 'Moving toward the other marker' alone does not guarantee convergence.

Hints to reveal in order

1. Add the two positions before and after a move.
2. Instead of tracking both numbers, track their midpoint and signed difference.

Extensions

- For fraction a , the signed difference multiplies by $1-2a$. Determine convergence for real a : unequal inputs converge exactly when $0 < a < 1$; $a=1/2$ meets in one round.
- Explore three connected markers with a specified averaging matrix, checking whether the average is actually conserved. General consensus needs conditions on the update matrix, not a visual impression.

Connection and its limits

Bridge type: exact-special-case

Preserves: Two marked quantities undergo the actual rational linear transformation, not a numerical analogy. Sum and difference give a change of coordinates that diagonalizes it.

Limits: One symmetric two-variable averaging rule does not prove convergence of every averaging network. The midpoint can remain fixed while other linear maps oscillate or expand.

Sources and inspection scope

[Sheldon Axler, Linear Algebra Done Right, fourth edition](#)

§3A linear maps and chapter 5 eigenvectors; §7B spectral discussion

Inspected: relevant definitions inspected; this map and iteration formula are proved directly

Prior use / what changes: Week 2 uses linear algebra over F_2 ; this family introduces real scalars, eigen-directions, and convergence. It is not a new toggle-board story.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-18 / Participation and launch

Stretch, shear, flatten: what happens to area?

MSC primary 15 - Linear and multilinear algebra; matrix theory | Secondary 51, 52

The opening challenge

A linear machine sends the horizontal unit arrow to u and the vertical unit arrow to v . Its full rule is $T(x,y)=x u+y v$: combine the chosen arrows using the same coefficients as the input. Their parallelogram is the transformed unit square. Find a machine that changes shape without changing area, one that reverses orientation, and one that flattens the square completely.

Learner choices: Choose image arrows, decide how to measure area, find examples with the same area but different shapes, and challenge a proposed inverse.

Prerequisite gates

Enter: O and V: construct parallelograms on a grid and compare areas by cutting or counting.

Explore: F and V: use signed coordinates and triangle areas where needed.

Explain: P: justify the shear's area preservation and distinguish reflection from collapse.

Prove: A and P: derive $ad-bc$ from coordinate geometry; L for the equivalence with invertibility and composition law.

Reading: Arrow diagrams suffice initially; two-by-two matrices optional.

Arithmetic: Whole-number coordinates and halves for triangle areas; signed multiplication for determinants.

Reasoning: Map both basis vectors from the same origin, and preserve the order of those vectors when discussing orientation.

Hard stop: General volume scaling in higher dimension or change of variables requires additional algebra/analysis, not merely many drawings.

Materials and preparation

10 minutes estimated prep. Squared paper, two colored arrow strips, transparent unit squares, scissors, and transformation cards for shear, reflection, and projection. A drawing alternative avoids cutting demands.

A flexible hour

0-10 build parallelograms; 10-15 explain how the two arrows determine every grid-point image; 15-35 compare the three machines; 35-40 walk around images in vertex order; 40-55 derive or test determinant rules and compose maps; 55-60 show one distinction that area alone misses.

Questions to explore

- Can a slanted parallelogram have area one?
- What changes when the two image arrows are exchanged?
- How can area reveal information loss?
- Can two different shapes have determinant one?

Satisfying stop: Three machines distinguished by area, orientation, and recoverability, with a concrete example showing area preservation is weaker than rigidity.

AD-18 / Facilitator reasoning

Stretch, shear, flatten: what happens to area?

The mathematical question

How does the determinant encode signed area change, and why does nonzero determinant distinguish invertibility from flattening?

A real two-by-two matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ sends the unit square to the parallelogram spanned by $(a,c), (b,d)$, with signed area $ad-bc$. Its ordinary area scale is $|ad-bc|$. Determinant zero means the two image directions are dependent.

Worked instance

Problem: Compare $S(x,y)=(x+y,y)$, $R(x,y)=(y,x)$, and $P(x,y)=(x,0)$.

Solution: S maps the square to vertices $(0,0), (1,0), (2,1), (1,1)$. Base and height are both one, so area remains one; its determinant is 1. R exchanges the basis arrows, retaining area one but changing signed area to -1. P sends both $(0,0)$ and $(0,1)$ to $(0,0)$, collapses the square to a segment, and has determinant zero. The inverse shear is $(u,v) \mapsto (u-v,v)$, whereas P cannot be inverted because distinct inputs coincide.

Boundary: S preserves area but sends the vertical unit edge to a diagonal of length $\sqrt{2}$. It therefore does not preserve lengths or make a rigid motion.

Hints to reveal in order

1. For a shear, cut a triangle from one side and move it to the other.
2. A reflection reverses the clockwise/counterclockwise vertex order even though its ordinary area stays the same.

Extensions

- Compose two shears and compare the resulting parallelogram with determinant multiplication. For S followed by $(x,y) \mapsto (x, x+y)$, the matrix is $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ and determinant remains one.
- Use integer image arrows to investigate lattice-point counts and fundamental parallelograms; introduce any connection to Pick's theorem separately with its polygon hypotheses.

Connection and its limits

Bridge type: exact-special-case

Preserves: Moving the square's two basis directions and completing the parallelogram is the exact action of a linear map. Cut-and-rearrange area arguments can precede matrices.

Limits: Preserving area does not imply preserving angles, distances, or the whole shape. Signed area records orientation and can be negative; physical area cannot.

Sources and inspection scope

[Sheldon Axler, Linear Algebra Done Right, fourth edition](#)

§7F volume discussion, especially 7.108-7.111; linear-map framework in §3A

Inspected: relevant volume discussion inspected; the two-dimensional examples use direct area arguments

Prior use / what changes: Week 1 shape work can supply manipulatives, but this family studies linear transformations and signed area rather than tiling reachability. It complements AD-17's spectral view.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-19 / Participation and launch

Take one linear arrow apart

MSC primary 16 - Associative rings and algebras | Secondary 15, 18

The opening challenge

Study $T(x,y,z)=(x+y,y+z)$ from Q^3 to Q^2 . Choose new coordinate directions so the machine becomes a collection of simple parts: directions killed, directions passed across unchanged, and target directions no input can reach. Show your new bases explicitly.

Learner choices: Choose convenient kernel and complement vectors, decide how to depict the blocks, compare different decompositions, and create another map with the same or a different block inventory.

Prerequisite gates

Enter: L: understand vectors, a linear map, and how images of a basis determine it.

Explore: L and A: solve a small homogeneous linear system and choose independent vectors.

Explain: P: separate killed directions, transmitted directions, and unreachable target directions.

Prove: L and P: choose complements to kernel and image and verify direct-sum decomposition and indecomposability of one-dimensional blocks.

Reading: Symbolic equations and small matrices are essential.

Arithmetic: Signed rational arithmetic and elementary linear systems.

Reasoning: Distinguish a basis change in the domain from one in the target; the two may be chosen independently.

Hard stop: Keep a genuine linear-algebra gate. A token-routing analogy is only motivation until linear combinations and basis changes are present.

Materials and preparation

5 minutes estimated prep. Graph paper, two labeled vector-space boxes, vector cards, and a matrix worksheet with space to record chosen bases. No specialized manipulative or software is required.

A flexible hour

0-10 test images of selected vectors; 10-15 define the three proposed simple blocks; 15-35 find bases and verify a decomposition; 35-40 reset with domain/target direction cards; 40-55 compare matrices and justify the classification; 55-60 report the block counts and which choices were nonunique. Do not require a general quiver theorem.

Questions to explore

- Which input changes are completely invisible?
- Can every target vector be reached?
- Do different chosen complements change the numbers of blocks?
- Why is this classification different from diagonalizing an operator?

Satisfying stop: An explicit basis choice makes the given machine visibly two transmitted directions and one killed direction.

AD-19 / Facilitator reasoning

Take one linear arrow apart

The mathematical question

How can a representation of a directed graph split into indecomposable pieces, and what information survives a change of bases?

A finite-dimensional representation of the one-arrow quiver is a linear map $T:V \rightarrow W$ over a field. Up to independent changes of bases, it decomposes into $\dim(\text{kernel } T)$ copies of $k \rightarrow 0$, $\text{rank}(T)$ copies of the identity $k \rightarrow k$, and $\dim(W) - \text{rank}(T)$ copies of $0 \rightarrow k$. These are the three indecomposable representations.

Worked instance

Problem: Decompose $T(x,y,z)=(x+y,y+z)$ into one-arrow indecomposables.

Solution: The kernel is spanned by $k=(1,-1,1)$. Choose $u=(1,0,0)$ and $v=(0,0,1)$, whose images are $(1,0)$ and $(0,1)$. The vectors u,v,k are independent and form a domain basis. Relative to that basis and the standard target basis, T has matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Thus the representation is two identity blocks $Q \rightarrow Q$ plus one killed block $Q \rightarrow 0$, with no unreachable-target block. Complement choices differ, but kernel dimension one and rank two fix the inventory.

Boundary: As endomorphisms under the same basis change on both sides, I and $2I$ are not similar over Q , despite equal rank: $P^{-1} I P$ is always I . Independent domain/target bases are an essential hypothesis here.

Hints to reveal in order

1. Solve $x+y=0$ and $y+z=0$ first.
2. Choose two input vectors whose images form the standard target basis, then add a kernel basis vector.

Extensions

- Try $S(x,y,z)=(x+y,2x+2y)$. Its rank is one, so the block counts are two killed, one transmitted, and one unreachable target direction.
- Add a second arrow $V \rightarrow W \rightarrow U$ and track the ranks of both arrows and their composite. More complicated quivers introduce new classification behavior; the one-arrow rank rule is not a universal solution.

Connection and its limits

Bridge type: exact-special-case

Preserves: The two spaces, their bases, and the specified linear map are an actual quiver representation, equivalently a module for the associated path algebra.

Limits: Dots and arrows without vector spaces would omit the mechanism. Rank classifies this single-arrow problem, not arbitrary quivers or linear operators under similarity.

Sources and inspection scope

[Pavel Etingof et al., Introduction to Representation Theory, 2011 notes](#)

§5.2, Example 5.8, printed p.81; §1.8 quiver representations

Inspected: full relevant decomposition text

Prior use / what changes: Week 2 contains image/kernel ideas over F_2 ; this advanced revisit changes the question to decomposition up to independent bases and its representation-theoretic meaning.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-20 / Participation and launch

Measure how matrix moves fail to commute

MSC primary 17 - Nonassociative rings and algebras | Secondary 16, 15

The opening challenge

Ordinary matrix multiplication is associative but can depend on order. Make a new operation $[A,B]=AB-BA$. Investigate whether this new operation is associative. If it fails, search for a three-term identity that always cancels, using E,F,H as a small laboratory.

Learner choices: Choose pairs to compare, choose a triple likely to reveal failure, organize a multiplication table, and decide whether an identity follows from examples or symbolic cancellation.

Prerequisite gates

Enter: L and F: multiply two-by-two matrices and subtract them, with a template if needed.

Explore: L: distinguish matrix product from the newly defined bracket and track nested operations.

Explain: P: give a concrete associativity failure and identify cancelling terms in Jacobi.

Prove: A and P: expand each double commutator using associativity of the original matrix product; L for identifying the span with trace-zero matrices.

Reading: Symbolic matrix and bracket notation is essential.

Arithmetic: Signed integer multiplication/addition; coefficients no larger than two in the main table.

Reasoning: There are two different multiplications in play. Use a color or box around each bracket calculation and keep parenthesization explicit.

Hard stop: Keep the matrix prerequisite. A story in which order matters would establish only noncommutativity, omitting the nonassociative Lie structure.

Materials and preparation

8 minutes estimated prep. Three matrix cards $E=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $F=\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $H=\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$; blank product/bracket tables; two colors for positive and negative expansion terms. Provide one worked ordinary matrix product, not a completed commutator table.

A flexible hour

0-10 refresh matrix products with chosen vectors; 10-15 introduce the bracket as a second operation; 15-35 build the small relation table; 35-40 reset and swap order cards physically; 40-55 test associativity and expand Jacobi; 55-60 display one failed law and one valid replacement. Calculations can be split between partners.

Questions to explore

- What is $[A,A]$?
- What happens when A and B are exchanged?
- Do $[H,[E,F]]$ and $[[H,E],F]$ agree?
- Where does each triple product cancel in the Jacobi sum?

Satisfying stop: An explicit nonassociativity witness and a checked Jacobi cancellation, with the two products clearly distinguished.

AD-20 / Facilitator reasoning

Measure how matrix moves fail to commute

The mathematical question

Why does the commutator of an associative algebra become a nonassociative Lie product, and what replaces associativity?

For rational two-by-two matrices define $[A,B]=AB-BA$. The bracket is bilinear, alternating, and satisfies $[[A,B],C]+[[B,C],A]+[[C,A],B]=0$. For $E=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $F=\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $H=\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, the relations are $[E,F]=H$, $[H,E]=2E$, $[H,F]=-2F$.

Worked instance

Problem: Compute the three listed brackets, find an associativity failure, and verify Jacobi on E,F,H .

Solution: $EF=\text{diag}(1,0)$ and $FE=\text{diag}(0,1)$, giving $[E,F]=H$. Direct products give $[H,E]=2E$ and $[H,F]=-2F$. Thus $[[H,E],F]=2H$ while $[H,[E,F]]=[H,H]=0$, so the bracket is not associative. For the cyclic Jacobi sum, $[[E,F],H]=0$, $[[F,H],E]=[2F,E]=-2H$, and $[[H,E],F]=2H$; their sum is zero. For arbitrary matrices, expansion pairs and cancels the same ordered triple products.

Boundary: A zero commutator does not mean both matrices are zero: every matrix commutes with the identity. Nonzero bracket measures failure to commute, not the size of the matrices.

Hints to reveal in order

1. Calculate EF and FE separately before subtracting.
2. Write all twelve terms of the three double commutators, then pair identical triple products with opposite signs.

Extensions

- Prove every rational trace-zero two-by-two matrix is a unique linear combination of E,F,H , so their closed bracket table describes the whole three-dimensional algebra.
- With polynomial differentiation as an explicit new prerequisite, compare D and multiplication by x : $[D,M_x]=I$. This is an operator-algebra continuation, not needed for the matrix investigation.

Connection and its limits

Bridge type: exact-special-case

Preserves: The new operation is the actual matrix commutator, and the three displayed matrices span the Lie algebra $\mathfrak{sl}(2(\mathbb{Q}))$. Calculations investigate its identities directly.

Limits: A physical sequence of large rotations is not automatically this bracket. The activity does not establish the Lie-group/Lie-algebra correspondence or classification of Lie algebras.

Sources and inspection scope

[Pavel Etingof et al., Introduction to Representation Theory](#)

§1.9, Definition 1.39, Example 1.40, and Example 1.46 $\mathfrak{sl}(2)$ relations

Inspected: full relevant definition and relations

Prior use / what changes: Week 3 compositions are reversible permutations; this is a substantially different advanced structure involving subtraction and a nonassociative operation. No elementary-equivalence claim is made.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-21 / Participation and launch

The collection that remembers both choices

MSC primary 18 - Category theory; homological algebra | Secondary 03, 68

The opening challenge

Choose one person card and one badge card, but their colors must match. Build a catalog remembering exactly which person and which badge were chosen. Another group will give each of its tokens a person and a compatible badge. Show that every token then has one forced place in your catalog.

Learner choices: Choose a layout, decide how to retain repeated-color identities, create an alternative group's assignments, and test whether a smaller catalog loses required information.

Prerequisite gates

Enter: O: match labels and keep two inventory roles distinct.

Explore: C: enumerate every compatible ordered pair without duplicates.

Explain: P: explain why each object carrying two compatible choices has one forced location in the pair collection.

Prove: P and X: functions, composition, and quantification over an arbitrary set X for the universal statement.

Reading: Names and color labels can be replaced by pictures; upper stage uses arrows/functions.

Arithmetic: Small counts; multiplication of fiber sizes is optional.

Reasoning: Preserve both individual identities, not just their shared color. The universal claim asks about all other compatible assignments, which is a genuine abstraction gate.

Hard stop: Constructing the pair list is a complete lower task; call the universal-property proof complete only when existence and uniqueness are both explained.

Materials and preparation

10 minutes estimated prep. Inventory A: Ada and Bo colored red, Cy colored blue. Inventory B: r red, s and t blue, u green. Draw one space for each compatible pair and keep a common color legend.

A flexible hour

0-10 freely match cards; 10-15 specify that both identities must survive; 15-35 create and test catalogs; 35-40 participants enact two projections from pair positions; 40-55 investigate the forced map from another inventory; 55-60 explain why one possible catalog entry per token is both available and unique.

Questions to explore

- Why are Ada-r and Bo-r different?
- What happens to the green badge?
- Can one blue catalog slot remember both Cy-s and Cy-t?
- What must be true before another token can be placed at all?

Satisfying stop: A four-entry catalog and a concrete explanation that preserving both choices forces a unique placement.

AD-21 / Facilitator reasoning

The collection that remembers both choices

The mathematical question

What makes a construction universal rather than merely a convenient list, and how do pullbacks encode compatible choices?

For functions $f:A \rightarrow C$ and $g:B \rightarrow C$, the pullback is $P = \{(a,b): f(a)=g(b)\}$, with its two projection maps. Given $u:X \rightarrow A$ and $v:X \rightarrow B$ with $f \circ u = g \circ v$, there is exactly one map $h:X \rightarrow P$ whose projections are u and v : $h(x) = (u(x), v(x))$.

Worked instance

Problem: Construct the pullback of the displayed inventories and explain its universal property for an arbitrary compatible assignment.

Solution: The catalog is $\{(Ada,r),(Bo,r),(Cy,s),(Cy,t)\}$. There are four pairs; badge u has no compatible person. If a token x is assigned person $u(x)$ and badge $v(x)$ of the same color, then $(u(x), v(x))$ is on the list, proving existence. Any catalog placement preserving both assignments must have exactly those two entries, proving uniqueness. This works for every token independently and therefore defines the unique required function.

Boundary: A three-entry catalog containing only one pair per color fails: there is no green pair, and red/blue repetitions require multiple distinct pairs. Equal labels do not erase identities.

Hints to reveal in order

1. Make a row for each person and list every badge with the same color.
2. A token's two assigned identities force its ordered pair; there is no further choice.

Extensions

- Change fiber sizes and prove the count is the sum over colors of $|A_{\text{color}}| \times |B_{\text{color}}|$; this counts all compatible pairs, not arbitrary pairings without replacement.
- Compare with an ordinary Cartesian product, obtained when both inventories map to a one-element color set. Then every pair is compatible.

Connection and its limits

Bridge type: faithful-representation

Preserves: Two labeled inventories map to a common color set. Compatible ordered pairs and the unique pairing map realize the full finite-set pullback, including its universal property.

Limits: Matching one object per color is not the pullback when labels repeat. The example does not establish all categorical limits or turn every commuting square into a pullback.

Sources and inspection scope

[Brendan Fong and David Spivak, Seven Sketches in Compositionality](#)

§3.5.3, pullbacks of sets and colored-pair example near PDF p.125

Inspected: full relevant construction and explanation

Prior use / what changes: Week 5 combines constraints but does not study a universal mapping property. This family must retain the uniqueness-of-map question to count as categorical content.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-22 / Participation and launch

Which loops can face moves erase?

MSC primary 18 - Category theory; homological algebra | Secondary 55, 15

The opening challenge

Mark any edges so every vertex touches an even number of marked edges. A move flips all three edges around a face whose card is present. Can you erase your pattern? Compare no face cards, only ABC, and both faces. Keep the graph unchanged while changing available faces.

Learner choices: Choose closed patterns, choose face moves, choose targets, invent a certificate that a target cannot be reached, and decide how to group mutually reachable patterns.

Prerequisite gates

Enter: O and C: flip edge markers and count parity at four vertices.

Explore: C and P: distinguish closed edge patterns from patterns with odd endpoints.

Explain: P: list all four cycles and compare which face flips are legal.

Prove: L and X: interpret vector spaces, boundary maps, image/kernel, and quotient for formal homology; finite parity proofs can precede notation.

Reading: Edge names and short move rules; oral launch suffices.

Arithmetic: Odd/even counts; binary addition means toggling twice cancels.

Reasoning: Track both a marked edge state and which faces are actually filled. An empty drawn triangular region is not automatically a legal face.

Hard stop: The concrete reachability result is exact. Claims about arbitrary surfaces or homotopy invariance require additional topology.

Materials and preparation

10 minutes estimated prep. Square A,B,C,D with diagonal AC, five edge counters, and removable face cards ABC and ACD. Use a different color for face availability than for marked edges.

A flexible hour

0-10 explore edge toggling; 10-15 define closed patterns and face moves; 15-35 catalog cycles and test erasing; 35-40 enact boundary cancellation with vertex gestures; 40-55 classify reachability for the three face choices; 55-60 explain one loop that changes status when a face is added.

Questions to explore

- Why does a face move preserve even incidence?
- Does the outer square count as a closed pattern?
- What is T_1 plus T_2 when the diagonal is toggled twice?
- Can a visible loop be equivalent to zero?

Satisfying stop: The same outer loop can or cannot be erased depending on available faces, with an exact reachability certificate.

AD-22 / Facilitator reasoning

Which loops can face moves erase?

The mathematical question

How does taking cycles modulo boundaries distinguish a closed pattern from a genuinely surviving homology class?

Over F_2 , a finite simplicial complex has boundary maps $C_2 \rightarrow C_1 \rightarrow C_0$ with consecutive composite zero. H_1 is $\ker(\partial_1)/\text{image}(\partial_2)$. In a square with diagonal AC , the cycle space has basis $T_1 = AB + BC + AC$ and $T_2 = AC + CD + DA$; filling chosen triangular faces changes the boundary subspace.

Worked instance

Problem: Classify closed patterns and homology classes with only face ABC available.

Solution: Let a mark $AB=BC$ and c mark $CD=DA$. Evenness at A forces $AC = a + c \pmod{2}$; the condition at C then follows. The four cycles are $0, T_1, T_2, T_1 + T_2$. The only available face boundary is T_1 , so classes are $\{0, T_1\}$ and $\{T_2, T_1 + T_2\}$. Thus H_1 has two elements, a one-dimensional F_2 space. With no faces all four classes differ; with both faces every cycle can be erased and H_1 is zero.

Boundary: The marked outer square is a cycle in all three versions, but its class is nonzero with only ABC filled and zero with both faces filled. 'Closed loop' and 'nonzero homology class' are not synonyms.

Hints to reveal in order

1. At B , AB and BC must have the same mark; at D , CD and DA must match.
2. Adding both triangle boundaries cancels AC and leaves the outer square.

Extensions

- Remove diagonal AC and all faces: only the empty pattern and outer square remain closed. Compare the surviving one-dimensional cycle space with the one-face version, without asserting they are identical complexes.
- Use signed integer edges to study oriented boundaries. The F_2 cancellation picture is exact for its coefficient field and should not silently be used to claim integer homology computations.

Connection and its limits

Bridge type: faithful-representation

Preserves: Marked edges are F_2 chains; endpoints with odd incidence form their boundary. Flipping the three edges of an available face adds that face's boundary. Reachability between closed patterns is equality of homology classes.

Limits: The legal faces are part of the data. A loop drawn in a plane does not automatically bound an available face, and this one complex does not prove topological invariance in general.

Sources and inspection scope

[Stacks Project. Complexes](#)

§12.13, chain-complex definition and homology formula after Lemma 12.13.3

Inspected: full relevant definitions; finite graph/face matrices checked directly

Prior use / what changes: Week 2 introduces parity and incidence; adding faces and quotienting cycles by face boundaries is the genuinely new question. Coordinate with topology families to avoid counting the identical complex twice.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-23 / Participation and launch

Why two dimensions are better than one count

MSC primary 19 - $\mathbb{K}$ -theory | Secondary 16, 15

The opening challenge

The ring R has two independent scalar controls: (r,s) scales the first space by r and the second by s . Classify finite pairs of rational vector spaces under maps respecting both controls. Then allow formal differences of classes. What information must survive, and which differences cannot be represented by a single actual module?

Learner choices: Choose pairs with equal total dimension, construct distinguishing ring actions, select stable additions, and decide what invariants should classify formal differences.

Prerequisite gates

Enter: L and X : vector spaces, direct sums, ring action, and module isomorphism for the genuine K_0 investigation. Two-color counting offers a preparatory subtask.

Explore: L : separate e_1 and e_2 components and compare dimension pairs.

Explain: P : show why total dimension forgets structure and why every pair is projective.

Prove: X and P : use the definition of projective as a direct summand of a free module, then verify the universal group-completion construction.

Reading: Symbolic definitions are essential for the K_0 claim; counters can carry the subsequent computations.

Arithmetic: Nonnegative integer dimension pairs and signed subtraction.

Reasoning: Distinguish a module, its isomorphism class, and a formal difference of classes. Negative coordinates need not be actual objects.

Hard stop: Participants who only manipulate counters finish a preliminary monoid investigation, not the full K_0 computation. The algebra gate is deliberate.

Materials and preparation

12 minutes estimated prep. Two-color counters, paired vector-space diagrams, and a one-page definition sheet for R , e_1, e_2 , projective modules, and group completion. No software or large computations.

A flexible hour

0-10 explore paired spaces and their two controls; 10-15 define module-respecting comparison; 15-35 derive the two dimensions and projectivity; 35-40 reset with two-color direct sums; 40-55 construct formal differences and the K_0 identification; 55-60 state the exact computed ring invariant and the limits of the counter model.

Questions to explore

- Why are $(Q,0)$ and $(0,Q)$ different despite equal total dimension?
- Which pairs are free as R -modules?
- Can adding the same pair erase a dimension mismatch?
- What could a negative coordinate of a class mean?

Satisfying stop: A correct computation $K_0(Q \times Q) = \mathbb{Z}^2$ with one actual projective nonfree module and one genuinely virtual class.

AD-23 / Facilitator reasoning

Why two dimensions are better than one count

The mathematical question

How does K_0 retain the independent ranks of projective modules, and why are formal differences needed even in a completely computable example?

For $R=Q \times Q$, every finitely generated R -module splits as $e_1 P \oplus e_2 P$, where $e_1=(1,0)$, $e_2=(0,1)$. It is classified by the pair of rational dimensions (a,b) , and is projective because it is a sum of the summands $e_1 R, e_2 R$ of R . $K_0(R)$, the group completion of projective isomorphism classes under direct sum, is $\mathbb{Z} \times \mathbb{Z}$.

Worked instance

Problem: Compute $[P]-[Q]$ for dimension pairs $P=(2,1)$, $Q=(1,2)$, and explain why $K_0(R)=\mathbb{Z}^2$.

Solution: The class difference is $(1,-1)$, so it is not the class of an actual module, whose coordinates are nonnegative. Every module splits into $e_1 P$ and $e_2 P$ because $e_1+e_2=1$ and $e_1 e_2=0$; respecting the R action preserves each component. Finite vector-space classification gives the monoid $\mathbb{N}^2$. Conversely every pair is a sum of $e_1 R, e_2 R$ and hence projective. Group completion sends formal differences to coordinate differences, which are additive and classify them, giving $\mathbb{Z}^2$.

Boundary: The modules $(Q,0)$ and $(0,Q)$ have equal total rational dimension but are not R -isomorphic: e_1 acts as identity on the first and zero on the second. One scalar dimension count would lose essential information.

Hints to reveal in order

1. Apply e_1 and e_2 separately; they sum to the identity and their product is zero.
2. A free module R^n has dimension pair (n,n) ; the two one-color modules are direct summands of R .

Extensions

- Classify free modules inside this monoid: precisely pairs (n,n) . A projective module with dimensions $(1,0)$ is not free, despite being a direct summand of R .
- For a general commutative monoid cancellation may fail: if $e+e=e$, its image in every group is zero. This warns against assuming every group completion embeds its starting monoid.

Connection and its limits

Bridge type: exact-special-case

Preserves: Paired vector spaces are genuine modules over $Q \times Q$. Two-color counters faithfully encode their isomorphism classes only after the idempotent decomposition is established. Formal signed pairs compute actual K_0 for this ring.

Limits: Counter subtraction by itself is only the group-completion mechanism. It does not constitute a lesson on arbitrary projective modules, higher K -groups, or algebraic K -theory generally.

Sources and inspection scope

[Charles Weibel, The K-book, chapter II](#)

§1 group completion; §2 definition of K_0 and product-ring calculation preceding Example 2.1.4, chapter-PDF p.6

Inspected: full relevant definition and product calculation

Prior use / what changes: No existing week covers this definition. The advanced gate prevents elementary counter bookkeeping from being counted as full K -theory coverage.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AD-24 / Participation and launch

Necklaces that resist dividing by four

MSC primary 20 - Group theory and generalizations | Secondary 05

The opening challenge

Make four-bead necklaces with two colors. Two count as the same if turning the ring makes them match; do not flip it over and do not exchange color names. Find every different necklace. Someone says there are sixteen strings and four turns, so the answer must be four. Decide whether that argument works.

Learner choices: Choose representatives, invent a catalog order, decide how to prove no necklace is missing, and search for the most and least symmetric patterns.

Prerequisite gates

Enter: O: arrange two colors on four positions and rotate a ring without turning it over.

Explore: C: group all sixteen labeled strings into rotation classes and avoid duplicates.

Explain: P: show why different orbit sizes invalidate division by four.

Prove: P: double-count pairs (rotation, fixed coloring); the general theorem needs finite group actions and orbit-stabilizer reasoning.

Reading: None for rings; binary strings optional for systematic recording.

Arithmetic: Count to sixteen, sum four fixed counts, and divide by four.

Reasoning: Keep rotation separate from reflection and preserve named colors; a marked top position helps compare labeled versus unlabeled states.

Hard stop: The six-orbit classification is complete without group terminology. General orbit-counting extensions require precise action rules and checked stabilizers.

Materials and preparation

10 minutes estimated prep. Four-position circular mats with a removable top marker, two-color counters, and sixteen small recording cards. Reversible bracelets are inappropriate unless flipping is explicitly discussed as a rule change.

A flexible hour

0-10 build and rotate patterns freely; 10-15 fix the equivalence rule; 15-35 catalog necklaces and inspect the divide-by-four claim; 35-40 rotate a four-person pattern; 40-55 count fixed colorings under each turn; 55-60 connect the two correct counts. The orbit catalog can be an earlier stop.

Questions to explore

- How many distinct rotations does 0000 have?
- How many does 0101 have?
- Why does every pattern fixed by a quarter-turn use one color only?
- Can the fixed-pattern average be explained rather than memorized?

Satisfying stop: A complete six-necklace catalog and a concrete explanation of why orbit sizes differ.

AD-24 / Facilitator reasoning

Necklaces that resist dividing by four

The mathematical question

How do stabilizers explain why dividing labeled configurations by the number of symmetries can fail, and why does averaging fixed configurations work?

For a finite group G acting on a finite set X , the number of orbits is $(1/|G|)$ times the sum over g of the number of configurations fixed by g . The cyclic group of four rotations acts on sixteen binary colorings of four labeled positions, producing six orbits.

Worked instance

Problem: Count binary four-bead necklaces up to rotation and verify Burnside's average.

Solution: Representatives are 0000, 1111, 0001, 0011, 0101, 0111. Their orbit sizes are 1, 1, 4, 4, 2, 4, totaling sixteen strings. The identity fixes sixteen colorings; each quarter-turn fixes only 0000 and 1111; the half-turn fixes four, determined by the first two bits. The average $(16+2+4+2)/4=6$ agrees. Each orbit contributes exactly four to the total fixed-pair count because its number of positions times its stabilizer size equals four.

Boundary: Dividing sixteen by four fails because not every orbit has four elements. For this particular binary length-four example, additionally allowing reflections happens not to reduce the count, but that coincidence must not be generalized.

Hints to reveal in order

1. Sort first by the number of beads of each color, then distinguish adjacent from opposite pairs.
2. Count colorings fixed by zero, one, two, or three quarter-turns separately.

Extensions

- Use three named colors on three beads. Rotations give $(27+3+3)/3=11$ classes; including reflections gives $(27+3+3+9+9+9)/6=10$. All-three-color necklaces expose the difference.
- Explore longer binary necklaces with a fixed bead count. Rotation-cycle lengths determine fixed configurations, but derive the constraints before applying a formula.

Connection and its limits

Bridge type: faithful-representation

Preserves: Four-bead rings and allowed rotations realize the exact group action. The same-versus-different rule is orbit equivalence, and symmetrical necklaces have nontrivial stabilizers.

Limits: Reflections are excluded from the core rule, and color names are fixed. Changing either convention changes the action; a symmetry story without an exact equivalence rule is not a well-defined counting problem.

Sources and inspection scope

[Thomas Judson, Abstract Algebra: Theory and Applications](#)

§14.3 Burnside's Counting Theorem; supporting orbit-stabilizer discussion in §14.1

Inspected: full relevant theorem discussion; finite fixed counts checked directly

Prior use / what changes: Week 3 studies permutation return times; this family studies actions on colorings and unequal stabilizers. Repeated rotations are shared tools, but the counting question and proof are new.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-01 / Participation and launch

Can the order of two turns change the result?

MSC primary 22 - Topological groups, Lie groups | Secondary 20, 53

The opening challenge

Start with the arrow pointing along positive z . Turn a quarter-turn about x , then about y . Reset completely and reverse the order. Can both results be right? Invent another pair of turns and predict before moving.

Learner choices: Learners choose turn pairs, starting arrows, and a recording method; they may ask a partner to execute their prediction.

Prerequisite gates

Enter: O: follow a spoken turn instruction and distinguish three fixed room directions.

Explore: V: record an arrow as right, forward, or up; optionally use signed coordinates.

Explain: P: show different final arrows under two orders, which is a complete counterexample.

Prove: L/D: rotation matrices and matrix exponentials for continuous generators.

Reading: Oral launch; a partner may record turn words.

Arithmetic: None for manipulation; signed unit vectors for checking.

Reasoning: Hold axes fixed in the room, not attached to the moving box; spatial memory is the main demand.

Hard stop: Stop before Lie brackets unless matrix multiplication and derivatives are available.

Materials and preparation

10 minutes estimated prep. A box marked with one large arrow, a paper compass showing fixed x, y, z axes, and two turn cards.

A flexible hour

0-10 freely rotate and describe the box; 10-15 establish right-hand positive turns with a demonstration; 15-35 compare chosen orders; 35-40 movement reset; 40-55 build a repeatable witness or compare planar turns; 55-60 share one checked sequence.

Questions to explore

- Which turns seem interchangeable?
- Does starting with a different arrow conceal the difference between the transformations?

Satisfying stop: A reproducible pair of order-sensitive turns and an explanation using the final arrow.

GA-01 / Facilitator reasoning

Can the order of two turns change the result?

The mathematical question

How does continuous rotational symmetry differ from simply permuting labels, and where does noncommutativity first appear?

Rotations of the plane about one center commute. In three dimensions, rotations about different fixed axes need not commute. Rotation through any real angle is allowed; quarter-turns supply an exactly checkable witness.

Worked instance

Problem: Use right-hand positive 90° rotations R_x and R_y on the starting vector $(0,0,1)$.

Solution: R_x first sends z to $-y$; R_y then leaves $-y$ fixed, so the result is $(0,-1,0)$. R_y first sends z to x ; R_x leaves x fixed, giving $(1,0,0)$. The different arrows prove $R_x R_y \neq R_y R_x$, with composition order explained verbally. Plane rotations through a then b both add $a+b$ to the initial angle, so they commute.

Boundary: Two turns about the same axis do commute. Also, testing just one vector can miss a difference: equality of a single image does not establish equality of transformations.

Hints to reveal in order

1. Keep the axis card on the table.
2. Track only the marked arrow before attempting all faces.

Extensions

- Find a three-turn sequence that returns the arrow but not the whole box.
- For matrix-ready learners, vary a small angle and compare the commutator with the identity; the limiting Lie bracket needs calculus.

Connection and its limits

Bridge type: exact-special-case

Preserves: A marked rigid box carries the usual rotation action of $SO(3)$; arrows record images of an initial direction.

Limits: The quarter-turn subgroup alone does not exhibit Lie algebra, covering-space theory, or all continuous rotations.

Sources and inspection scope

[Peter Woit. Quantum Theory. Groups and Representations](#)

Chapters 4-6; orthogonal groups and rotations

Inspected: Relevant text and definitions inspected; box example derived directly

Prior use / what changes: Week 3 explored permutation composition; this uses continuous physical rotations and fixed axes. Actual use of related instances remains unknown.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-02 / Participation and launch

Can opposite fence heights always be different?

MSC primary 26 - Real functions | Secondary 55

The opening challenge

Draw a fence top that meets at its seam without a jump; corners are allowed, jumps are not. Try to keep the red marker higher than the blue one through half a turn, with the markers always opposite.

Learner choices: Choose the profile, starting position, direction of motion, and an attempted escape from equality.

Prerequisite gates

Enter: O: compare which of two markers is higher and follow opposite positions.

Explore: V: read a closed profile represented on a strip with joined ends.

Explain: P: argue that the higher marker must become lower after half a turn.

Prove: P/X: intermediate value reasoning or its real-analysis proof.

Reading: Oral; no numerical labels needed initially.

Arithmetic: None for comparison; fractions only for the optional half-turn parameter.

Reasoning: Track two points simultaneously and distinguish observed equality from an existence argument.

Hard stop: Higher-dimensional Borsuk-Ulam is not proved by this circle argument.

Materials and preparation

12 minutes estimated prep. A paper strip with a continuous wavy line whose end heights agree; a duplicated strip and two linked markers half a period apart.

A flexible hour

0-10 compare moving marker heights freely; 10-15 establish the continuous seam and opposition; 15-35 try to defeat equality; 35-40 stand and exchange places; 40-55 explain the exchange argument or test missing assumptions; 55-60 share a conjecture and its reason.

Questions to explore

- What has swapped after half a turn?
- Can six separate posts avoid matching opposite heights?

Satisfying stop: The half-turn argument that defeats every continuous attempted fence.

GA-02 / Facilitator reasoning

Can opposite fence heights always be different?

The mathematical question

How can continuity guarantee an exact equality when no formula for the shape is known?

For every continuous real-valued function h on a circle, some antipodal points have equal values:

$g(t) = h(t) - h(t+1/2)$ is continuous and $g(1/2) = -g(0)$.

Worked instance

Problem: Use six post heights 0,4,1,3,0,2 in circular order, with straight ramps joining neighbors. Find one matching pair.

Solution: At the start the difference between opposite points is $0-3=-3$. One sixth of a turn later it is $4-0=4$. On these straight ramps the difference is linear, so equality occurs $3/7$ of the way along that first sixth, at $t=1/14$ of a turn. The shared height is $12/7$. More generally a sign reversal and continuity guarantee some equality without calculating it.

Boundary: With only the six isolated posts available, opposite differences are $-3,4,-1$ and their negatives: none is zero. Continuous ramps are essential.

Hints to reveal in order

1. Ignore the exact heights and record only higher, same, lower.
2. Start and finish with the markers at each other's initial positions.

Extensions

- Bracket the crossing and halve the bracket, separating guaranteed location error from measurement error.
- Ask about matching two continuous measurements on a sphere; state the higher theorem as a continuation, not a proved result.

Connection and its limits

Bridge type: exact-special-case

Preserves: Sliding opposite markers samples a continuous periodic function; exchanging their roles reverses the sign of the difference.

Limits: A finite list of isolated posts does not satisfy the continuous-domain hypothesis, and a model cannot resolve tiny physical height differences exactly.

Sources and inspection scope

[Jiří Lebl, Basic Analysis I](#)

§3.3, intermediate value theorem; circle deduction supplied here

Inspected: Theorem inspected; complete special-case argument supplied

Prior use / what changes: New continuous strand; no matching worksheet in Weeks 1-10. The advisory atlas example suggested this mechanism; this exact ramp instance is separately checked.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-03 / Participation and launch

Can infinitely many points fit under very short covers?

MSC primary 28 - Measure and integration | Secondary 03, 40

The opening challenge

Your opponent names a first, second, third, and later point. Before hearing later points, promise a covering rule that spends less than one unit of total length, even if the list never ends.

Learner choices: Learners allocate the budget, place covers, allow overlaps, and choose adversarial point lists.

Prerequisite gates

Enter: O/C: cover named dots and compare total used strip lengths.

Explore: F: repeatedly halve a length budget and discuss a listed infinity.

Explain: P: explain why the n th listed point is covered and why the total stays below the requested budget.

Prove: X: definition of outer measure; geometric-series bound suffices for this theorem.

Reading: Short numbered labels; oral explanation with an adult recorder works.

Arithmetic: Halves and geometric sums; optional enumeration of rational pairs.

Reasoning: Separate every individual point receiving a cover from finitely many observed covers.

Hard stop: Uncountability and nonmeasurability require new arguments; do not infer either from a sparse-looking picture.

Materials and preparation

10 minutes estimated prep. A number line, named point cards, paper strips of halving widths, and a visible total-length budget.

A flexible hour

0-10 cover five chosen points using little paper; 10-15 state the never-ending challenge; 15-35 invent and test budget rules; 35-40 reset; 40-55 explain one rule for all numbered positions or list rational numbers; 55-60 report precisely what was proved.

Questions to explore

- Must each interval have the same width?
- Could the points be dense and still admit your rule?

Satisfying stop: A budget rule that covers every numbered point and an exact total-length bound.

GA-03 / Facilitator reasoning

Can infinitely many points fit under very short covers?

The mathematical question

Why is counting points different from measuring length, and what does a countable covering argument actually prove?

A countable set $\{p_1, p_2, \dots\}$ in $\mathbb{R}$ has outer measure zero: for every $\varepsilon > 0$, cover p_n by an open interval of length $\varepsilon/2^{n+1}$; the sum of lengths is $\varepsilon/2 < \varepsilon$.

Worked instance

Problem: Cover $p_n = 1/n$, $n \geq 1$, using total length below $1/100$.

Solution: Center an open interval of length $(1/100)/2^{n+1}$ at each $1/n$. Every listed point lies inside its assigned interval. The total interval length is $(1/100)(1/4 + 1/8 + \dots) = 1/200$, below the requested budget. Overlaps only reduce union length. Replace $1/100$ by any positive ε to establish zero outer measure for this listed set.

Boundary: Assigning every listed point its own interval of one fixed positive width makes the sum of assigned lengths diverge. This particular budget proof therefore uses shrinking widths, despite possible overlaps. It does not exclude other finite covers of special sets such as $\{1/n\}$.

Hints to reveal in order

1. Reserve some budget for later points.
2. Give each new point half as much as the previous point.

Extensions

- Enumerate rational numbers by numerator-denominator diagonals and apply the same rule, despite density.
- Contrast with the uncountable Cantor set in GA-29; a small cover does not establish countability.

Connection and its limits

Bridge type: exact-special-case

Preserves: Each chosen point is assigned a genuine interval; a rule, rather than an impossible completed physical construction, specifies all intervals.

Limits: Finite paper strips only illustrate initial stages. Zero outer measure does not assert that the set is empty or countable.

Sources and inspection scope

[Sheldon Axler, Measure, Integration & Real Analysis](#)

§2A, outer measure and countable sets

Inspected: Relevant definitions and countable-cover argument inspected

Prior use / what changes: No direct prior-week instance; connects to infinite-series reasoning rather than routine counting. Actual use is unrecorded.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-04 / Participation and launch

Follow a square root around a circle

MSC primary 30 - Functions of a complex variable | Secondary 22, 55

The opening challenge

Move the target once around its circle. Keep a root arrow whose doubled angle always points at the target, and never jump your root. Where does it finish? Choose a different route or starting root and try again.

Learner choices: Choose direction, number of target circuits, intermediate checkpoints, and whether to display both roots.

Prerequisite gates

Enter: O/V: track angles and paired opposite arrows on a circle.

Explore: F/A: half angles and complex multiplication, or a supplied double-angle rule.

Explain: P: explain why a chosen root cannot jump between opposite alternatives.

Prove: A/P: express $w(t)/e^{it/2}$ as a continuous value in $\{1, -1\}$.

Reading: Short labels; proof extension uses function notation.

Arithmetic: Halving angles; optional multiplication with i .

Reasoning: Distinguish the moving target, both candidate roots, and the tracked choice.

Hard stop: Holomorphic continuation on general domains requires complex analysis; angle tracking alone does not establish it.

Materials and preparation

8 minutes estimated prep. Two circle diagrams, a target arrow, red and blue root arrows, and four quarter-turn cards.

A flexible hour

0-10 learn the squaring action with free arrow play; 10-15 state the no-jump rule; 15-35 follow and record routes; 35-40 movement reset; 40-55 explain one-turn failure and two-turn recovery; 55-60 share a route certificate.

Questions to explore

- Why are there two roots at every target point?
- Could you exchange roots halfway without breaking continuity?

Satisfying stop: Explain why a no-jump root needs two target turns to return.

GA-04 / Facilitator reasoning

Follow a square root around a circle

The mathematical question

Can a locally consistent choice of root fail to return to its starting value after a closed journey?

Along $z=e^{it}$, $0 \leq t \leq 2\pi$, the continuous root starting at $w=1$ is $w=e^{it/2}$, so it ends at -1 . A continuous single-valued square root cannot be defined on the entire unit circle.

Worked instance

Problem: Start the target at 1 and the chosen root at 1. Follow the target through $1, i, -1, -i, 1$ counterclockwise.

Solution: The tracked roots have arguments $0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ$; their squares have exactly the target arguments. The ending root is -1 . Any continuous competing choice differs from this one by a sign, and that sign cannot change continuously while restricted to $\{1, -1\}$. Thus one circuit cannot yield a continuous periodic root choice. Two target circuits advance the root 360° and restore it.

Boundary: Allow a jump from one root to the other and a single-valued rule such as a principal branch becomes possible on a cut diagram, but it is discontinuous at the cut.

Hints to reveal in order

1. Start at target 1 with roots 1 and -1 .
2. Each quarter-turn of the target advances the tracked root by an eighth-turn.

Extensions

- For cube roots, one target circuit cyclically permutes three choices; three circuits restore them.
- Investigate a loop that does not wind around zero; label general winding results as later topology.

Connection and its limits

Bridge type: exact-special-case

Preserves: Moving arrows realize complex squaring: double the angle and square the length. Tracking one root preserves continuity along a path.

Limits: The construction proves a circle obstruction, not all monodromy theory or the general existence theorem for holomorphic roots.

Sources and inspection scope

[Jiří Lebl, Guide to Cultivating Complex Analysis](#)

Example 10.2.6, continuation of a square root; §1.2 polar multiplication

Inspected: Relevant example inspected; elementary circle proof supplied

Prior use / what changes: Week 3 cycles are an algebraic comparison; the new obstruction concerns continuous root selection around a loop.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-05 / Participation and launch

Can an averaging landscape hide a peak?

MSC primary 31 - Potential theory | Secondary 05, 35

The opening challenge

Keep the end dots at 0 and 9. Choose the two middle heights so each equals the average of its neighbors. Then invent a connected landscape with fixed boundary and try to hide a height above every boundary value.

Learner choices: Choose a graph, boundary values, and proposed solution; exchange designs to seek a hidden peak.

Prerequisite gates

Enter: C/F: add small values and divide by two; colors can first compare heights.

Explore: A: solve two simultaneous linear equations or use balancing.

Explain: P: chase an assumed interior maximum along neighbors to a fixed boundary.

Prove: P/L: connected-graph argument; continuous mean-value proof needs analysis.

Reading: One short averaging rule; adult can record equations.

Arithmetic: Whole numbers and halves; selected example uses integers.

Reasoning: Track fixed versus changeable vertices and apply the rule simultaneously.

Hard stop: Continuum harmonicity and Green functions require calculus and limits.

Materials and preparation

8 minutes estimated prep. Four dots joined in a path, endpoint cards 0 and 9, movable interior number cards, and optional extra graph templates.

A flexible hour

0-10 play with averaging pairs; 10-15 distinguish fixed and interior dots; 15-35 solve and invent landscapes; 35-40 reset; 40-55 chase maxima or compare two solutions; 55-60 present one impossibility argument.

Questions to explore

- If a highest interior dot equals its neighbors' average, what must those neighbors be?
- What happens if a component has no fixed boundary?

Satisfying stop: A solved landscape and the reason a connected averaging graph cannot hide a new highest value.

GA-05 / Facilitator reasoning

Can an averaging landscape hide a peak?

The mathematical question

How can a local averaging rule force a global maximum principle and uniqueness?

On a finite connected graph with a nonempty fixed boundary, each interior value equals the arithmetic mean of its neighbors. Any strict excess over all boundary values would propagate through equal maximal neighbors to the boundary, a contradiction.

Worked instance

Problem: Solve the path 0-a-b-9 with averaging at a and b.

Solution: The equations $2a=b$ and $2b=a+9$ give $a=3$ and $b=6$. They can also be found by equal spacing along the path. For uniqueness, subtract any two solutions: boundary differences are zero and interior differences still average neighbors. A positive maximum forces every neighbor equal; connectedness propagates it to a zero boundary, impossible. Apply the same argument to a negative minimum.

Boundary: A disconnected component with no boundary can have any constant value. It therefore defeats uniqueness and can support a peak above the boundary elsewhere.

Hints to reveal in order

1. An average cannot exceed every number being averaged.
2. Subtract two proposed solutions and inspect the biggest difference.

Extensions

- Give three unequal boundary values and compare how changing one influences interiors.
- Compare the discrete proof with mean values on every circle for a true harmonic function; use GA-35 for continuous polynomial examples.

Connection and its limits

Bridge type: shared-mechanism

Preserves: The learner solves an exact discrete Dirichlet problem and uses maximum propagation.

Limits: This is a discrete counterpart to classical potential theory; no convergence to a harmonic function on a continuum domain is claimed.

Sources and inspection scope

[John Hunter, Notes on Partial Differential Equations](#)

§§2.1 and 2.3, harmonic mean-value and maximum principles

Inspected: Continuous statements inspected; finite graph proof supplied independently

Prior use / what changes: Week 10 networks supply graph familiarity; here edges constrain values rather than routes. This instance is new and unpiloted.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-06 / Participation and launch

When two complex branches meet

MSC primary 32 - Several complex variables and analytic spaces | Secondary 14, 30

The opening challenge

Choose any complex z and find every w satisfying $z^2 + w^2 = 0$. Can your partner move continuously through solutions? What changes if both coordinates must be real? Try changing the equation to $z^2 + w^2 = 1$.

Learner choices: Choose parameter paths, real or complex restrictions, and a branch to track; design a slice that conceals or reveals solutions.

Prerequisite gates

Enter: A/X: complex numbers, $i^2 = -1$, and factoring a quadratic; elementary age access is not claimed.

Explore: V/A: use two complex-coordinate diagrams and parameterize solution pairs.

Explain: P: verify factorization and prove that the two branches meet only at zero.

Prove: X: complex manifolds and the local graph definition for a full singularity proof.

Reading: Symbolic text; a facilitator may separate algebra from geometric interpretation.

Arithmetic: Signed numbers and complex multiplication; no calculus for the factorization.

Reasoning: Remember that each complex coordinate has two real coordinates.

Hard stop: Stop before claiming the origin is singular solely because one gradient vanishes; use the explicit two-branch local structure.

Materials and preparation

10 minutes estimated prep. Two Argand diagrams labeled z and w , paired colored markers, and a table for recording solution pairs.

A flexible hour

0-10 play with multiplication by i ; 10-15 introduce the equation and paired diagrams; 15-35 parameterize branches and compare slices; 35-40 reset; 40-55 study the constant-one equation or discuss local structure; 55-60 state a proved algebraic fact.

Questions to explore

- What does multiplication by i do geometrically?
- Does a drawing containing one isolated real point establish an isolated complex zero?

Satisfying stop: An exhaustive parameterization and a clear explanation of why the real and complex pictures differ.

GA-06 / Facilitator reasoning

When two complex branches meet

The mathematical question

How does a zero set in two complex variables develop a singularity, and why can a real slice hide most of it?

For the polynomial $F(z,w)=z^2+w^2$ on $\mathbb{C}^2$, $F=0$ is the union of two complex lines $w=iz$ and $w=-iz$. They meet only at the origin. In contrast its real slice consists only of $(0,0)$.

Worked instance

Problem: Describe all solutions of $z^2+w^2=0$ and test $z=1+i$.

Solution: Factor as $(w-iz)(w+iz)=0$. Since complex numbers have no zero divisors, $w=iz$ or $w=-iz$. For $z=1+i$ the roots are $-1+i$ and $1-i$. Their squares are $-2i$, while $z^2=2i$. The branches coincide only when $iz=-iz$, hence $z=w=0$. For real coordinates, both squares are nonnegative, so their sum vanishes only at the origin.

Boundary: The real slice is just one point, yet every neighborhood of the complex origin contains infinitely many nonzero solutions. Real pictures cannot certify complex isolation.

Hints to reveal in order

1. Factor z^2+w^2 over $\mathbb{C}$.
2. For nonzero z , divide by z^2 and solve for w/z .

Extensions

- Replace zero by one: at any solution the complex gradient $(2z, 2w)$ is nonzero, so the holomorphic implicit-function theorem gives a smooth complex curve; this last step needs that theorem.
- Compare $z^2-w^3=0$ using $z=t^3, w=t^2$; parameterization alone is not a proof of local nonsingularity.

Connection and its limits

Bridge type: exact-special-case

Preserves: Actual complex polynomial equations and their solution branches are investigated; this is a genuine analytic variety.

Limits: Factoring this example does not prove Hartogs extension, general singularity classification, or that all scalar holomorphic zero sets behave the same way.

Sources and inspection scope

[Jiří Lebl, Tasty Bits of Several Complex Variables](#)

§6.5, Definition 6.5.5 and Example 6.5.6, intersecting complex manifolds

Inspected: Relevant definition and example inspected; plus-sign variant factored directly

Prior use / what changes: No existing week supplies this complex-variable content; this deliberately keeps a substantial algebra prerequisite.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-07 / Participation and launch

Invent a polynomial that triples an angle

MSC primary 33 - Special functions | Secondary 30, 41

The opening challenge

An angle machine receives only $x = \cos \theta$, not θ itself. Build a polynomial that outputs $\cos(3\theta)$. Can different angles with the same input demand different outputs? Choose checks that might expose an error.

Learner choices: Choose test angles, expand through the recurrence, or derive with angle formulas; compare independently obtained answers.

Prerequisite gates

Enter: V/F: read horizontal coordinates on a unit circle and multiply an angle.

Explore: A: expand simple polynomials and substitute fractions.

Explain: P/A: derive T_2 and T_3 from recurrence and check a trigonometric derivation.

Prove: A/P: induction with the cosine addition identity.

Reading: Short symbolic instructions; graphing labels can be read aloud.

Arithmetic: Signed fractions, multiplication, and cubic expressions.

Reasoning: Distinguish the angle from its horizontal coordinate; multiple angles may share a cosine.

Hard stop: Bessel functions, gamma integrals, and approximation theory need additional calculus or analysis.

Materials and preparation

8 minutes estimated prep. Unit-circle sheet, angle marker, graph paper, and recurrence cards $T_0=1$ and $T_1=x$.

A flexible hour

0-10 explore projections while turning arrows; 10-15 state the polynomial-machine task; 15-35 derive and challenge candidates; 35-40 reset; 40-55 graph the machine or compose multipliers; 55-60 share a derivation rather than only a table.

Questions to explore

- Why does squaring x help double an angle?
- What happens when one angle multiplier feeds another?

Satisfying stop: A correct triple-angle polynomial with one explanation and one adversarial check.

GA-07 / Facilitator reasoning

Invent a polynomial that triples an angle

The mathematical question

How can the same family of functions be generated by a geometric action and by an algebraic recurrence?

Chebyshev polynomials satisfy $T_n(\cos \theta) = \cos(n\theta)$, $T_0 = 1$, $T_1 = x$, and $T_{n+1} = 2xT_n - T_{n-1}$ for $n \geq 1$. These identities define genuine polynomial angle multipliers.

Worked instance

Problem: Find T_3 and compute its output at $x = 1/2$.

Solution: The recurrence gives $T_2 = 2x^2 - 1$ and $T_3 = 2x(2x^2 - 1) - x = 4x^3 - 3x$. At $x = 1/2$ the output is $1/2 - 3/2 = -1$, matching $\cos(3 \cdot 60^\circ)$. To see why the same x is enough, cosine addition gives $\cos((n+1)\theta) + \cos((n-1)\theta) = 2\cos \theta \cos(n\theta)$, so induction constructs a polynomial in x alone.

Boundary: Do not identify $T_3(x)$ with $\cos(3x)$. The input is the cosine of an angle, not the angle itself; at $x = 0$ the two expressions are 0 and 1.

Hints to reveal in order

1. First build T_2 from $2x \cdot x - 1$.
2. Use $T_3 = 2xT_2 - T_1$.

Extensions

- Prove $T_m(T_n(x)) = T_{(mn)}(x)$ first on $[-1, 1]$, then as polynomials.
- Study how T_n oscillates with equal extreme heights; connect to minimax approximation only after defining the error criterion.

Connection and its limits

Bridge type: exact-special-case

Preserves: Circle projections and polynomial substitution represent the same exact function on $-1 \leq x \leq 1$.

Limits: A measured projection is approximate; the recurrence and trig identity establish equality. This does not cover all special functions.

Sources and inspection scope

[NIST Digital Library of Mathematical Functions](#)

Equation 18.5.1; recurrence relations in §18.9

Inspected: Displayed identities inspected; $n=2,3$ examples expanded directly

Prior use / what changes: Week 3 repeated actions is a conceptual comparison; polynomial functions and projections create new substance.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-08 / Participation and launch

Can a motion wait and still obey its rate law?

MSC primary 34 - Ordinary differential equations | Secondary 26

The opening challenge

Start at height zero and obey $\text{speed} = 2\sqrt{\text{height}}$ at every time. Can you stay still for a while and then leave? Choose your own waiting time and defend the complete motion, including the moment you start.

Learner choices: Learners select waiting times, propose other departures, and try to disprove each other's solution at the join.

Prerequisite gates

Enter: A/D: functions, square roots, and derivatives of quadratics.

Explore: V: draw a function with a chosen waiting time and inspect the join.

Explain: D/P: check both pieces, the initial condition, and the derivative at the join.

Prove: X: local Lipschitz condition and a uniqueness theorem.

Reading: Symbolic task; oral discussion can support interpretation.

Arithmetic: Squares, square roots, and nonnegative differences.

Reasoning: Distinguish a rate law from a step-by-step update rule and check the exceptional join.

Hard stop: Do not lower this to a coin-moving game and call it an ODE proof; derivatives are central.

Materials and preparation

5 minutes estimated prep. Graph paper, pencils, and three blank time-height axes; optional graphing tool.

A flexible hour

0-10 sketch possible height-time motions; 10-15 clarify that slope is the required speed; 15-35 test formulas and join points; 35-40 reset; 40-55 compare with $y' = y$ or inspect Lipschitz ratios; 55-60 explain the failure of uniqueness.

Questions to explore

- What slope does the equation demand at zero?
- Why does knowing the slope at one instant not settle the whole future?

Satisfying stop: Two fully checked different futures for the same initial data.

GA-08 / Facilitator reasoning

Can a motion wait and still obey its rate law?

The mathematical question

When does a differential equation determine a unique future, and how can a plausible rate rule fail?

For nonnegative y and $t \geq 0$, the initial-value problem $y' = 2\sqrt{y}$, $y(0) = 0$ has $y_c(t) = 0$ for $t \leq c$ and $(t-c)^2$ for $t \geq c$, for every $c \geq 0$. The right side is not locally Lipschitz at $y = 0$.

Worked instance

Problem: Check the motion that waits until $c = 2$ and then follows $y = (t-2)^2$.

Solution: For $t < 2$ both y' and $2\sqrt{y}$ are zero. For $t > 2$, $y' = 2(t-2)$, and nonnegativity of $t-2$ gives $2\sqrt{(t-2)^2} = 2(t-2)$. At $t = 2$, the right difference quotient is $h^2/h = h \rightarrow 0$ and the left quotient is zero, so the derivative exists and equals zero. The initial height is zero. Waiting forever is another solution with exactly the same initial condition.

Boundary: For $y' = y$, $y(0) = 0$, the analogous delayed exponential cannot leave zero while remaining differentiable and satisfying the equation; local Lipschitz uniqueness applies. Failure of one hypothesis permits, but does not force, nonuniqueness.

Hints to reveal in order

1. After leaving at time c , try a translated square.
2. Compute the derivative at c from both sides, not just away from c .

Extensions

- Calculate $[2\sqrt{h-0}]/h = 2/\sqrt{h}$ to see why no uniform local Lipschitz constant exists.
- Compare finite-time blow-up in GA-34: uniqueness and existence for all time are separate properties.

Connection and its limits

Bridge type: exact-special-case

Preserves: Learners choose and verify actual differentiable solutions of an initial-value problem.

Limits: An arrow picture suggests motion but does not verify the equation. This counterexample does not say all non-Lipschitz equations are nonunique.

Sources and inspection scope

[Gerald Teschl, Ordinary Differential Equations and Dynamical Systems](#)

§2.2, Theorem 2.2 and Problem 2.6 on local Lipschitz continuity

Inspected: Existence/uniqueness hypotheses inspected; waiting solutions checked directly

Prior use / what changes: No existing worksheet covers rate-law uniqueness; this is intentionally a calculus-gated extension upward.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-09 / Participation and launch

A wave that travels and a shape that fades

MSC primary 35 - Partial differential equations | Secondary 42

The opening challenge

Here are two shape rules: slide $\sin x$ to the right, or keep it in place while multiplying its height by $e^{(-t)}$. Choose times to compare. Which differential equation does each obey, and can either obey the other's equation?

Learner choices: Choose time slices, derivative order, and a point where a proposed equation is likely to fail.

Prerequisite gates

Enter: V/A: read graphs and understand a formula depending on space and time.

Explore: D: differentiate sine and exponential while holding the other variable fixed.

Explain: D/P: substitute into each PDE and check initial/boundary conditions.

Prove: X: general existence or convergence requires PDE analysis.

Reading: Symbolic formulas; a partner can manage the graph while another differentiates.

Arithmetic: Coordinates and trigonometric values; exponentials for the fading profile.

Reasoning: Separate position x from time t and distinguish time derivative orders.

Hard stop: Without derivatives, stop at comparing graphs and label the PDE connection as supplied information.

Materials and preparation

10 minutes estimated prep. A sine-wave transparency that can slide, fixed axes, and three printed or sketched fading profiles.

A flexible hour

0-10 slide and resize profiles; 10-15 distinguish x and t ; 15-35 verify the two rules and boundary data; 35-40 reset; 40-55 superpose solutions or test a wrong equation; 55-60 share a diagnostic calculation.

Questions to explore

- Does a peak move or only shrink?
- Why does a wave need an initial velocity as well as an initial shape?

Satisfying stop: One complete PDE substitution and a clear graph-based distinction between motion and decay.

GA-09 / Facilitator reasoning

A wave that travels and a shape that fades

The mathematical question

What differences between partial differential equations can be discovered from exact solutions before learning a general solution method?

On the whole line, $u(x,t)=\sin(x-t)$ solves $u_{tt}=u_{xx}$. On $[0,\pi]$ with zero endpoint values, $v(x,t)=e^{(-t)}\sin x$ solves $v_t=v_{xx}$. Partial differentiation verifies both.

Worked instance

Problem: Verify both formulas and test whether $\sin(x-t)$ solves the heat equation.

Solution: For u , $u_t=-\cos(x-t)$, $u_{tt}=\sin(x-t)$, and $u_{xx}=-\sin(x-t)$, so the wave equation holds. The heat equation would require $-\cos(x-t)=-\sin(x-t)$, which fails, for example at $x=t=0$. For v , both v_t and v_{xx} equal $-e^{(-t)}\sin x$. Also $v(0,t)=v(\pi,t)=0$ and $v(x,0)=\sin x$. Equal initial shapes therefore do not imply equal evolution under different laws.

Boundary: The traveling wave on the whole line does not satisfy fixed zero boundary conditions at 0 and π for every t . Changing the domain and boundary data changes the problem.

Hints to reveal in order

1. For u , compute u_t first, then u_{tt} .
2. Check equality symbolically before substituting one convenient point.

Extensions

- Show that $\sin x \cos t$ is a standing wave with zero endpoint displacement and zero initial velocity.
- Add two exact solutions of the same linear PDE and explain why addition works; it generally fails for nonlinear equations.

Connection and its limits

Bridge type: exact-special-case

Preserves: Learners manipulate exact continuum solution profiles and distinguish translation from amplitude decay.

Limits: Sliding paper is a representation, not a physical heat experiment; arbitrary token-spreading rules are not claimed to converge to these PDEs.

Sources and inspection scope

[John Hunter, Notes on Partial Differential Equations](#)

§5.1 and §7.1; heat and wave solutions compared

Inspected: Relevant equations and d'Alembert discussion inspected; selected formulas differentiated directly

Prior use / what changes: Week 9 billiards involves trajectories; this concerns functions spread over space, so it is not a renamed orbit activity.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-10 / Participation and launch

Program a repeating orbit by choosing its itinerary

MSC primary 37 - Dynamical systems and ergodic theory | Secondary 39

The opening challenge

Choose a three-letter pattern of left and right halves. Find a point that repeats that itinerary forever under T . Work backward if forward guessing stalls. Avoid landing exactly at the midpoint while coding the halves.

Learner choices: Choose itineraries, forward or backward reasoning, and a candidate starting point to challenge.

Prerequisite gates

Enter: F/A: double fractions, subtract from two, and follow a two-piece rule.

Explore: A: compose affine inverse branches and solve one linear equation.

Explain: P: verify a complete orbit and exclude smaller periods.

Prove: P/X: general symbolic coding and chaos require precise dynamical definitions.

Reading: Short symbolic rules; two colored interval halves support reading.

Arithmetic: Signed fractions; denominators seven in the checked example.

Reasoning: Track whether a letter names the current half or an inverse branch; use one convention consistently.

Hard stop: No assertion of randomness or chaos follows merely from an irregular plot.

Materials and preparation

8 minutes estimated prep. Number line $[0, 1]$, two branch cards, fraction markers, and graph paper; no computer required.

A flexible hour

0-10 iterate easy fractions; 10-15 fix the itinerary convention; 15-35 engineer periodic points; 35-40 reset; 40-55 compare periods or nearby starts; 55-60 present an exact orbit rather than a decimal trace.

Questions to explore

- How can one target value have two possible previous values?
- Does a word of length three always force least period three?

Satisfying stop: Construct and verify one prescribed repeating orbit.

GA-10 / Facilitator reasoning

Program a repeating orbit by choosing its itinerary

The mathematical question

Can we design periodic behavior in a nonlinear system by solving backward rather than guessing starting points?

The tent map on $[0,1]$ is $T(x)=2x$ for $x \leq 1/2$ and $T(x)=2-2x$ for $x \geq 1/2$. Its inverse branches are $L(y)=y/2$ and $R(y)=1-y/2$; branch words can specify exact periodic orbits.

Worked instance

Problem: Find a point with repeating current-half itinerary LRR.

Solution: Writing the three-step return using inverses gives $x=(1-(1-x/2)/2)/2=1/4+x/8$, hence $x=2/7$. Forward verification gives $2/7 \rightarrow 4/7 \rightarrow 6/7 \rightarrow 2/7$; the halves are left,right,right. The three values are distinct, so the least period is three. Inverse algebra found a candidate; checking the branch inequalities made it a valid solution.

Boundary: The word LLL yields the fixed point zero rather than least period three. Points landing at $1/2$ have a coding ambiguity unless a convention is imposed, so our core avoids them.

Hints to reveal in order

1. Solve $x=L(R(R(x)))$ for the word LRR.
2. Verify which half contains each intermediate value after solving.

Extensions

- Find the starting point for RRL. Answer: $4/7 \rightarrow 6/7 \rightarrow 2/7 \rightarrow 4/7$, the same orbit as LRR with a different starting point. Verify the half labels at each step.
- Compare nearby rational starts for several steps; distinguish finite sensitivity evidence from a theorem about chaos.

Connection and its limits

Bridge type: exact-special-case

Preserves: A number-line point is the actual state of a continuous interval map; legal moves are evaluations of T .

Limits: Three observed steps do not prove chaotic dynamics. Finite decimal arithmetic can manufacture periodic orbits.

Sources and inspection scope

[Gerald Teschl, Ordinary Differential Equations and Dynamical Systems](#)

§§11.3-11.5, tent map and symbolic dynamics

Inspected: Tent-map definition and symbolic mechanism inspected; itinerary solved directly

Prior use / what changes: Week 4 modular orbits were finite cyclic dynamics; here the state interval is continuous and the map folds it.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-11 / Participation and launch

How much does an addition machine have to reveal?

MSC primary 39 - Difference and functional equations | Secondary 26

The opening challenge

The machine always obeys $f(x+y)=f(x)+f(y)$, and $f(1)=3$. Choose an input whose output you think is forced. Convince a skeptical partner using only the rule. Then try an input your argument cannot reach.

Learner choices: Choose inputs, chains of decompositions, and adversarial output proposals; decide which domain the machine accepts.

Prerequisite gates

Enter: C: combine whole-number inputs under one stated rule.

Explore: F/A: use negatives and fractions and name an unknown output.

Explain: P: force integer and rational values using repeated addition.

Prove: P/X: extending the rational linearity conclusion to all real inputs uses a stated regularity assumption, such as continuity; constructing nonstandard additive extensions is a separate algebraic task. The restricted irrational domain here uses $\mathbb{Q}$ -linear independence.

Reading: Short rules; symbolic follow-up can be adult recorded.

Arithmetic: Addition, division, signed fractions.

Reasoning: Distinguish values forced by a universal rule from guesses suggested by examples.

Hard stop: Do not claim the rational proof covers arbitrary real inputs without a continuity argument.

Materials and preparation

5 minutes estimated prep. Input-output cards, blank cards for learner-selected inputs, and a rule card.

A flexible hour

0-10 invent whole-number machines; 10-15 impose the addition rule; 15-35 force zero, negative, and fractional outputs; 35-40 reset; 40-55 examine irrational inputs or prove the rational theorem; 55-60 classify forced and unproved claims.

Questions to explore

- What must $f(0)$ be?
- Why does a repeated-addition proof reach $2/3$ but not $\sqrt{2}$?

Satisfying stop: One forced fractional output and a correct statement of the domain on which the proof works.

GA-11 / Facilitator reasoning

How much does an addition machine have to reveal?

The mathematical question

Does a rule about combining inputs determine an entire function, or do hidden domain assumptions matter?

If $f: \mathbb{Q} \rightarrow \mathbb{R}$ is additive, then $f(q) = qf(1)$. On $\mathbb{R}$, continuity at one point forces the same form; additivity alone allows nonstandard functions. The core proves only the rational statement.

Worked instance

Problem: Determine $f(2/3)$ and decide whether $f(\sqrt{2})=5$ is already impossible from the rational argument.

Solution: Additivity gives $f(2)=6$ and $3f(2/3)=f(2)=6$, so $f(2/3)=2$. On the restricted domain $D=\{a+b\sqrt{2}: a, b \text{ rational}\}$, define $f(a+b\sqrt{2})=3a+5b$. Representation is unique because $\sqrt{2}$ is irrational, and coordinatewise addition verifies additivity. Thus $f(1)=3$ and $f(\sqrt{2})=5$ coexist on D . This is an exact counterexample to extending the rational conclusion by algebra alone.

Boundary: If continuity is additionally required on D , rational numbers approaching $\sqrt{2}$ force $f(\sqrt{2})=3\sqrt{2}$, contradicting 5. Continuity is a mathematical condition, not a property established by a smooth-looking sketch.

Hints to reveal in order

1. Use $f(0)=f(0)+f(0)$.
2. Three copies of the input $2/3$ add to two.

Extensions

- For arbitrary $f(1)=c$, prove $f(m/n)=cm/n$ with n positive.
- Replace additivity by positive multiplicativity $f(x+y)=f(x)f(y)$; logarithms turn it into an additive equation when those tools are available.

Connection and its limits

Bridge type: exact-special-case

Preserves: An actual function obeys $f(x+y)=f(x)+f(y)$; learners infer values from the equation rather than from a finite pattern table.

Limits: The elementary core does not construct a pathological additive function on all of $\mathbb{R}$ or prove the general regularity theorem.

Sources and inspection scope

[Daniel Reem, Remarks on the Cauchy functional equation and variations of it](#)

Introduction and §2, regularity assumptions for additive functions

Inspected: Relevant discussion inspected; rational proof and restricted-domain counterexample supplied

Prior use / what changes: No prior-week equivalent; differs from guessing patterns because the universal functional equation is the evidence.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-12 / Participation and launch

Smaller pieces: a finite total or an endless climb?

MSC primary 40 - Sequences, series, summability | Secondary 26, 28

The opening challenge

Two machines pay $1, 1/2, 1/4, \dots$ or $1, 1/2, 1/3, \dots$. Choose a target total. Can you guarantee that either machine will pass it, without adding every term individually?

Learner choices: Choose targets, grouping schemes, and an opponent's proposed upper bound.

Prerequisite gates

Enter: O/F: compare halves and collect successive pieces.

Explore: F/M: use reciprocals, powers of two, and equal-size blocks.

Explain: P: give a construction exceeding a chosen target.

Prove: P: quantified unboundedness, contrasting a geometric remainder bound.

Reading: Fraction labels; oral grouping with an adult recorder is possible.

Arithmetic: Unit fractions, doubling, and simple lower bounds.

Reasoning: Track a bound for a whole block without needing its exact sum.

Hard stop: A graph leveling off over finitely many terms is not evidence sufficient for convergence.

Materials and preparation

10 minutes estimated prep. Fraction strips or drawn rectangles, cards $1/n$, and blank target cards.

A flexible hour

0-10 assemble early payments; 10-15 compare the two rules; 15-35 search for bounds and blocks; 35-40 reset; 40-55 prove one general claim or estimate a target-crossing index; 55-60 distinguish experiment from guarantee.

Questions to explore

- Could a million tiny payments still contribute a substantial amount?
- What feature gives the geometric machine an exact remaining debt?

Satisfying stop: A grouping argument that defeats every proposed finite harmonic bound.

GA-12 / Facilitator reasoning

Smaller pieces: a finite total or an endless climb?

The mathematical question

How can terms become tiny while their accumulated sum still exceeds every bound?

Geometric partial sums have explicit shrinking remainders. Harmonic partial sums are unbounded: each block from $2^{(k-1)+1}$ through 2^k contains $2^{(k-1)}$ terms at least $1/2^k$, contributing at least $1/2$.

Worked instance

Problem: Prove that the first 16 harmonic terms exceed 3, without computing their exact sum.

Solution: Separate 1; then blocks $\{1/2\}$, $\{1/3, 1/4\}$, $\{1/5, \dots, 1/8\}$, and $\{1/9, \dots, 1/16\}$. Each block after the initial 1 contributes at least $1/2$, so the sum is at least 3. The second block contributes strictly more than $1/2$, hence the total exceeds 3. Continuing gives $H_{(2^k)} \geq 1 + k/2$, exceeding any fixed target for sufficiently large k . In contrast the geometric sum through $1/2^k$ is $2 - 1/2^k$, always below 2.

Boundary: Both sequences of payments tend to zero. That necessary condition for series convergence therefore cannot decide which total converges.

Hints to reveal in order

1. Group harmonic terms by doubling the final denominator.
2. Replace every term in a block by its smallest term to get a lower bound.

Extensions

- Give a guaranteed, deliberately loose index that exceeds 10: $k=19$ and $N=2^{19}$ suffice.
- Average the partial sums of $1-1+1-\dots$; explain why their limit $1/2$ is a summability convention rather than ordinary convergence.

Connection and its limits

Bridge type: exact-special-case

Preserves: Lengths and grouped fractions are actual partial sums; the infinite claim is supported by a rule producing arbitrarily many blocks.

Limits: Physical strip width limits terminate a demonstration, not the mathematical sequence. Nothing here licenses rearranging a conditionally convergent series.

Sources and inspection scope

[Jiří Lebl, Basic Analysis I](#)

§§2.5-2.6, geometric and harmonic series

Inspected: Relevant series statements inspected; block proof supplied

Prior use / what changes: Year-1 figurate and fraction work may support entry; exact prior use of infinite series is unrecorded. GA-03 uses a geometric cover for a different question.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-13 / Participation and launch

Fit a curve inside the narrowest error band

MSC primary 41 - Approximations and expansions | Secondary 49

The opening challenge

Choose any straight line and try to keep its vertical error from the parabola as small as possible at every x . Tilting is allowed. Can your opponent prove you cannot beat their band width?

Learner choices: Choose slope and intercept, diagnostic points, and a proposed certificate of optimality.

Prerequisite gates

Enter: O/V: compare a line's vertical discrepancy from a supplied parabola.

Explore: F/A: use $y=ax+b$ and absolute errors at three x -values.

Explain: P: combine inequalities to force a lower bound and verify attainment.

Prove: A/P: direct proof suffices here; general equioscillation requires approximation theory.

Reading: Short graph labels; symbolic proof can follow an oral search.

Arithmetic: Halves, signed values, linear equations.

Reasoning: Distinguish vertical error from perpendicular distance and worst error from average error.

Hard stop: Calculus is not needed for this instance; least-squares extensions require a new objective and possibly integration.

Materials and preparation

10 minutes estimated prep. Graph of x^2 on $[-1,1]$, transparent movable lines, and an adjustable vertical-width band.

A flexible hour

0-10 experiment with lines; 10-15 define maximum vertical error; 15-35 optimize and challenge candidates; 35-40 reset; 40-55 build a three-point proof or change the objective; 55-60 report the best value and its certificate.

Questions to explore

- Why might a tilted line help one endpoint but hurt the other?
- Which three locations are enough to force a lower bound?

Satisfying stop: A complete lower bound plus an attaining line, not just the prettiest fit.

GA-13 / Facilitator reasoning

Fit a curve inside the narrowest error band

The mathematical question

How can a few unavoidable errors certify the best approximation everywhere on an interval?

For $f(x)=x^2$ on $[-1,1]$, the unique affine function minimizing maximum absolute error is $p(x)=1/2$, with error $1/2$. Three alternating extreme errors certify the optimum; the card supplies a direct proof.

Worked instance

Problem: Prove that no affine line beats the constant $1/2$.

Solution: Suppose $p=ax+b$ has error at most E . At zero, $|b|\leq E$. At the endpoints, $|1-b-a|\leq E$ and $|1-b+a|\leq E$, implying $|1-b|\leq E$ by averaging. Thus $1\leq |b|+|1-b|\leq 2E$, so $E\geq 1/2$. The constant $1/2$ attains this bound because $0\leq x^2\leq 1$. Equality forces $b=1/2$; endpoint inequalities then force $a=0$, proving uniqueness.

Boundary: Passing through more selected points need not improve worst error between them. Also a least-squares best constant for x^2 on this interval is $1/3$, so a different error criterion gives a different answer.

Hints to reveal in order

1. Inspect $x=-1,0,1$.
2. Average the two endpoint residuals before comparing with the center.

Extensions

- Repeat on $[0,1]$: $p=x-1/8$ has maximum error $1/8$, certified at $0,1/2,1$; verify by completing the square.
- Let students choose a continuous polygonal target and derive the best constant from its maximum and minimum.

Connection and its limits

Bridge type: exact-special-case

Preserves: Moving a line and its uniform-width error band is precisely a best-uniform approximation problem.

Limits: Measured fit suggests a candidate; endpoint and center inequalities establish optimality over all affine competitors, not just sampled lines.

Sources and inspection scope

[NIST Digital Library of Mathematical Functions](#)

§3.11(i), minimax approximation and alternating extrema

Inspected: Relevant criterion inspected; affine example proved independently

Prior use / what changes: New approximation family; related to optimization in Week 1 but the domain is a continuous interval.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-14 / Participation and launch

Two different motions with the same snapshots

MSC primary 42 - Harmonic analysis on Euclidean spaces | Secondary 65

The opening challenge

One arrow completes one turn per second; another completes five. Photograph their heights four times per second. Can you tell which motion produced the photographs? Choose one extra photograph that would help.

Learner choices: Choose starting phases, sample schedules, alternative speeds, and a diagnostic extra time.

Prerequisite gates

Enter: O/V: compare arrow heights at regularly spaced snapshots.

Explore: F: quarter-turn timing and integer extra rotations.

Explain: A/P: use trigonometric periodicity to prove equality at all sample times.

Prove: X: band-limited reconstruction requires Fourier analysis and precise hypotheses.

Reading: Short time labels; formulas optional until explanation.

Arithmetic: Integer turns, quarters, and optional exact sine values.

Reasoning: Separate actual motion from its sampled record and allow multiple compatible explanations.

Hard stop: Do not promise reconstruction without an explicit frequency restriction.

Materials and preparation

8 minutes estimated prep. Two clock-face arrows, four snapshot cards, and a strip of equally spaced time marks.

A flexible hour

0-10 compare free rotations; 10-15 fix sample times and equal initial phase; 15-35 build identical records from different motions; 35-40 movement reset; 40-55 prove the ambiguity or redesign sampling; 55-60 share both explanations of one record.

Questions to explore

- How many extra complete turns occur between photographs?
- Would photographing only the height lose more information than recording the whole arrow?

Satisfying stop: Two different exact motions that produce the same entire regular snapshot record.

GA-14 / Facilitator reasoning

Two different motions with the same snapshots

The mathematical question

What information is lost by sampling, and why can exact agreement at every recorded time still conceal different signals?

The signals $f(t)=\sin(2\pi t)$ and $g(t)=\sin(10\pi t)$ agree at $t=n/4$ for every integer n , since their angles differ by $2\pi n$. They differ between those sample times.

Worked instance

Problem: List samples at $t=0, 1/4, 1/2, 3/4$, and distinguish the signals at $t=1/8$.

Solution: Both sample lists are $0, 1, 0, -1$. More generally, $10\pi(n/4) - 2\pi(n/4) = 2\pi n$, so equality persists for every integer snapshot index. At $t=1/8$, $f=\sin(\pi/4)=\sqrt{2}/2$ and $g=\sin(5\pi/4)=-\sqrt{2}/2$, so one extra height observation separates these two candidates. A physical device adds measurement uncertainty; the mathematical result uses exact times.

Boundary: That extra sample only distinguishes the two proposed signals. It does not determine the correct signal among every continuous function; infinitely many can fit any finite record.

Hints to reveal in order

1. Compare angles modulo one full turn at each sampling time.
2. Look halfway between two adjacent sample times.

Extensions

- Find all integer frequencies giving the same samples as frequency one at rate four.
- Record both horizontal and vertical coordinates, then compare the remaining ambiguity with height-only sampling.

Connection and its limits

Bridge type: exact-special-case

Preserves: Rotating arrows represent continuous sinusoidal signals; sampling is an exact restriction of those functions to specified times.

Limits: This does not prove a general sampling theorem, and four values never determine an unrestricted continuous function.

Sources and inspection scope

[Daniel Stroock, Topics in Fourier Analysis](#)

§§1-2, periodic exponentials and Fourier representation

Inspected: Periodic-frequency framework inspected; aliasing example checked directly

Prior use / what changes: Week 4 periodicity is a prerequisite comparison; the new question concerns information lost by observing continuous motion discretely.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-15 / Participation and launch

Four brightnesses, four hidden patterns

MSC primary 43 - Abstract harmonic analysis | Secondary 15, 42

The opening challenge

Make a four-square brightness picture. Rebuild it using only scaled copies of these four sign patterns. Can a partner recover your hidden coefficients? Then swap the two columns and predict which coefficients change sign.

Learner choices: Choose target arrays, coefficients, reconstruction order, and a translation of the square labels.

Prerequisite gates

Enter: C/F: add brightness values and signed contributions; halves may occur.

Explore: L: treat four-entry lists as vectors and use dot products.

Explain: P: verify orthogonality and reconstruct the original pattern.

Prove: L/X: group characters for the general finite-abelian theorem.

Reading: Four short pattern cards; negative contributions should be explained orally.

Arithmetic: Signed addition, multiplication, and division by four.

Reasoning: Keep one ordering: 00,01,10,11; negative coefficients are corrections, not negative physical light.

Hard stop: Stop before nonabelian or continuous transforms without representation theory or integration.

Materials and preparation

10 minutes estimated prep. Four sign-pattern cards, counters for positive brightness, red counters for negative corrections, and a two-by-two grid.

A flexible hour

0-10 combine sign patterns freely; 10-15 fix coordinates and scaling; 15-35 encode and decode pictures; 35-40 reset; 40-55 predict column swaps or justify the formula; 55-60 share a reconstructed pattern.

Questions to explore

- Why does each nonconstant sign card have average zero?
- What happens when two different sign cards are multiplied entry by entry and summed?

Satisfying stop: Encode and decode one picture, then predict a symmetry without recomputing everything.

GA-15 / Facilitator reasoning

Four brightnesses, four hidden patterns

The mathematical question

Why do a group's symmetries supply the right coordinates for decomposing a pattern?

On $G=(\mathbb{Z}/2\mathbb{Z})^2$, the four character patterns $(1,1,1,1)$, $(1,1,-1,-1)$, $(1,-1,1,-1)$, and $(1,-1,-1,1)$ are mutually orthogonal. Coefficients of any real four-entry pattern are one quarter of its dot products with these patterns.

Worked instance

Problem: Decompose $(4,2,0,2)$ in the displayed basis.

Solution: The dot products are 8,4,0,4, so the coefficients are 2,1,0,1. Reconstruction gives $(2,2,2,2)+(1,1,-1,-1)+(1,-1,-1,1)=(4,2,0,2)$. Each basis vector has squared length four, and distinct ones have dot product zero, explaining the division by four. Swapping columns translates by 01; it changes the signs of the third and fourth coefficients, giving $(2,4,2,0)$.

Boundary: Brightness data need not give nonnegative coefficients. Requiring only positive pattern weights would forbid many valid reconstructions and change the mathematical problem.

Hints to reveal in order

1. Add all four brightnesses to find the constant contribution.
2. A sign card cancels every other sign card under a dot product.

Extensions

- Construct an array requiring four nonzero coefficients.
- Use three binary coordinates to build eight Walsh patterns; explain why the signs multiply according to parity.

Connection and its limits

Bridge type: exact-special-case

Preserves: A two-by-two brightness array is a function on an actual finite group, and the sign cards are its characters.

Limits: The finite transform is exact but does not establish Haar integration or harmonic analysis on infinite groups.

Sources and inspection scope

[Rohan Dandavati, The Fourier Transform on Finite Groups: Theory and Computation](#)

§3, Theorems 3.5-3.8

Inspected: Finite-character basis argument inspected; four-entry arithmetic checked directly

Prior use / what changes: Week 2 used binary toggles; this uses real brightness, orthogonal decomposition, and translation. The object and question are different.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-16 / Participation and launch

Slide two windows and predict their overlap graph

MSC primary 44 - Integral transforms, operational calculus | Secondary 26, 42

The opening challenge

Slide the short window $[t-1, t]$ across the long window $[0, 2]$. Before tracing the answer, predict its entire overlap-length graph. Choose lengths that make the graph a triangle, a trapezoid, or something else.

Learner choices: Choose strip lengths, direction, observation points, and a claimed graph to challenge.

Prerequisite gates

Enter: O: slide strips and compare overlap lengths.

Explore: F/V: read positions, lengths, and a graph against displacement.

Explain: A/P: locate each change in the endpoint formula and explain every interval.

Prove: I: integrate indicator products, or accept length as their integral.

Reading: Minimal reading; endpoint labels 0, 1, 2, 3.

Arithmetic: Subtraction of lengths and optional halves.

Reasoning: The moving interval is $[t-1, t]$, so its right endpoint, not its center, is t .

Hard stop: General convolution and Laplace identities require integrability and convergence conditions.

Materials and preparation

8 minutes estimated prep. One fixed strip of length two, one transparent strip of length one, ruler, and blank t -versus-overlap axes.

A flexible hour

0-10 play with sliding overlaps; 10-15 define t precisely; 15-35 draw and justify graphs; 35-40 reset; 40-55 reverse roles or stack two-window operations; 55-60 explain a corner or plateau.

Questions to explore

- At what t does overlap begin and end?
- Why does unequal window length create a flat top?

Satisfying stop: A complete overlap graph justified by endpoint cases.

GA-16 / Facilitator reasoning

Slide two windows and predict their overlap graph

The mathematical question

How does convolution turn two shapes into a third, and how can geometry reveal the formula before integration?

For interval indicators $f=1_{[0,2]}$ and $g=1_{[0,1]}$, $(f*g)(t)$ is the length of $[0,2] \cap [t-1,t]$. It rises linearly, has a plateau, and falls linearly.

Worked instance

Problem: Find the full convolution of the length-two and length-one indicators.

Solution: The overlap length is $\max(0, \min(2, t) - \max(0, t-1))$. Therefore it is zero for $t \leq 0$, t for $0 \leq t \leq 1$, one for $1 \leq t \leq 2$, $3-t$ for $2 \leq t \leq 3$, and zero for $t \geq 3$. At $t=1/2, 3/2, 5/2$ the values are $1/2, 1, 1/2$. These cases cover every real displacement, so they prove the whole graph. Single endpoints have zero length and do not change the values.

Boundary: Simply sliding $[t, t+1]$ instead changes the parameter and shifts the graph. Forgetting the reflection/shift convention can produce a correct-looking but incorrectly positioned convolution.

Hints to reveal in order

1. The overlap starts at the larger left endpoint.
2. The overlap ends at the smaller right endpoint; subtract only when the result is positive.

Extensions

- For lengths $a \geq b > 0$, derive a plateau of height b and support $[0, a+b]$.
- Integrate the overlap graph: its area is two, the product of the original interval lengths; then ask why this extends to nonnegative integrable functions.

Connection and its limits

Bridge type: exact-special-case

Preserves: Sliding a reflected-and-shifted interval computes the actual continuum convolution integral of two indicator functions.

Limits: A sampled overlap table is only partial evidence; interval endpoints give the exact piecewise formula. This is not merely finite multiplication.

Sources and inspection scope

[Stephen Boyd, EE102 Lecture 3: The Laplace Transform](#)

Slides 3-27 through 3-29, convolution and its transform

Inspected: Convolution definition inspected; overlap integral derived directly

Prior use / what changes: No current-week equivalent; connects continuous geometry to transforms with inexpensive reusable materials.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-17 / Participation and launch

An equation that listens to its own average

MSC primary 45 - Integral equations | Secondary 47

The opening challenge

Choose f and λ . Find a graph whose height is $f(x)$ plus λ times its own average height. Can turning the feedback dial destroy all solutions, or create infinitely many?

Learner choices: Choose feedback values and forcing functions; predict the number of solutions before calculating.

Prerequisite gates

Enter: A/I: functions, definite integrals, and the mean over a unit interval.

Explore: A: introduce m as an unknown number and solve a linear equation.

Explain: P/I: integrate both sides, then substitute the resulting function back.

Prove: L/X: compact operators for general Fredholm theory.

Reading: Symbolic text; graphs support the meaning of an average.

Arithmetic: Fractions and linear equations; $\int_0^1 x \, dx = 1/2$.

Reasoning: Separate necessary compatibility from sufficient construction; verify both.

Hard stop: A finite-group averaging puzzle is an analogy unless the integral formulation is actually used.

Materials and preparation

5 minutes estimated prep. Graph paper and cards selecting $f(x)=x$ or $x-1/2$ and $\lambda=0, 1/2, 1, 2$.

A flexible hour

0-10 compare means of simple lines; 10-15 state the feedback equation; 15-35 solve selected pairs; 35-40 reset; 40-55 investigate $\lambda=1$ and compatibility; 55-60 explain one transition in the number of solutions.

Questions to explore

- What equation must the unknown mean satisfy?
- If the mean equation works, how do you recover the whole graph?

Satisfying stop: Three fully verified examples showing unique, absent, and infinitely many solutions.

GA-17 / Facilitator reasoning

An equation that listens to its own average

The mathematical question

When does feedback through an integral produce one solution, no solution, or infinitely many?

On $[0, 1]$, solve $u(x) = f(x) + \lambda m$ where $m = \int_0^1 u(t) dt$. Integrating gives $(1 - \lambda)m = \int f$. This rank-one integral equation is exactly reducible to one scalar compatibility condition.

Worked instance

Problem: Solve first $f=x$, $\lambda=1/2$, then compare $\lambda=1$ for $f=x$ and $f=x-1/2$.

Solution: For $\lambda=1/2$, $m=1/2+m/2$ gives $m=1$ and $u=x+1/2$; its integral is indeed one. With $f=x$ and $\lambda=1$, integration demands $0=1/2$, so no solution exists. With $f=x-1/2$ and $\lambda=1$, the forcing has zero mean. Every $u=x-1/2+C$ has mean C and satisfies the equation; conversely every solution must have this form, giving exactly a one-parameter family.

Boundary: The singular dial value does not always mean no solution. It means the forcing must satisfy a compatibility condition; when it does, uniqueness fails.

Hints to reveal in order

1. Name the mean m before trying to solve for every $u(x)$.
2. At $\lambda=1$, inspect the mean of f rather than dividing by zero.

Extensions

- Choose $f=x^2$ and compute the compatibility using its mean $1/3$.
- Replace the constant kernel by $x \cdot t$, so the unknown moment is $\int t u(t) dt$; derive another scalar condition.

Connection and its limits

Bridge type: exact-special-case

Preserves: The unknown is an actual function and its continuum mean is part of its defining equation.

Limits: The simple kernel is a genuine special case; it does not prove the Fredholm alternative for arbitrary compact operators.

Sources and inspection scope

[Michael O'Neil, Integral Equations and Fast Algorithms](#)

Fredholm Alternative, Theorem 3, printed p.54

Inspected: Theorem statement inspected; rank-one examples solved independently

Prior use / what changes: New calculus-level family; graph averaging in GA-05 introduces a related mechanism but has different unknowns and solvability conditions.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-18 / Participation and launch

Which constant best represents a whole landscape?

MSC primary 46 - Functional analysis | Secondary 41, 49

The opening challenge

Replace this two-level landscape by one flat height. Each part charges its width times the square of its vertical error. Choose a replacement that costs least. Then change the rule to charge only the largest error.

Learner choices: Choose candidate heights, compare cost rules, and redesign the widths to favor a different answer.

Prerequisite gates

Enter: F/V: compare the widths and heights of two rectangular pieces.

Explore: A: square discrepancies and combine weighted costs.

Explain: P/A: complete the square to prove global optimality.

Prove: I/X: interpret the cost as an L^2 integral and extend to Hilbert spaces.

Reading: Short labels and a cost rule; a recorder can handle algebra.

Arithmetic: Quarters, squares, signed differences.

Reasoning: Keep width weights: a small high region must not count equally with a wide low region.

Hard stop: General function-space conclusions require norms, integrals, and completeness.

Materials and preparation

8 minutes estimated prep. A unit-width profile with height zero over three quarters and height four over one quarter; movable horizontal line.

A flexible hour

0-10 inspect and approximate the landscape; 10-15 define the weighted squared cost; 15-35 optimize by trials and algebra; 35-40 reset; 40-55 compare maximum error or derive a general weighted mean; 55-60 state which definition of best was used.

Questions to explore

- Why should the wide low region matter more?
- Can the same constant win under two different error rules?

Satisfying stop: An optimal constant with a completed-square certificate and a second optimum under another distance.

GA-18 / Facilitator reasoning

Which constant best represents a whole landscape?

The mathematical question

How does a choice of distance between functions change the meaning of best approximation?

For $f=0$ on $[0, 3/4)$ and $f=4$ on $[3/4, 1]$, its squared L^2 distance to constant c is $(3/4)c^2 + (1/4)(4-c)^2 = (c-1)^2 + 3$. Thus $c=1$ uniquely minimizes this distance.

Worked instance

Problem: Find the squared-error best constant and the maximum-error best constant.

Solution: The squared-error cost is $c^2 - 2c + 4 = (c-1)^2 + 3$, so $c=1$ uniquely gives the minimum three. Under maximum error, the cost is $\max(|c|, |4-c|)$, at least two by the triangle inequality $4 \leq |c| + |4-c|$. Height $c=2$ attains two and is the unique minimizer. Thus different legitimate distances select different flat representatives.

Boundary: Using the unweighted average of the two displayed heights gives two, which ignores their unequal widths and fails for squared error. Requiring exact equality to the profile has no constant solution.

Hints to reveal in order

1. Write the two rectangular contributions separately.
2. Expand the total and try to complete one square.

Extensions

- For heights a, b occupying proportions p and $1-p$, derive $c=pa+(1-p)b$.
- Explain orthogonality: the residual $f-1$ has integral zero, so its inner product with every constant vanishes.

Connection and its limits

Bridge type: exact-special-case

Preserves: A step-function space with weighted squared distance is an exact finite-dimensional subspace of $L^2[0, 1]$; constant approximation is orthogonal projection.

Limits: This example does not prove infinite-dimensional projection existence, completeness, or general Banach-space behavior.

Sources and inspection scope

[Sheldon Axler, Measure, Integration & Real Analysis](#)

§8B, orthogonal projection; 8.34 and 8.43

Inspected: Projection definitions and characterization inspected; step-function calculation supplied

Prior use / what changes: GA-13 minimizes uniform error for a parabola; this changes both the space and the error geometry rather than repeating the same fit.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-19 / Participation and launch

A tiny graph with a large derivative

MSC primary 47 - Operator theory | Secondary 26, 65

The opening challenge

Design a function that stays extremely close to zero but whose derivative is at least one somewhere. Your opponent proposes a universal multiplier C between graph size and derivative size. Can your function break that promise?

Learner choices: Choose n , an error budget, and a claimed bound; test alternative function families.

Prerequisite gates

Enter: A/D: polynomial functions and the power rule.

Explore: V: compare graphs on one fixed scale and inspect endpoint slopes.

Explain: P: use a parameter n to defeat any proposed amplification bound.

Prove: P/L: definition of a bounded linear operator.

Reading: Symbolic formulas; graph explanations can precede notation.

Arithmetic: Powers, reciprocals, and inequalities.

Reasoning: Do not change the graph scale secretly; separate visual height from derivative height.

Hard stop: Without derivatives, observations remain motivation. Infinite-dimensional spectral theory is a separate hard stop.

Materials and preparation

5 minutes estimated prep. Graph paper with a fixed vertical scale, polynomial cards $n=1,2,5,10$, and optional plotting software.

A flexible hour

0-10 compare low-degree graphs; 10-15 define both supremum sizes; 15-35 challenge bounds using polynomials; 35-40 reset; 40-55 prove unboundedness or change the norm; 55-60 state exactly what failed.

Questions to explore

- Where is x^n largest on $[0,1]$?
- What happens if we measure the derivative as part of input size?

Satisfying stop: One exact tiny-input witness and a parameter argument defeating every fixed bound.

GA-19 / Facilitator reasoning

A tiny graph with a large derivative

The mathematical question

Can a transformation amplify arbitrarily small input errors, and why does the chosen norm matter?

Differentiation $D: C^1[0, 1] \rightarrow C[0, 1]$ is unbounded when both spaces use the supremum norm. The polynomials $f_n(x) = x^n/n$ have norm $1/n$ while $Df_n = x^{n-1}$ has norm one.

Worked instance

Problem: Find a polynomial with maximum height at most $1/100$ and derivative equal to one at $x=1$.

Solution: Take $f(x) = x^{100}/100$. For $0 \leq x \leq 1$, its height is between zero and $1/100$. Its derivative is x^{99} , equaling one at $x=1$. More generally $\|Df_n\|_\infty / \|f_n\|_\infty = n$, so for any fixed C choose an integer $n > C$ and the inequality $\|Df\|_\infty \leq C \|f\|_\infty$ fails. The sequence f_n tends uniformly to zero while its derivatives do not tend uniformly to zero.

Boundary: With input norm $\|f\|_\infty + \|f'\|_\infty$, differentiation has bound one by definition. The operator did not change; the topology used to judge small input changed.

Hints to reveal in order

1. Divide x^n by n .
2. Choose n larger than the opponent's proposed constant.

Extensions

- Compare $f_n = x^n$ with $f_n = x^n/n$ and explain why only the second tends uniformly to zero.
- Explore finite differences on a noisy table, clearly separating an error-amplification model from a proved algorithmic bound.

Connection and its limits

Bridge type: exact-special-case

Preserves: The input objects are actual functions and differentiation is a linear operator; supremum discrepancy measures their size.

Limits: This establishes one unbounded operator between specified normed spaces, not a general theorem about spectra or numerical differentiation.

Sources and inspection scope

[Sheldon Axler, Measure, Integration & Real Analysis](#)

Chapter 10, bounded linear maps and operator norms

Inspected: Definitions inspected; unbounded-differentiation witness supplied

Prior use / what changes: New operator-theory family. GA-18 compares norms for optimization; here the question is continuity of a transformation.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-20 / Participation and launch

Save quadratic effort with a steady journey

MSC primary 49 - Calculus of variations and optimal control; optimization | Secondary 26

The opening challenge

Reach position one from zero in one time unit. Each time segment charges its duration times speed squared. Choose your speeds and see whether rushing early and resting later can cost less than moving steadily.

Learner choices: Choose two-speed schedules, permit backward motion, and invent smooth challengers.

Prerequisite gates

Enter: F/V: compare journeys using constant speeds on two equal time intervals.

Explore: A: square speeds and calculate weighted cost.

Explain: P/A: prove the two-speed optimum by completing a square.

Prove: D//P: the general continuous piecewise- C^1 path proof applies the fundamental theorem of calculus on each piece and adds.

Reading: Short schedule labels initially; symbolic integral extension.

Arithmetic: Halves, signed speeds, and squares.

Reasoning: Distinguish fixed duration, endpoint displacement, and total traveled distance; reversing is allowed.

Hard stop: The universal result for continuous piecewise- C^1 paths needs calculus; checking finitely many schedules alone does not prove it.

Materials and preparation

8 minutes estimated prep. A unit time-position diagram, two half-time cards, speed cards, and optional string curves.

A flexible hour

0-10 act or draw varied journeys; 10-15 define the artificial cost; 15-35 optimize two-speed schedules; 35-40 reset; 40-55 prove the smooth theorem or design another objective; 55-60 share an optimality certificate.

Questions to explore

- If the first-half speed rises, what must the second-half speed do?
- Why is finding a stationary trajectory insufficient by itself?

Satisfying stop: A complete best-two-speed argument; calculus-ready learners may finish the global proof.

GA-20 / Facilitator reasoning

Save quadratic effort with a steady journey

The mathematical question

Why can local changes to a whole trajectory prove that one motion minimizes a functional?

Among continuous piecewise- C^1 paths $y:[0,1]\rightarrow\mathbb{R}$ with $y(0)=0$ and $y(1)=1$, the cost $E=\int_0^1 (y')^2 dt$ is at least one, with equality only for $y(t)=t$. Use a finite partition into C^1 pieces and expand $\int (y'-1)^2 = E-1$; values of the derivative at joins do not affect the integral.

Worked instance

Problem: Optimize two half-time speeds and then verify $y=t+a\ t(1-t)$.

Solution: The endpoint condition forces speeds $1+a$ and $1-a$. Their cost is $[(1+a)^2+(1-a)^2]/2=1+a^2$, minimized only at $a=0$. The resulting journeys are continuous piecewise- C^1 competitors. For the smooth family $y=t+a\ t(1-t)$, derivative is $1+a(1-2t)$; integrating its square gives $1+a^2/3$. For every permitted path, apply the fundamental theorem on each piece to get $\int y'=1$. Thus $E=1+\int (y'-1)^2 \geq 1$. Equality makes the continuous squared integrand zero on each piece, so the derivative is one there. Continuity at joins and $y(0)=0$ then give $y=t$ throughout.

Boundary: Minimizing total distance $\int |y'|$ gives many minimizers: every nondecreasing path with these endpoints has distance one. Uniqueness depends on the squared-speed objective.

Hints to reveal in order

1. Write the two speeds as $1+a$ and $1-a$.
2. Subtract the constant speed one and square the difference.

Extensions

- Use three unequal time intervals and derive a weighted variance identity.
- Change endpoints to a,b over duration $T>0$: the minimum is $(b-a)^2/T$, attained by constant speed.

Connection and its limits

Bridge type: exact-special-case

Preserves: A chosen time-position curve is the actual competitor in a variational problem; its velocity determines the defined cost.

Limits: Quadratic effort is a mathematical objective, not a claim about human metabolic energy. This proof uses more than merely stationarity.

Sources and inspection scope

[UCL MATH0043. Calculus of Variations](#)

§2, functional viewpoint and Euler-Lagrange setup

Inspected: Functional setup inspected; global squared-cost proof supplied independently

Prior use / what changes: Week 1 optimized discrete packing; this optimizes a continuous function. It is new even though both use lower-bound-plus-construction reasoning.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-21 / Participation and launch

The shortest route that touches a mirror

MSC primary 51 - Geometry | Secondary 49

The opening challenge

Travel from A to B while touching the marked mirror segment exactly once at M. Choose M to make the total route shortest. Your partner may challenge your choice with any other point on the segment.

Learner choices: Choose and adjust M, use measurement or reflection, and shorten the mirror to test the solution.

Prerequisite gates

Enter: O/V: move a contact point and compare string lengths.

Explore: V/F: reflect a point and use equal triangles.

Explain: P: apply the triangle inequality to every possible contact point.

Prove: A: optional coordinate solution; calculus only for certain constrained extensions.

Reading: Oral launch; coordinate labels for exact computation.

Arithmetic: None for geometric proof; fractions and square roots for exact length.

Reasoning: Track the reflected endpoint while preserving original path lengths.

Hard stop: Multiple obstacles or curved mirrors require new arguments; do not reuse the straight-mirror theorem blindly.

Materials and preparation

10 minutes estimated prep. Paper with $A=(1,2)$, $B=(5,4)$, mirror $y=0$ from $x=0$ to 4; ruler, string, and tracing paper.

A flexible hour

0-10 compare candidate string routes; 10-15 fix legal contact points; 15-35 discover reflection; 35-40 reset; 40-55 produce a universal proof or change the segment; 55-60 explain a certificate.

Questions to explore

- What stays equal when B is reflected?
- Could the best infinite-mirror route miss the permitted segment?

Satisfying stop: A shortest route certified without testing every mirror point.

GA-21 / Facilitator reasoning

The shortest route that touches a mirror

The mathematical question

How can a geometric transformation turn a constrained broken path into an unconstrained straight one?

For A and B in the same open half-plane of a mirror line, reflecting B to B' makes $|AM| + |MB| = |AM| + |MB'|$. The straight crossing gives the minimum whenever it lies in the allowed mirror segment.

Worked instance

Problem: Find the optimal M on the specified segment and its exact route length.

Solution: $B' = (5, -4)$. The segment from A to B' has parametrization $(1+4t, 2-6t)$, meeting $y=0$ at $t=1/3$ and $M = (7/3, 0)$, which lies in $[0, 4]$. Its length is $\sqrt{4^2 + 6^2} = 2\sqrt{13}$. Every legal route unfolds to $A \rightarrow M \rightarrow B'$ and has length at least $|AB'|$ by the triangle inequality. The exhibited crossing attains the bound, so it is globally optimal.

Boundary: If the allowed mirror ends at $x=2$, $M=7/3$ is illegal. The previous lower bound still holds, but its equality construction is unavailable; a new constrained optimization is required.

Hints to reveal in order

1. Draw B' below the mirror at the same perpendicular distance as B.
2. A bent path from A to B' cannot beat their straight-line distance.

Extensions

- Vary A and B and express M as a weighted average of their horizontal coordinates using their heights.
- For the shortened mirror, calculus or convexity proves the minimum occurs at $x=2$; the route then has length $\sqrt{5+5}$.

Connection and its limits

Bridge type: exact-special-case

Preserves: A chosen contact point determines a real Euclidean broken path; unfolding preserves its length.

Limits: The minimum for an infinite mirror need not be legal for a finite mirror. Physical light behavior is motivation, not needed for the proof.

Sources and inspection scope

[Anton Petrunin, Euclidean Plane and Its Relatives](#)

§5D-F, reflection and distance; triangle-inequality framework

Inspected: Reflection facts inspected; constrained instance derived directly

Prior use / what changes: Week 9 used billiard unfolding and periodicity; this adds a continuous optimization and a finite-mirror obstruction. Avoid reusing the earlier trajectory instance.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-22 / Participation and launch

Split four pegs so their hulls meet

MSC primary 52 - Convex and discrete geometry | Secondary 15

The opening challenge

Choose four distinct positions. Color each red or blue, using both colors, so the red convex hull and blue convex hull share a point. Challenge another group with an arrangement that seems hard.

Learner choices: Choose positions, partition, intersection witness, and a case missed by a proposed universal explanation.

Prerequisite gates

Enter: O: sort four pegs into two colors and draw segments or triangular hulls.

Explore: V: recognize inside a triangle versus four convex corners.

Explain: P: explain the complete geometric case split, including collinearity.

Prove: L/P: linear dependence gives the general R^d theorem.

Reading: Oral; a partner may trace boundaries.

Arithmetic: None for the first cases; fractions for a coordinate certificate.

Reasoning: A hull of one point is that point; hulls include their boundaries.

Hard stop: Do not assume every four-point arrangement is a convex quadrilateral.

Materials and preparation

8 minutes estimated prep. Four movable pegs or dots, two colors, thread, and a transparent grid.

A flexible hour

0-10 make and discuss hulls; 10-15 establish the intersection task; 15-35 challenge arrangements; 35-40 reset; 40-55 organize a proof or try three points; 55-60 explain one difficult case.

Questions to explore

- What if one peg is inside the triangle of the others?
- Can three corners of a triangle always be split successfully?

Satisfying stop: A successful partition for an adversarial arrangement and a proof that handles its geometric type.

GA-22 / Facilitator reasoning

Split four pegs so their hulls meet

The mathematical question

What tiny configuration forces an intersection between two independently chosen convex hulls?

Any four points in $\mathbb{R}^2$ can be partitioned into two nonempty groups whose convex hulls intersect. The core uses distinct points and analyzes convex quadrilaterals, a point inside a triangle, and collinear degeneracy.

Worked instance

Problem: Use $(0,0), (4,0), (0,4), (1,1)$. Find a partition and certify the intersection.

Solution: Put $(1,1)$ alone and the other three together. Since $(1,1) = 1/2(0,0) + 1/4(4,0) + 1/4(0,4)$, with nonnegative coefficients totaling one, it lies in the triangular hull. More generally, if one point lies in or on the triangle of the other three, put it alone. This includes three collinear points plus one off their line: choose the middle collinear point alone. Otherwise four distinct noncollinear points form a convex quadrilateral, whose diagonal segments intersect, giving two pairs. If all four are collinear, choose an inner point alone; it lies between the extreme points in the other group. These cases cover every arrangement of four distinct points.

Boundary: For three noncollinear points, every nonempty partition is one vertex versus the opposite edge, and those hulls are disjoint. The number four is therefore necessary in the plane.

Hints to reveal in order

1. For four convex corners, try opposite corners together.
2. A one-peg hull is allowed.

Extensions

- Choose five points in three dimensions and seek a partition using a tetrahedron and an interior point.
- Translate intersection witnesses into two convex combinations and relate them to affine dependence.

Connection and its limits

Bridge type: exact-special-case

Preserves: Peg subsets and stretched outlines are literal convex hulls; choosing a partition asks the exact planar Radon question.

Limits: Rubber bands show an idealized hull approximately. Four pegs do not prove the higher-dimensional theorem by experiment.

Sources and inspection scope

[Daniel Hug and Wolfgang Weil, A Course on Convex Geometry](#)

§1.2, Theorem 1.2.1, Radon's theorem

Inspected: Statement and proof framework inspected; planar cases supplied

Prior use / what changes: Week 1 tilings are different convex/discrete geometry. No identical existing task was found.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-23 / Participation and launch

Return home with your arrow pointing somewhere new

MSC primary 53 - Differential geometry | Secondary 58

The opening challenge

Start at A with your arrow pointing toward B. Follow the shortest arc $A \rightarrow B \rightarrow C \rightarrow A$. Along each leg keep the arrow's angle to the forward tangent unchanged. When you return, compare with the original arrow without resetting it at the corners.

Learner choices: Choose an initial tangent direction, reverse the route, or compare a retraced route; predict before carrying the arrow.

Prerequisite gates

Enter: O/V: follow three great-circle arcs and keep a chosen angle to forward travel.

Explore: V: identify tangent directions along equator and meridians.

Explain: P/V: track exact coordinate directions leg by leg.

Prove: D/X: parallel transport equations and general Gauss-Bonnet.

Reading: Oral route with labeled landmarks; coordinates optional.

Arithmetic: Right angles; no length calculations in the octant proof.

Reasoning: Separate the sphere's tangent plane from surrounding three-dimensional space.

Hard stop: Do not infer transport from compass direction alone; the legal rule must be demonstrated.

Materials and preparation

15 minutes estimated prep. A ball marked with equator, north pole, and two meridians 90° apart; a tangent arrow and route diagram. Mark A,B,C and use different colors for the forward tangent and transported arrow. Keep the arrow fixed at each corner before starting the next arc.

A flexible hour

0-10 explore tangent directions on the ball; 10-15 demonstrate transport on one arc; 15-35 complete and record routes; 35-40 reset; 40-55 verify exact vectors or relate angle excess; 55-60 share the route-dependent result.

Questions to explore

- Can an arrow remain tangent while staying fixed in ordinary space?
- Why is retracing an arc different from completing a triangle?

Satisfying stop: One precisely described route and a verified changed arrow.

GA-23 / Facilitator reasoning

Return home with your arrow pointing somewhere new

The mathematical question

How can locally straight transport around a curved surface retain a record of the route?

On the unit sphere, parallel transport along a great-circle arc keeps the angle to its forward tangent constant. Around the positively oriented octant triangle $A=(1,0,0)$, $B=(0,1,0)$, $C=(0,0,1)$, the transported tangent arrow rotates by $\pi/2$.

Worked instance

Problem: Transport initial vector $+y$ from A around the specified octant.

Solution: Along $A \rightarrow B$, the forward tangent rotates from $+y$ to $-x$, so the arrow reaches B as $-x$. Along $B \rightarrow C$ the vector $-x$ is constant, tangent to the sphere, and perpendicular to the great-circle plane, hence parallel. At C it is opposite the forward tangent $+x$ for $C \rightarrow A$. Keeping that opposite relation makes the ending arrow $+z$ at A . Thus $+y$ has become $+z$, a $+90^\circ$ rotation in A 's tangent plane. The octant area is $4\pi/8 = \pi/2$, matching spherical holonomy.

Boundary: Walking out and exactly retracing the same arc restores the arrow. Curvature does not make every closed itinerary produce nonzero rotation.

Hints to reveal in order

1. On the equator, track the arrow that points along travel.
2. On $B \rightarrow C$ the x -direction stays tangent and perpendicular to travel.

Extensions

- Reverse the triangle and predict -90° .
- Use a different longitude separation α with two meridians and an equator arc; the spherical lune argument predicts transport angle α .

Connection and its limits

Bridge type: exact-special-case

Preserves: A tangent arrow follows an explicitly defined connection along actual spherical great-circle arcs.

Limits: A hand-held arrow is approximate. General holonomy/curvature theorems need differential geometry; an arbitrary 'keep pointing north' rule is different transport.

Sources and inspection scope

[Anton Petrunin and Sergio Zamora Barrera, What Is Differential Geometry?](#)

Chapters 16-17; Proposition 17.10 and octant transport example in semisolutions

Inspected: Relevant transport and spherical-area discussion inspected; vector tracking supplied

Prior use / what changes: New curved-geometry strand; plane billiards from Week 9 cannot exhibit this transport effect.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-24 / Participation and launch

Can a loop be reshaped into a segment?

MSC primary 54 - General topology | Secondary 55

The opening challenge

Can you continuously reshape a loop into a segment without cutting, gluing, or making different points become the same point? Invent a test that every successful reshaping would have to pass.

Learner choices: Choose where to puncture each shape, propose candidate reshaping rules, and challenge a test using endpoints.

Prerequisite gates

Enter: O: trace a loop and segment and compare pieces after a marked cut.

Explore: P: reason about all possible images of an interior point.

Explain: P: formulate the puncture contradiction using preservation of connectedness.

Prove: X: definitions of homeomorphism and connectedness for a formal proof.

Reading: Oral launch with pictures.

Arithmetic: None; counting one or two components.

Reasoning: Separate allowable continuous reversible deformation from cutting, gluing, or identifying points.

Hard stop: A drawing with crossings needs explicit decisions about which strands are joined.

Materials and preparation

5 minutes estimated prep. One closed string loop and one open string segment, paper diagrams, and removable dot markers.

A flexible hour

0-10 freely bend strings; 10-15 specify reversible no-identification moves; 15-35 invent invariants; 35-40 reset; 40-55 explain the puncture test or compare another pair; 55-60 state an impossibility with its legal-move assumptions.

Questions to explore

- Does an endpoint behave like a middle point?
- If a map is reversible, can two different string points merge?

Satisfying stop: A complete impossibility argument that does not rely on visual resemblance.

GA-24 / Facilitator reasoning

Can a loop be reshaped into a segment?

The mathematical question

What property survives every homeomorphism even when lengths and angles are discarded?

Deleting a point from a circle leaves a connected space, while deleting an interior point from a closed interval disconnects it. A homeomorphism restricts to a homeomorphism of the corresponding punctured spaces.

Worked instance

Problem: Prove that $[0,1]$ is not homeomorphic to the unit circle.

Solution: Suppose a homeomorphism existed. Take the point $1/2$ in the interval and its image p on the circle. Removing them preserves the homeomorphism. But $[0,1]$ without $1/2$ separates into two nonempty relatively open pieces $[0,1/2)$ and $(1/2,1]$. A circle without p is connected: rotating p to $(1,0)$, the remaining circle is the continuous image of the interval $0 < t < 2\pi$ under $t \mapsto (\cos t, \sin t)$. This contradicts preservation of connectedness.

Boundary: Deleting an endpoint of $[0,1]$ leaves it connected, so one careless cut does not distinguish the spaces. The argument deliberately chooses an interior point and uses the circle's every-point property.

Hints to reveal in order

1. Mark a point in the interior of the segment.
2. Its corresponding point on a loop would have to behave the same after deletion.

Extensions

- Compare a Y-shaped graph with a segment by deleting the branch point; three components appear.
- Investigate a figure-eight, specifying that its crossing is one joined point, and compare it with two disjoint circles.

Connection and its limits

Bridge type: exact-special-case

Preserves: Idealized strings represent a circle and interval as topological spaces, and deleting a point tests an invariant.

Limits: Physical cutting removes a short length; the proof concerns deleting one mathematical point. No general classification of topological spaces is claimed.

Sources and inspection scope

[Romvar Sharifi. Point-Set Topology](#)

§3.1, connectedness and Proposition 3.1.8

Inspected: Continuous-image principle inspected; puncture argument supplied

Prior use / what changes: Week 1 reconfiguration involved legal moves on discrete states; this introduces reversible continuous maps of spaces.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-25 / Participation and launch

How many corridors must close to erase every loop?

MSC primary 55 - Algebraic topology | Secondary 05

The opening challenge

Close as few corridors as possible so every room remains reachable but no cycle survives. Choose your closures. Can a partner prove that using fewer closures could never work?

Learner choices: Choose the spanning tree, a deletion algorithm, and a graph that seems to have more cycles than the formula predicts.

Prerequisite gates

Enter: O/C: trace routes and count rooms and corridors.

Explore: P: delete a cycle edge without disconnecting the graph.

Explain: P: combine a termination algorithm with the tree-edge count.

Prove: X: fundamental groups or homology for the algebraic interpretation.

Reading: Oral; vertex labels help record chosen deletions.

Arithmetic: Counting and $E - V + 1$.

Reasoning: Distinguish a cycle from a backtracking walk and preserve all vertices.

Hard stop: Cycle rank does not count every simple cycle; avoid calling both quantities 'number of loops'.

Materials and preparation

8 minutes estimated prep. Four room dots A,B,C,D in a square, corridors AB,BC,CD,DA,AC, and removable edge strips.

A flexible hour

0-10 explore routes and cycles; 10-15 define connected and cycle-free; 15-35 optimize closures; 35-40 reset; 40-55 prove the edge count or change the graph; 55-60 share two different optimal closure sets.

Questions to explore

- Can removing an edge on a cycle disconnect the graph?
- Why are the two small triangles and the outside square not three independent cycles?

Satisfying stop: A minimal closure plan and an explanation of why all optimal plans remove the same number.

GA-25 / Facilitator reasoning

How many corridors must close to erase every loop?

The mathematical question

Why does a connected graph have an invariant number of independent cycles even when it has many visible loop routes?

For a finite connected graph with V vertices and E edges, every spanning tree has $V-1$ edges. Therefore exactly $E-V+1$ edges must be removed to leave a connected graph with no cycles.

Worked instance

Problem: Solve the four-room square with diagonal AC.

Solution: Here $V=4$ and $E=5$, so two deletions are required. Delete DA and AC, leaving the path A–B–C–D. One deletion would leave four edges on four vertices; a connected acyclic graph on four vertices has only three edges, so one cannot suffice. To prove the general procedure, an edge on a cycle has an alternate route around that cycle, so deleting it preserves connectivity. Repetition ends at a spanning tree. Removing a leaf inductively proves every finite tree has $V-1$ edges.

Boundary: The original graph has three simple cycles: ABC, ACD, and the outer square. Its independent cycle count is only two; listing cycles and finding a cycle basis are different questions.

Hints to reveal in order

1. Keep deleting one cycle edge until no cycle remains.
2. A tree can be built by attaching one new vertex with one new edge.

Extensions

- Allow several connected components: the cycle rank becomes $E-V+c$, where c is the component count.
- Fill one triangle with a two-dimensional face; ask why its boundary loop is now contractible, introducing a new dimension.

Connection and its limits

Bridge type: exact-special-case

Preserves: Rooms and corridors form the actual one-dimensional cell complex; deleting selected edges asks for its cycle rank.

Limits: This computes a graph invariant, not higher-dimensional homology. Crossing corridors are joined only when a vertex is explicitly marked.

Sources and inspection scope

[Allen Hatcher, Algebraic Topology](#)

Chapter 0, graphs and homotopy type; §1.1 fundamental group of graphs

Inspected: Graph discussion inspected; elementary spanning-tree proof supplied

Prior use / what changes: Week 10 studied Euler/postman routes. This asks about spanning trees and independent cycles, not traversing all corridors.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-26 / Participation and launch

Can a coloring prove a knot cannot untie?

MSC primary 57 - Manifolds and cell complexes | Secondary 20, 55

The opening challenge

Color the trefoil's arcs so each crossing has either one color or three, and use more than one color overall. Can the same kind of coloring survive every legal diagram move? Could the knot then become a plain circle?

Learner choices: Choose colorings, local moves to test, and a proposed invariant-breaking move.

Prerequisite gates

Enter: O/C: follow an arc through crossings and use three colors.

Explore: P: check local color completion under moves; modular arithmetic is optional.

Explain: P: distinguish a nonconstant coloring from the single-arc unknot.

Prove: M/P: the formula supplies a compact proof of all three move identities.

Reading: Oral rule; short crossing labels.

Arithmetic: None for coloring; remainders modulo three for the algebraic extension.

Reasoning: An arc stops at undercrossings, not at every crossing; tracing deserves time.

Hard stop: General knot equivalence requires Reidemeister's theorem; do not present local checks alone as its proof.

Materials and preparation

15 minutes estimated prep. A clearly drawn standard trefoil with three arcs labeled A,B,C, an unknot, three pencils, and local move sketches. Mark an explicit overpass at every crossing and stop each arc only at undercrossings. Future student sheets need independently checked trefoil and local-move drawings.

A flexible hour

0-10 trace knot arcs and manipulate a closed cord; 10-15 learn the crossing rule; 15-35 color and test moves; 35-40 reset; 40-55 explain the obstruction or check algebraic identities; 55-60 separate proof from failed untying attempts.

Questions to explore

- Why must a nonconstant coloring be required?
- What changes if a strand is allowed to pass through another?

Satisfying stop: A nonconstant trefoil coloring and one checked local reason it survives legal moves.

GA-26 / Facilitator reasoning

Can a coloring prove a knot cannot untie?

The mathematical question

How can a local diagram rule provide a global obstruction to deformation?

Fox 3-coloring assigns residues 0,1,2 to diagram arcs so that $2 \cdot \text{over} = \text{under}_1 + \text{under}_2 \pmod{3}$ at each crossing. Equivalently, all three colors agree or all differ. Reidemeister moves preserve colorings bijectively.

Worked instance

Problem: Color the standard three-crossing trefoil and explain why the unknot is excluded.

Solution: Assign $A=0, B=1, C=2$. At every crossing the three colors are distinct, satisfying $2b=a+c$ modulo three. The unknot has only one arc, so every coloring is constant. For move invariance define $a*b=2b-a$: $a*a=a$, $(a*b)*b=a$, and $(a*b)*c=(a*c)*(b*c)$, since both sides are $a-2b+2c$. These are the type I, II, and III local color-completion identities. The induced bijections preserve constant colorings and hence nonconstant ones. Together with Reidemeister's theorem this obstructs unknotting.

Boundary: Every diagram admits constant colorings, so allowing those alone proves nothing. Changing a crossing is not a Reidemeister move and can alter knot type.

Hints to reveal in order

1. Give the trefoil arcs three different colors.
2. For a crossing with overcolor b and incoming undercolor a , the outgoing undercolor must be $2b-a$ modulo three.

Extensions

- Count all trefoil colorings: choose two arc colors freely and the third is forced, giving nine total and six nonconstant.
- Find a nontrivial knot with no nonconstant 3-coloring; treat an example as a new sourced investigation rather than assuming the test is complete.

Connection and its limits

Bridge type: exact-special-case

Preserves: Arcs and crossings are those of genuine knot diagrams; allowed Reidemeister moves preserve the embedded-knot type.

Limits: The invariant can prove two knots different but matching colorability does not prove sameness. Diagrams need unambiguous over/under data.

Sources and inspection scope

[Louis Kauffman, Knots](#)

Three Colored Trefoil, Figures 12-14 and theorem

Inspected: Coloring rule and move-invariance argument inspected

Prior use / what changes: New knot-theory content; Week 2 parity invariants offer a reasoning comparison but not the same object or obstruction.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-27 / Participation and launch

What happens when a lake rises through a saddle?

MSC primary 58 - Global analysis, analysis on manifolds | Secondary 53, 54

The opening challenge

Raise the water level c . A point floods exactly when $x^2 - y^2 \leq c$. Choose test points and sketch the flooded regions at $-1/4, 0, 1/4$. At which height can the upper and lower lakes first meet?

Learner choices: Choose test points, routes through flooded ground, and candidate levels where the topology changes.

Prerequisite gates

Enter: V/A: plot inequalities or test coordinates in a supplied contour map.

Explore: A: rewrite the inequality and use symmetry.

Explain: P: prove disconnectedness by the sign of y and connectedness by explicit paths.

Prove: D/L/X: gradient, Hessian signature, and Morse theory for the general theorem.

Reading: Coordinate labels; oral flooded/not-flooded decisions before symbolic proof.

Arithmetic: Squares and signed quarters.

Reasoning: Track connected regions rather than just the number of contour curves.

Hard stop: A sampled height map cannot certify that no hidden critical point exists.

Materials and preparation

12 minutes estimated prep. Graph paper showing $[-1, 1]^2$, transparent shaded sublevels, and optional clay saddle for comparison.

A flexible hour

0-10 explore a saddle or its contours; 10-15 define sublevel flooding; 15-35 map the three levels; 35-40 reset; 40-55 prove the connection change or classify derivatives; 55-60 report one topological event.

Questions to explore

- Can a flooded path cross $y=0$ when c is negative?
- What makes radial travel toward the origin safe when $c \geq 0$?

Satisfying stop: An exact reason the two regions cannot join below zero and do join at zero.

GA-27 / Facilitator reasoning

What happens when a lake rises through a saddle?

The mathematical question

How can a local critical point change the topology of an entire sublevel set?

The quadratic $f(x,y)=x^2-y^2$ has a nondegenerate critical point of index one at the origin. Within $[-1,1]^2$, its sublevel sets at $c=-1/4, 0, 1/4$ exhibit two components, a point connection, and a connected region.

Worked instance

Problem: Explain the component change for the three stated levels.

Solution: For $c=-1/4$, every flooded point satisfies $y^2 \geq x^2 + 1/4$, so y cannot be zero. The upper and lower regions are separated. Each is connected: move vertically toward $y=1$ or $y=-1$, which only decreases f , then travel along that boundary within $|x| \leq \sqrt{3}/2$. At $c=0$ the two wedges $|y| \geq |x|$ meet at the origin. For $c=1/4$, every flooded point can connect radially to the origin because multiplying f by t^2 keeps it $\leq 1/4$, so the set is connected. Gradient $(2x, -2y)$ vanishes only at zero; Hessian $\text{diag}(2, -2)$ has one negative direction.

Boundary: A function such as $x^4 - y^4$ has a similar-looking drawing but a degenerate Hessian at zero. The Morse lemma cannot be invoked just from visual resemblance.

Hints to reveal in order

1. At $y=0$, $f=x^2$ cannot be negative.
2. Use $f(tx, ty) = t^2 f(x, y)$ for $0 \leq t \leq 1$.

Extensions

- Compare a bowl $x^2 + y^2$, where a component is born as c crosses zero.
- Build a piecewise-linear landscape and track component mergers, labeling it a different model before connecting it to persistence.

Connection and its limits

Bridge type: exact-special-case

Preserves: Flooding means the precise inequality $f \leq c$, and the quadratic is the actual local Morse saddle model.

Limits: A clay landscape is a representation. The square has corners, so we do not apply the compact smooth-manifold slab theorem to its boundary.

Sources and inspection scope

[Patrick Gillespie, Morse Theory Notes](#)

Lemma 1.12; Theorem 2.9; §3 passing critical levels

Inspected: Relevant local normal form and topology-change statements inspected; square-domain instance proved directly

Prior use / what changes: New global-analysis entry; no prior week studies critical points or sublevel topology.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-28 / Participation and launch

A bracket or a tangent: which promise survives?

MSC primary 26 - Real functions | Secondary 65, 37

The opening challenge

Find a number whose square is two, using questions that answer only “too large” or “too small.” Decide what information to keep after every question.

Learner choices: Choose test points, compare equal-halving with other choices, and set a target interval width before computing.

Prerequisite gates

Enter: F: halve rational intervals and compare squares.

Explore: A: evaluate polynomial signs; D is required only for Newton slopes.

Explain: P: explain why the selected half still contains some root.

Prove: P/A: inductive interval bound using the intermediate value theorem; D for the Newton counterexample.

Reading: Short numeric records; a facilitator may read procedural instructions.

Arithmetic: Fractions and squares; cubic values and signed division in the extension.

Reasoning: Keep an invariant rather than merely collecting plausible decimal digits.

Hard stop: Do not introduce tangent iteration without derivatives, or claim continuity from a finite plot.

Materials and preparation

8 minutes estimated prep. Number lines, blank bracket records, fraction cards, optional calculator and graphing tool.

A flexible hour

0-10 guess and square; 10-15 agree on a certificate; 15-35 maintain brackets; 35-40 reset; 40-55 investigate the Newton cycle if derivative-ready, otherwise prove the halving bound; 55-60 compare promises.

Questions to explore

- Can a good-looking guess lack a proved error bound?
- Does an endpoint sign change prove a root if the function jumps?

Satisfying stop: A root-containing interval of chosen width and an explanation of why it is trustworthy.

GA-28 / Facilitator reasoning

A bracket or a tangent: which promise survives?

The mathematical question

How can a root-finding method certify an answer, and why can a fast local method fail?

For continuous f on $[a,b]$ with opposite endpoint signs, bisection preserves a root-containing interval and halves its width each step. Newton iteration has no corresponding unconditional convergence guarantee; for $f(x)=x^3-2x+2$ it cycles $0 \rightarrow 1 \rightarrow 0$.

Worked instance

Problem: Start with $[1,2]$ for x^2-2 and perform four bisections. Compare with Newton on x^3-2x+2 from zero.

Solution: The tested midpoints are $3/2, 5/4, 11/8, 23/16$; their square-minus-two signs are $+, -, -, +$. The remaining interval is $[11/8, 23/16]$, width $1/16$. Its midpoint $45/32$ is within $1/32$ of $\sqrt{2}$. Generally the width after n bisections is 2^{-n} . For the cubic, Newton gives $x-(x^3-2x+2)/(3x^2-2)$: at zero it gives one, and at one it gives zero. Both derivative denominators are nonzero, yet iteration never reaches a root.

Boundary: The discontinuous function equal to -1 for $x < 0$ and $+1$ for $x \geq 0$ changes sign across $[-1, 1]$ without any zero. Continuity is essential to the bracket certificate.

Hints to reveal in order

1. Keep both a lower and an upper witness.
2. For the Newton example, calculate the next value from zero and then from one.

Extensions

- Bracket a root of the cubic on $[-2, -1]$ and compare its reliable progress with the cycle.
- Design a hybrid that accepts a Newton step only when it preserves a maintained bracket; specify fallback rules.

Connection and its limits

Bridge type: exact-special-case

Preserves: Learners execute genuine algorithms on real continuous functions, with a maintained existence certificate.

Limits: A graph that appears to cross an axis supplies intuition; the sign condition plus continuity supplies the mathematical existence argument.

Sources and inspection scope

[MIT 2.086, Numerical Computation for Mechanical Engineers, Unit VI](#)

§29, bisection discussion and §29.2.5 Newton pathologies, printed pp.432-435

Inspected: Both methods and their differing guarantees inspected; chosen arithmetic verified independently

Prior use / what changes: Comparison and extension of AP-06: the Newton cubic x^3-2x+2 and its $0 \leftrightarrow 1$ cycle are deliberate cross-volume reuse. Here the main additional work is a bisection invariant and explicit error certificate. GA-02 supplies the underlying continuity principle.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-29 / Participation and launch

Remove almost all the length; keep uncountably many points

MSC primary 28 - Measure and integration | Secondary 03, 40

The opening challenge

Repeatedly remove the open middle third of every surviving interval. Before cutting, predict which endpoints survive and whether all the length can disappear while points remain.

Learner choices: Choose addresses for surviving points, invent a proposed list of addresses, and challenge another group to find an omitted address.

Prerequisite gates

Enter: O/C: cut or shade the open middle third repeatedly, keeping endpoints.

Explore: F: compute interval counts, lengths, and a shrinking total budget.

Explain: P: distinguish number of pieces, length, and number of points.

Prove: P/F: use geometric limits and diagonalization; no derivatives are needed.

Reading: Facilitator can narrate the cut rule; address strings must be tracked accurately.

Arithmetic: Powers of two and three, fractions, and a geometric sum.

Reasoning: Coordinate a nested construction with two different quantifiers: every length budget and every proposed list.

Hard stop: Uncountability needs an infinite-sequence argument. “Too many dots to count” is only an observation.

Materials and preparation

10 minutes estimated prep. Long paper strips, three-color pencils, interval templates, and 0/2 address cards.

A flexible hour

0-10 explore cutting; 10-15 state endpoint and nesting rules; 15-35 compute covers and encode locations; 35-40 reset; 40-55 compare length with listing and diagonalize; 55-60 share separate certificates.

Questions to explore

- Why does zero total length not mean the empty set?
- Where does the argument use infinitely many stages?

Satisfying stop: A small-length cover and an independently explained missed-address argument.

GA-29 / Facilitator reasoning

Remove almost all the length; keep uncountably many points

The mathematical question

How can a set have zero length and still contain more points than any list?

The middle-thirds Cantor set C is the intersection of its closed construction stages. Stage n has 2^n intervals of length 3^{-n} , so $\text{outer measure}(C)=0$. Infinite addresses using ternary digits 0 and 2 give an injection from all binary sequences into C .

Worked instance

Problem: Compute the stage-four cover, locate $1/4$ in C , and explain why no list exhausts C .

Solution: Stage four has 16 closed intervals of length $1/81$, total $16/81$. At stage n the cover total is $(2/3)^n$, below every positive budget for sufficiently large n ; hence outer measure is zero. If using open covers, enlarge the finitely many stage- n closed intervals slightly, adding arbitrarily little total length. The repeating ternary address $0.020202\dots$ has value $(2/9)/(1-1/9)=1/4$ and avoids every deleted middle third. Every infinite $0/2$ address determines a point by nested intervals. Two such addresses differing first at digit k give values separated by at least $1/3^k$, so they cannot coincide. If all such addresses were listed, changing the diagonal digits produces another address missing from the list.

Boundary: After any finite number of stages the retained length is positive. Endpoints such as $1/3$ survive because the removed middle thirds are open; using closed deletions changes the construction.

Hints to reveal in order

1. At stage n multiply the number of intervals by their individual length.
2. Against a list of infinite addresses, change the n th digit of the n th address.

Extensions

- Find surviving rational points from repeating $0/2$ addresses.
- Compare GA-03: countable sets have zero outer measure, but the converse fails by this example.

Connection and its limits

Bridge type: exact-special-case

Preserves: Finite stages supply actual covers of the limiting set; nested interval addresses specify actual real points.

Limits: A finite paper model always retains positive total length and cannot display all points or prove an infinite statement by itself.

Sources and inspection scope

[Sheldon Axler, Measure, Integration & Real Analysis](#)

§2D, Cantor set construction and results 2.76-2.79

Inspected: Construction, zero outer measure, and uncountability inspected; stage counts and chosen address checked

Prior use / what changes: New measure/cardinality synthesis. Paper cutting resembles earlier hands-on work, but the infinite limiting object and two distinct proofs are new.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-30 / Participation and launch

Two shortest routes around a cylinder

MSC primary 53 - Differential geometry | Secondary 55, 51

The opening challenge

Join the short edges into a cylinder. Connect a bottom-rim point to the top point halfway around. Find two visibly different routes and decide whether one can be shortened.

Learner choices: Choose a cut line, a winding direction, and extra windings; use another cut if a promising path crosses the first seam.

Prerequisite gates

Enter: O: trace alternative routes around a rolled paper surface.

Explore: V: unroll and compare straight segments to translated copies.

Explain: P/V: explain why every winding choice has a straight lower bound.

Prove: A/P: minimize over integer lifts with the distance formula.

Reading: Labels and short diagrams suffice; no calculus reading.

Arithmetic: Squares, square roots, integer multiples, and comparison of absolute values.

Reasoning: Separate local straightness from being globally shortest.

Hard stop: The informal paper argument proves this developable case; arbitrary curved-surface geodesics need further differential geometry.

Materials and preparation

10 minutes estimated prep. A 6-by-4 rectangle, tape, ruler, string, and a tiled strip of three or more copied rectangles.

A flexible hour

0-10 explore string routes; 10-15 fix the surface-only rule; 15-35 unroll and measure; 35-40 reset; 40-55 compare lifted routes and prove the minimum; 55-60 show two equal winners.

Questions to explore

- Why might one unrolled rectangle hide a shortest route?
- Can a locally straight route lose to another route with different winding?

Satisfying stop: Two constructions of length five and a lower-bound argument excluding every shorter surface route.

GA-30 / Facilitator reasoning

Two shortest routes around a cylinder

The mathematical question

How does a surface's global topology affect shortest paths even when its local geometry is flat?

On the lateral surface of a circular cylinder of circumference six and height four, A and B lie on opposite generators at opposite rims. Unrolling lifts B to $(3+6k, 4)$, $k \in \mathbb{Z}$, from $A=(0,0)$. Straight segments give candidates of length $\sqrt{((3+6k)^2+16)}$; the minimum is five, attained twice.

Worked instance

Problem: Use circumference six, height four, and a half-circumference displacement. Find all shortest routes.

Solution: In the unrolled strip, B has copies $(3+6k, 4)$. A path with that lift has length at least the straight segment distance $\sqrt{((3+6k)^2+16)}$. Each straight segment lies between heights zero and four and rolls to an allowed path, so the bound is attained for each winding choice. The smallest possible $|3+6k|$ is three, attained only at $k=0$ and $k=-1$. Both routes therefore have length $\sqrt{25}=5$. The next candidates have horizontal displacement nine and length $\sqrt{97}>5$.

Boundary: A straight line to a distant lift is locally a geodesic but is not globally shortest. A chord through the cylinder's interior does not compete under the stated rules.

Hints to reveal in order

1. Repeat the rectangle sideways before drawing.
2. A path ending at B can lift to any copy of B, but its endpoint height is always four.

Extensions

- Move B to horizontal offset two; show that the unique shortest lift has displacement two.
- Vary the offset continuously and locate exactly where two shortest routes exchange priority.

Connection and its limits

Bridge type: exact-special-case

Preserves: Paper unrolling preserves intrinsic lengths on the actual cylinder surface, giving a continuous geometric proof.

Limits: Routes must stay on the lateral surface. Travel through the solid cylinder or across end caps is a different optimization problem.

Sources and inspection scope

[Anton Petrunin and Sergio Zamora Barrera, What Is Differential Geometry?](#)

Chapters 14-15 on length and shortest paths; §18G cylinder as a length-preserving image of a plane domain

Inspected: Unrolling/length-preserving cylinder model inspected; lift minimization checked independently

Prior use / what changes: New continuous geometry. GA-23 studies curvature on a sphere; this cylinder is locally flat while its global wrapping still matters.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-31 / Participation and launch

Everybody agrees in pairs: is there a common choice?

MSC primary 52 - Convex and discrete geometry | Secondary 54, 90

The opening challenge

Each person marks an allowed interval for a meeting location. Every pair can agree somewhere. Can the entire group agree? Try hard to produce a failure before making a rule.

Learner choices: Choose interval endpoints, choose which pairs to test, then change the space from a line to a plane or remove the no-gaps rule.

Prerequisite gates

Enter: O/C: overlap strips representing allowed locations on a common line.

Explore: V: draw intervals or clipped half-planes on a coordinate grid.

Explain: P: identify the latest left endpoint and earliest right endpoint.

Prove: P: complete the finite interval proof and construct a planar counterexample.

Reading: Read short endpoint labels; symbols for inequalities are needed only in the planar extension.

Arithmetic: Ordering integers; signed coordinates for the counterexample.

Reasoning: Distinguish one common witness from a different witness for every pair.

Hard stop: A general proof of planar Helly needs additional convexity reasoning, such as Radon; do not present the interval proof as that proof.

Materials and preparation

10 minutes estimated prep. Transparent strips on a number line, endpoint clips, grid paper, and optional three translucent half-plane sheets.

A flexible hour

0-10 free overlap play; 10-15 state the pairwise condition; 15-35 search and prove on a line; 35-40 reset; 40-55 build planar or nonconvex failures; 55-60 identify the hypothesis doing the work.

Questions to explore

- Why is checking that every interval overlaps one favorite interval insufficient?
- What changes when allowed regions have holes or live in a plane?

Satisfying stop: A complete interval proof and a counterexample showing precisely why its scope matters.

GA-31 / Facilitator reasoning

Everybody agrees in pairs: is there a common choice?

The mathematical question

When can small groups of convex constraints certify simultaneous feasibility?

For a finite family of nonempty closed bounded intervals, pairwise intersection implies a common point. In $\mathbb{R}^2$ pairwise intersection of convex sets is insufficient; finite Helly requires every triple to intersect, for families with at least three members.

Worked instance

Problem: Find the common part of $[0,4], [2,6], [3,5], [1,7]$. Then give three planar convex sets intersecting pairwise but not all together.

Solution: The largest left endpoint is three and the smallest right endpoint is four, so the common interval is $[3,4]$. In general, let L be the largest left endpoint and U the smallest right endpoint. The two intervals attaining them intersect by hypothesis, forcing $L \leq U$; every interval contains $[L, U]$. For the plane use $A = \{x \geq 0\}$, $B = \{y \geq 0\}$, $C = \{x + y \leq -1\}$. Pair witnesses are $(0,0)$ for A, B ; $(0, -1)$ for A, C ; and $(-1, 0)$ for B, C . A common point would have $x + y \geq 0$ and ≤ -1 , impossible. Intersect each set with $[-2, 2]^2$ if bounded physical windows are preferred.

Boundary: The nonconvex sets $\{0, 1\}, \{1, 2\}, \{0, 2\}$ intersect pairwise but have empty triple intersection even on a line. Convexity, not merely dimension, matters.

Hints to reveal in order

1. Which interval starts furthest right? Which ends furthest left?
2. In the plane try three constraints that each prohibit a different direction.

Extensions

- Use GA-22 Radon partitions to investigate a proof of planar Helly.
- Design a three-window puzzle where pair witnesses misleadingly suggest global agreement.

Connection and its limits

Bridge type: exact-special-case

Preserves: Interval choices are actual one-dimensional convex sets, and their shared feasibility is exactly the dimension-one Helly theorem.

Limits: The planar triple theorem is a further theorem. Pairwise classroom trials do not prove it, and arbitrary nonconvex sets need not obey the interval result.

Sources and inspection scope

[Daniel Hug and Wolfgang Weil. A Course on Convex Geometry](#)

§1.2, Theorems 1.2.1-1.2.3 on Radon and Helly

Inspected: Finite-dimensional convex intersection statements inspected; interval proof and planar counterexample checked independently

Prior use / what changes: New convex-feasibility family. GA-22 supplies a possible later proof tool; no prior worksheet is assumed.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-32 / Participation and launch

Cut a Möbius band and follow the seam

MSC primary 57 - Manifolds and cell complexes | Secondary 54

The opening challenge

Build one band with matching ends and one with a half-turn. Trace an edge all the way back to your start. Predict what will happen if each band is cut exactly along its middle.

Learner choices: Choose tracking colors, draw a prediction diagram, and decide which observations would distinguish one component from two.

Prerequisite gates

Enter: O: trace an edge and make a prediction before cutting.

Explore: C: label upper/lower halves and follow paired seam labels.

Explain: P: explain why crosswise gluing connects the two halves.

Prove: P: use the quotient diagram to prove boundary counts and the cut surface type.

Reading: Colored arrows can replace written labels; oral reasoning is sufficient at entry.

Arithmetic: Counting one or two circuits; no algebra is required for the physical proof.

Reasoning: Distinguish a connected strip from the number of boundary loops and from orientability.

Hard stop: General surface classification is beyond a single band. A dramatic cutting demonstration alone is not a classification proof.

Materials and preparation

12 minutes estimated prep. Long paper strips, tape, two marker colors, scissors or precut strips, and large matching seam diagrams.

A flexible hour

0-10 handle and trace both bands; 10-15 record predictions; 15-35 cut and follow seams; 35-40 reset; 40-55 rebuild the explanation on flat diagrams; 55-60 present a proof another group can check without cutting.

Questions to explore

- Does one connected object necessarily have one boundary loop?
- Which part of the seam rule changes the result of the center cut?

Satisfying stop: A seam diagram that correctly predicts the outcome of the cut and distinguishes two kinds of counting.

GA-32 / Facilitator reasoning

Cut a Möbius band and follow the seam

The mathematical question

How can gluing rules determine connectedness, boundary, and orientability before a surface is physically built? Compare the annulus quotient $(0,s) \sim (1,s)$ with the Möbius quotient $(0,s) \sim (1,-s)$ of $[0,1] \times [-1,1]$. The Möbius band has one boundary component. Cutting along $s=0$ produces one annulus; the corresponding annulus cut produces two annuli.

Worked instance

Problem: Predict boundary loops and connected pieces before and after a center cut, using the two seam rules.

Solution: In the annulus, the top edge returns to the top and the bottom to the bottom, giving two boundary loops. Cutting the center leaves U glued to U and L to L, so there are two separate annuli. In the Möbius band, following the top edge crosses to the bottom and only then returns, giving one boundary loop. After the center cut, U joins L and L joins U: the two rectangular pieces form one cyclic strip. Traversing both pieces reverses the transverse coordinate twice, so the combined strip has matching ends and is an annulus. It is connected but has two boundary loops.

Boundary: Do not infer “two pieces” from two visible edges after cutting. Connected components and boundary components count different objects; the cut Möbius example separates them.

Hints to reveal in order

1. Name the upper half U and lower half L before attaching the seam.
2. After a half-turn, the end of U attaches to L. Follow one more seam crossing.

Extensions

- Replace the half-turn by several half-turns and predict seam behavior from parity, keeping knotting questions separate.
- Draw the center circle and explain why flattening a band onto it forgets boundary and orientability information.

Connection and its limits

Bridge type: exact-special-case

Preserves: Paper represents these exact quotient surfaces. Tracking glued edges proves connectedness beyond the empirical surprise of one cut.

Limits: The embedding in three-dimensional space may be twisted or linked. This activity classifies the resulting surfaces, not their knotting or a numerical twist count.

Sources and inspection scope

[Allen Hatcher, Algebraic Topology](#)

Chapter 0, opening deformation-retraction examples, printed p.2; Möbius band and core circle

Inspected: Core-circle surface model inspected; boundary and cutting conclusions derived independently by the explicit seam tracking below

Prior use / what changes: New surface investigation. Week 5 used geometric transformations; here identification of whole edges creates a space rather than moving a planar figure.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-33 / Participation and launch

The transform must remember how the system started

MSC primary 44 - Integral transforms, operational calculus | Secondary 34, 93

The opening challenge

A system moves toward level one according to $y' = 1 - y$. Predict its behavior when it starts empty and when it starts at level two. Can an algebraic shortcut preserve that difference?

Learner choices: Choose starting levels, sketch predictions, and choose a constant or fading input before calculating.

Prerequisite gates

Enter: D/A: read a rate equation and compare initial slopes.

Explore: I/A: use the definition, integration by parts, and a short transform table.

Explain: P/D: substitute the recovered trajectory into the original equation and initial condition.

Prove: I/P: justify the derivative rule, including the boundary term at infinity.

Reading: Symbolic mathematical reading; a shared table reduces memory demand.

Arithmetic: Partial fractions and exponential algebra.

Reasoning: Track the same initial datum through time-domain and transformed descriptions.

Hard stop: Stop before this family unless derivatives and improper integrals are meaningful; forcing an elementary version would lose the central mechanism.

Materials and preparation

8 minutes estimated prep. Graph paper or plotting tool, a table with transforms of $1, e^{-t}, t, e^{-t}$, and equation cards with different initial values.

A flexible hour

0-10 sketch initial slopes and qualitative paths; 10-15 establish the transform definition; 15-35 solve and compare initial conditions; 35-40 reset; 40-55 investigate a fading input and verify by differentiation; 55-60 explain the indispensable boundary term.

Questions to explore

- Where exactly is the starting value stored in the transformed equation?
- Can two solutions have the same forcing but different trajectories?

Satisfying stop: Two correctly distinguished initial-value solutions and one direct time-domain verification.

GA-33 / Facilitator reasoning

The transform must remember how the system started

The mathematical question

How does an integral transform convert a differential equation into algebra without losing initial data?

For the one-sided Laplace transform $Y(s) = \int_0^\infty e^{-st} y(t) dt$, integration by parts gives $L(y') = sY - y(0)$ under convergence and vanishing-boundary hypotheses. Applied to $y' + y = 1$, the initial value selects $y(t) = 1 + (y(0) - 1)e^{-t}$.

Worked instance

Problem: Solve $y' + y = 1$ first with $y(0) = 0$, then with $y(0) = 2$; finally change the forcing to e^{-t} and start at zero.

Solution: For the constant forcing and initial value a , $sY - a + Y = 1/s$, so $Y = 1/[s(s+1)] + a/(s+1)$. Inversion gives $y = 1 - e^{-t} + a e^{-t}$. Thus the first two trajectories are $1 - e^{-t}$ and $1 + e^{-t}$: both approach one, from opposite sides. For forcing e^{-t} and zero initial value, $(s+1)Y = 1/(s+1)$, hence $Y = 1/(s+1)^2$ and $y = t e^{-t}$. Differentiating gives $y' = (1-t)e^{-t}$, so $y' + y = e^{-t}$ and $y(0) = 0$. These transforms converge for real $s > 0$.

Boundary: Omitting $-y(0)$ incorrectly assigns the zero-start trajectory to every initial condition. A transform expression without its domain of convergence is also incomplete.

Hints to reveal in order

1. Keep $-y(0)$ attached to the transformed derivative.
2. After inversion, check both the differential equation and the initial value.

Extensions

- Find when $t e^{-t}$ reaches its maximum, and relate that time to input versus decay.
- Replace the decay coefficient one by $k > 0$ and determine the steady level for constant input.

Connection and its limits

Bridge type: exact-special-case

Preserves: The transformed equation and recovered continuous trajectory solve exactly the same initial-value problem.

Limits: This is a calculus-gated activity. The algebraic table game alone does not establish the transform identity or convergence.

Sources and inspection scope

[Stephen Boyd, EE102 Laplace Transform](#)

Slides 3-19 and 3-20, differentiation and initial-condition example; transform table

Inspected: Derivative transform and role of initial conditions inspected; selected equations and inverses verified directly

Prior use / what changes: New transform family. GA-16 develops convolution geometrically; this offers a later route to transformed differential equations and system response.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-34 / Participation and launch

A well-behaved rate law can still explode

MSC primary 34 - Ordinary differential equations | Secondary 26

The opening challenge

A quantity grows at a rate equal to its square. Starting at one, could it double over and over before time one? Predict the times before looking at the candidate formula.

Learner choices: Choose positive, zero, or negative starting values; choose target heights and compare their arrival times.

Prerequisite gates

Enter: A/F: compare values of a supplied rational trajectory and mark doubling times.

Explore: D/A: interpret the rate equation and differentiate the proposed solution.

Explain: P: explain why an infinite value cannot be inserted as a real-valued continuation.

Prove: D/P: verify the solution and use local uniqueness to establish its maximal interval.

Reading: Symbolic fractions and rate notation; discussion can precede formal theorem reading.

Arithmetic: Rational expressions, signed values, and geometric time intervals.

Reasoning: Separate uniqueness, boundedness, and duration of existence.

Hard stop: Without derivatives, stop at the exact rational-function investigation; identifying it as the solution of the rate equation requires calculus.

Materials and preparation

8 minutes estimated prep. Time-number lines, value cards, graph paper or plotting software, and optional slope-field images.

A flexible hour

0-10 predict growth and inspect candidate curves; 10-15 define the rate equation; 15-35 verify and find doubling times; 35-40 reset; 40-55 vary initial values and discuss continuation; 55-60 distinguish the three existence questions.

Questions to explore

- Does a finite rate at every finite y prevent the solution from escaping to infinity?
- Why can a formula valid beyond a vertical asymptote fail to continue the original trajectory?

Satisfying stop: A directly verified solution, an exact finite blow-up time, and a clear reason continuation fails.

GA-34 / Facilitator reasoning

A well-behaved rate law can still explode

The mathematical question

Why does local existence and uniqueness not guarantee that a trajectory exists for all future time?

For $y' = y^2$ with $y(0) = a > 0$, $y(t) = a/(1 - at)$ is the unique solution on its maximal forward interval $0 \leq t < 1/a$ and diverges as t approaches $1/a$. The polynomial rate law is locally Lipschitz, so this is finite-time blow-up without a local uniqueness failure.

Worked instance

Problem: Verify the solution for $a=1$, calculate repeated doubling times, and compare $a=1/2, 0, -1$.

Solution: Differentiating $y = a/(1 - at)$ gives $y' = a^2/(1 - at)^2 = y^2$, with $y(0) = a$. For $a=1$, height 2^n is reached at $t = 1 - 2^{-n}$, so all finite doublings occur before one. The solution tends to $+\infty$ as $t \rightarrow 1$ from below and cannot extend continuously as a real-valued solution through that time. For $a=1/2$ the blow-up time is two. For $a=0$ the solution is constantly zero. For $a=-1$ it is $-1/(1+t)$, defined for all $t \geq 0$ and approaching zero from below.

Boundary: The expression $1/(1-t)$ also satisfies the equation on $t > 1$, but that disconnected branch does not pass through the initial point and is not a continuation across $t=1$. Local uniqueness remains valid.

Hints to reveal in order

1. Differentiate $a/(1 - at)$ before trying to solve the equation from scratch.
2. For $a=1$, solve $1/(1 - t) = 2^n$ for the arrival time.

Extensions

- Compare $y' = y$, whose exponential solution grows without finite-time blow-up.
- For $y' = y^p$ with positive initial data and $p > 1$, investigate how the blow-up time depends on p .

Connection and its limits

Bridge type: exact-special-case

Preserves: The algebraic trajectory is an exact solution of a genuinely continuous differential equation; finite doubling times reveal its limiting behavior.

Limits: A step-by-step numerical table can suggest growth, but cannot establish either finite-time divergence or the exact differential law.

Sources and inspection scope

[Gerald Teschl, Ordinary Differential Equations and Dynamical Systems](#)

§2.2, local existence/uniqueness and Problem 2.9 on finite-time existence

Inspected: Local theorem and blow-up exercise inspected; explicit solution and maximal forward intervals checked

Prior use / what changes: New counterpart to GA-08: the waiting-time example loses uniqueness at zero, whereas this smooth rate law retains local uniqueness but loses global existence.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-35 / Participation and launch

A circle's average can equal its center exactly

MSC primary 31 - Potential theory | Secondary 35, 26

The opening challenge

Place four equally spaced points on a circle and evaluate $x^2 - y^2$ at each. Does their average know the value at the center? Move the circle and try to break your guess.

Learner choices: Choose center, radius, and starting angle; then choose altered quadratics and design a sampling test likely to expose failure.

Prerequisite gates

Enter: F/V: evaluate a supplied quadratic at four marked circle points.

Explore: A: move the center, change the radius, and pair opposite points.

Explain: P/A: prove the four-point cancellation and propose a general coefficient condition.

Prove: I: integrate the trigonometric parametrization; D for the Laplacian connection.

Reading: Coordinates and short formulas; calculus notation only at the advanced gate.

Arithmetic: Signed squares and averages; trigonometric identities for the continuous calculation.

Reasoning: Turn sampled evidence into a structural identity with stated scope.

Hard stop: General mean-value and maximum principles for arbitrary harmonic functions need analysis; a finite sampling experiment is not their proof.

Materials and preparation

8 minutes estimated prep. Coordinate grids, circle templates, value tables, and optional plotting software.

A flexible hour

0-10 evaluate and compare circles; 10-15 distinguish samples from a continuous average; 15-35 find symmetry cancellations; 35-40 reset; 40-55 integrate if ready or classify four-point identities; 55-60 state exactly what was proved.

Questions to explore

- What cancels between opposite points?
- Could a different function pass our four-point test but fail the full circle-average test?

Satisfying stop: A proved symmetry identity; calculus-ready learners additionally prove the actual continuous mean formula.

GA-35 / Facilitator reasoning

A circle's average can equal its center exactly

The mathematical question

What makes harmonic functions obey exact averaging laws on a continuum of surrounding points?

The harmonic polynomial $h(x,y)=x^2-y^2$ equals its mean over every circle centered at (a,b) . More generally a quadratic $Ax^2+Bxy+Cy^2+Dx+Ey+K$ has this property exactly when $A+C=0$, equivalently its Laplacian is zero.

Worked instance

Problem: Test center $(1,2)$, radius two, then prove the mean identity and find a misleading four-point test.

Solution: The cardinal points $(3,2), (-1,2), (1,4), (1,0)$ have h -values $5, -3, -15, 1$, averaging $-3=h(1,2)$.

Parametrization gives $h=a^2-b^2+2r(a \cos \theta - b \sin \theta)+r^2(\cos^2 \theta - \sin^2 \theta)$. Every extra term has circle mean zero, proving the identity for all centers and radii. For a general quadratic the circle mean is its center value plus $(A+C)r^2/2$; hence the stated criterion. But $q=x^2y^2$ on a circle centered at zero has value zero at all four cardinal points while its actual circle mean is $r^4/8 > 0$ for $r > 0$.

Boundary: The four cardinal samples for q falsely suggest the mean-value identity. Changing the starting angle can expose the failure; a universal proof still needs an exact argument.

Hints to reveal in order

1. Use $x=a+r \cos \theta$ and $y=b+r \sin \theta$.
2. The full-circle means of $\cos \theta$ and $\sin \theta$ vanish; the means of their squares both equal one-half.

Extensions

- Use the Laplacian to classify more harmonic polynomials and then test their mean values.
- Explain directly why x^2-y^2 has no interior local maximum, and compare with $-x^2-y^2$.

Connection and its limits

Bridge type: exact-special-case

Preserves: The activity investigates actual functions on the plane and their actual circle integrals. Symmetric point groups expose why exact cancellation occurs.

Limits: Four successful sample values do not establish an integral identity for an arbitrary function. The continuous mean and the discrete test must remain distinct.

Sources and inspection scope

[John K. Hunter, Notes on Partial Differential Equations](#)

§2.1, Theorems 2.1-2.2 on mean values; §2.3 maximum principle

Inspected: Mean-value statements and smoothness/domain hypotheses inspected; quadratic classification derived independently

Prior use / what changes: GA-05 introduced a finite network averaging mechanism. This is the deliberately separate continuous family and does not claim that network experiments prove harmonic-function theorems.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

GA-36 / Participation and launch

A fixed point exists-but will repeated feedback find it?

MSC primary 54 - General topology | Secondary 26, 37

The opening challenge

A machine turns a number between zero and one into another number in that interval. Can some input come out unchanged? If you repeatedly feed the output back in, must you find it?

Learner choices: Choose a continuous graph inside the square, choose a starting value, and try to create settling, alternating, and cycling behavior.

Prerequisite gates

Enter: O/V: match a curve with the diagonal and follow feedback arrows.

Explore: F/A: iterate simple affine maps and compare errors.

Explain: P: explain the endpoint sign argument without differentiating.

Prove: P/A: prove the interval theorem by continuity and derive geometric error decay in the chosen contraction.

Reading: Graph labels and a short iteration rule; no derivative notation is necessary.

Arithmetic: Fractions, signed differences, powers of three.

Reasoning: Distinguish an object's existence, uniqueness, and accessibility by a chosen process.

Hard stop: A general contraction theorem needs completeness and a sequence-limit proof; the finite computation alone establishes neither.

Materials and preparation

7 minutes estimated prep. Unit-square grids, diagonal lines, arrow cards, and a table for iterates and errors.

A flexible hour

0-10 graph and arrow play; 10-15 define fixed points; 15-35 compare two machines; 35-40 reset; 40-55 prove existence and investigate error decay; 55-60 separate the conclusions.

Questions to explore

- Why does staying inside the unit interval force an intersection with the diagonal?
- What property prevents two different fixed points in the shrinking-distance example?

Satisfying stop: An existence argument and two exact iterations proving that existence and convergence are different claims.

GA-36 / Facilitator reasoning

A fixed point exists-but will repeated feedback find it?

The mathematical question

What separates a topological existence theorem from a convergent algorithm?

Every continuous $f: [0,1] \rightarrow [0,1]$ has a fixed point because $f(x)-x$ changes from nonnegative to nonpositive.

Iterating f need not converge: $f(x)=1-x$ has fixed point $1/2$ but cycles from zero. A contraction adds a quantitative convergence mechanism.

Worked instance

Problem: Start at zero and compare $f(x)=(x+1)/3$ with $g(x)=1-x$. Prove that every continuous self-map of the closed interval $[0,1]$ has some fixed point.

Solution: For f , solving $x=(x+1)/3$ gives $x=1/2$. The iterates from zero are $1/3, 4/9, 13/27, 40/81$. Their errors satisfy $x_n - 1/2 = (x_0 - 1/2)/3^n$, so after four steps the error is $1/162$ and tends to zero. For g , the only fixed point is also $1/2$, but zero and one alternate forever. For any continuous interval self-map, $d(x)=f(x)-x$ is continuous, $d(0) \geq 0$, and $d(1) \leq 0$. An endpoint zero is already a fixed point; otherwise the intermediate value theorem gives an interior zero.

Boundary: On the incomplete space $(0,1)$, $f(x)=x/2$ shrinks distances but has no fixed point in that space: iteration approaches the omitted endpoint zero. The domain hypotheses cannot be discarded.

Hints to reveal in order

1. Compare $f(x)$ with x at the two endpoints.
2. For $f(x)=(x+1)/3$, subtract $1/2$ from both sides of the iteration equation.

Extensions

- Use $f_q(x)=1/2+q(x-1/2)$ with $-1 < q < 1$. It maps $[0,1]$ into $[0,1]$ and has error multiplier q about its fixed point $1/2$. Compare the bound $|x_n - 1/2| = |q|^n |x_0 - 1/2|$, including alternating errors when $q < 0$.
- Find continuous self-maps with an interval of fixed points, then examine which uniqueness hypothesis fails.

Connection and its limits

Bridge type: exact-special-case

Preserves: The learner's feedback map is an actual continuous interval self-map. Its fixed point and iterates are exact mathematical objects.

Limits: A crossing in a coarse sketch is evidence, while continuity and endpoint inequalities prove existence. Existence by itself supplies no universal iteration guarantee.

Sources and inspection scope

[Jiří Lebl, Basic Analysis I](#)

§3.3 intermediate value theorem; §7.6, Theorem 7.6.2 and Exercise 7.6.4 on contractions and completeness

Inspected: Existence and contraction hypotheses inspected; both explicit iterations and the omitted-endpoint example checked

Prior use / what changes: New fixed-point synthesis. GA-02 and GA-28 also use continuity; GA-10 studies iteration through symbolic orbits. Return visits should foreground a different question, not repeat the same calculation.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-01 / Participation and launch

Which shore will the wandering token reach?

MSC primary 60 - Probability theory and stochastic processes | Secondary 65

The opening challenge

Put the token on 1 of the track 0-1-2-3-4. Heads moves right one space, tails left one. Stop when it reaches 0 or 4. Which starting places make reaching 4 easier? Choose a start and investigate.

Learner choices: Choose starts, organize trials, decide whether to compare counts or proportions, and propose a track change to test.

Prerequisite gates

Enter: O/C: move one place following a coin result and recognize endpoints.

Explore: C/F: tally outcomes from repeated starts; compare fractions.

Explain: F/A: an interior winning chance is the average of its two neighbors.

Prove: A/P: solve the finite averaging equations, and justify eventual absorption.

Reading: None with oral rules; a recorder may label the table.

Arithmetic: Small counts; quarters and halves for the worked example.

Reasoning: Separate individual paths from repeated experiments; hold the starting location fixed.

Hard stop: Martingale stopping theorems and Brownian limits require new probability prerequisites.

Materials and preparation

5 minutes estimated prep. Draw five numbered spaces, provide a token and coin, and prepare a start/end tally sheet. A partner can flip while another directs moves.

A flexible hour

0-10 explore coin paths freely; 10-15 agree on independent tosses and stopping; 15-35 collect starts and endpoint records; 35-40 walk a human-size track; 40-55 build a probability-height drawing or longer track; 55-60 share a conjecture and its evidence.

Questions to explore

- Could starting one place farther right make the winning chance smaller?
- What must the chance at 2 have to do with the chances at 1 and 3?

Satisfying stop: A complete five-position chance map, supported by a finite averaging argument.

AP-01 / Facilitator reasoning

Which shore will the wandering token reach?

The mathematical question

How can a random path have an exactly predictable endpoint probability, and why is that probability harmonic?

An independent fair nearest-neighbor random walk on $0, 1, \dots, N$, stopped at 0 or N , hits N with probability i/N from i . The first-step equations and boundary values determine the finite answer.

Worked instance

Problem: Find the exact probability of reaching 4 before 0 when starting at 1.

Solution: Write $p_0=0, p_4=1$ and $2p_i=p(i-1)+p(i+1)$. Consecutive differences are equal, so $p_1=1/4, p_2=1/2, p_3=3/4$. Absorption occurs almost surely: from any interior state, four consecutive heads suffice; each independent block has probability $1/16$, so survival through k blocks is at most $(15/16)^k$. This justifies the endpoint interpretation.

Boundary: A coin with heads probability $3/4$ breaks equal averaging. Starting at 1 on 0-1-2 then wins with probability $3/4$, not $1/2$.

Hints to reveal in order

1. Give endpoints height 0 and height 1; leave interior heights blank.
2. Make every interior height the average of its neighbors. Equal spacing works.

Extensions

- Use track 0-6 and explain why doubling a start from 1 to 2 doubles its endpoint probability.
- With algebra, solve the biased recurrence; with advanced probability, compare discrete harmonic functions and diffusion.

Connection and its limits

Bridge type: exact-special-case

Preserves: A token position is precisely the state of the stopped Markov chain. Independent fair coin flips implement its transition rule.

Limits: Short experimental frequencies are estimates. This is not Brownian motion or a proof of a diffusion limit.

Sources and inspection scope

[Grinstead and Snell, Introduction to Probability](#)

Chapters 4 and 12.2

Inspected: relevant exposition inspected; finite instance independently derived below

Prior use / what changes: Week 1 extensions include random walks on tiling graphs; this uses absorbing boundaries and harmonic probabilities. Actual use of this new instance is unknown.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-02 / Participation and launch

The clue that changes the bag

MSC primary 60 - Probability theory and stochastic processes | Secondary 62

The opening challenge

Bag A contains two red counters and one blue; bag B contains one red and two blue. Choose A or B by a fair coin, then draw one counter uniformly without looking. You see red. Which bag is now more likely? Arrange all possible bag-and-counter outcomes before making your claim.

Learner choices: Choose a representation of outcomes, design a different pair of bags, and compare information from red, blue, or a different reporting rule.

Prerequisite gates

Enter: O/C: match a card to a bag and remove outcomes inconsistent with a clue.

Explore: C/F: compare surviving case counts and fractions.

Explain: P: explain why retained cases have equal probability.

Prove: F/A: derive Bayes probabilities with unequal bag-selection weights.

Reading: Labels can be colors or pictures; rules can be spoken.

Arithmetic: Counting six outcomes; halves, thirds and sixths.

Reasoning: Keep the process producing the clue separate from the clue itself.

Hard stop: Continuous Bayesian inference needs densities and integration; a finite case is already complete mathematics.

Materials and preparation

10 minutes estimated prep. Two opaque cups A and B, each with three labeled counters. A contains RRBlue; B contains RBlueBlue. Provide six outcome cards A1,A2,A3,B1,B2,B3.

A flexible hour

0-10 inspect and label counters; 10-15 rehearse the two random choices; 15-35 build and filter outcome cards; 35-40 swap investigator and dealer roles; 40-55 change the clue-producing rule or bag-selection coin; 55-60 describe what information matters.

Questions to explore

- Why are the bags equally likely before drawing but not after seeing red?
- Would hearing “this bag contains red” give the same information as seeing a randomly drawn red?

Satisfying stop: Two different clue mechanisms produce different correct answers, explained with surviving cards.

AP-02 / Facilitator reasoning

The clue that changes the bag

The mathematical question

How does evidence update probabilities, and why must its selection mechanism be specified?

Conditional probability on a finite equally likely sample space is the fraction of surviving outcomes. Posterior odds depend on prior selection and evidence likelihood, not only on the visible counter.

Worked instance

Problem: With a fair bag choice and uniform draw, compute $P(A \text{ given red})$. Then compare an announcer who merely reports whether the chosen bag contains any red.

Solution: There are six equally likely outcomes. Three are red: two from A and one from B. Therefore the posterior chance of A is $2/3$. Both bags contain red, so the announcer always gives the second report; that report leaves the chance at $1/2$. The identical word red does not make the experiments equivalent.

Boundary: If A is selected with probability $1/3$ and B with probability $2/3$, red has contributions $2/9$ from each bag; the posterior chance of A is then $1/2$.

Hints to reveal in order

1. List bag identity and counter identity together, not just the two possible colors.
2. Cover every blue-draw card and compare the bag labels still visible.

Extensions

- Choose integer bag contents making a red observation leave the prior odds unchanged; equal red fractions do so.
- Use a tree with unequal prior weights and derive posterior odds = prior odds $\times$ likelihood ratio.

Connection and its limits

Bridge type: exact-special-case

Preserves: Bag choices and individually labeled counter draws constitute an explicit finite experiment. Filtering cards implements conditioning exactly.

Limits: A posterior describes this specified model; it does not establish that a physical bag was chosen uniformly.

Sources and inspection scope

[Grinstead and Snell. Introduction to Probability](#)

Chapters 4 and 12.2

Inspected: relevant exposition inspected; finite instance independently derived below

Prior use / what changes: The year-1 collection contains probability activities. This exact evidence-selection investigation is new; do not count a changed color story as a new mechanism.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-03 / Participation and launch

Could the treatment labels have landed this way?

MSC primary 62 - Statistics | Secondary 05, 60

The opening challenge

Four people received scores 0,1,3,4. Before the activity, exactly two were chosen uniformly for training. The trained pair scored 3 and 4. If training changed nobody's score, how often could random labels make the trained average at least this much higher?

Learner choices: Choose a systematic listing method, predict which assignments are most extreme, and decide before counting whether to test improvement only or any difference.

Prerequisite gates

Enter: O/C: assign two labels and compare two groups.

Explore: F: compute two-number means and enumerate six assignments.

Explain: P: justify equal assignment probabilities under the stated design.

Prove: F/P: relate tail counts to a prespecified statistic and null model.

Reading: Short labels; a partner can read the experimental story.

Arithmetic: Small sums divided by two; sixths.

Reasoning: Distinguish observed outcomes, hypothetical assignments, and the null assumption.

Hard stop: Observational causal inference requires additional assumptions; random shuffling does not manufacture randomized assignment.

Materials and preparation

8 minutes estimated prep. Four outcome cards numbered 0,1,3,4, two treatment clips, and a six-row record sheet. Use a fictional paper-airplane training example, not health claims.

A flexible hour

0-10 freely group the cards; 10-15 distinguish the six equally likely assignments; 15-35 calculate the difference for every assignment; 35-40 physically switch two group signs; 40-55 compare one-sided and two-sided questions; 55-60 state a conclusion with its exact assumption.

Questions to explore

- Why do we move labels while leaving scores fixed?
- Does a difference that occurs once in six assignments prove that training works?

Satisfying stop: A complete exact reference distribution with a carefully qualified conclusion.

AP-03 / Facilitator reasoning

Could the treatment labels have landed this way?

The mathematical question

How can randomized assignment support an exact inference without assuming a normal population?

Under the sharp null that treatment changes no participant's outcome, outcomes remain fixed while treatment labels follow the actual assignment mechanism. A prespecified tail of the randomization distribution gives an exact finite test.

Worked instance

Problem: Find the one-sided and two-sided exact tail proportions for observed trained scores {3,4}.

Solution: Trained pairs {0,1},{0,3},{0,4},{1,3},{1,4},{3,4} give mean differences $-3, -1, 0, 0, 1, 3$. Only one of six is at least 3, giving $1/6$. Two have absolute difference at least 3, giving $2/6$. Neither is small at a conventional 0.05 threshold. The tail direction must not be chosen after noticing which result looks better.

Boundary: If training was offered to the strongest participants rather than assigned uniformly, these six assignments are not the actual randomization distribution; this calculation cannot establish a treatment effect.

Hints to reveal in order

1. Choose the two trained scores; the other pair is forced.
2. Subtract untrained mean from trained mean and order the six results.

Extensions

- Use six outcome cards with three treatment labels; enumerate the 20 assignments and study attainable tail proportions.
- Compare random sampling, which supports population generalization, with random assignment, which supports the treatment comparison; one does not automatically supply the other.

Connection and its limits

Bridge type: exact-special-case

Preserves: Assigning two treatment labels uniformly among four fixed outcome cards reproduces a complete randomized experiment under its sharp null.

Limits: These invented scores are a mathematical example, not evidence about a real treatment. A test does not compute the probability that the null is true.

Sources and inspection scope

[Allen Downey, Think Stats, third edition](#)

Section 9.2 supports permutation-test mechanics; the randomized-assignment/sharp-null version here is independently proved

Inspected: relevant exposition inspected; finite instance independently derived below

Prior use / what changes: No matching inference design is documented in Weeks 1-10. This is not another dice-fairness lesson.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-04 / Participation and launch

A sampling rule that always misses

MSC primary 62 - Statistics | Secondary 60

The opening challenge

Estimate the average value of all eight hidden cards. Team A draws two identifiers uniformly from all eight; team B draws from the first four only. Replace each card after recording it, including between the two draws in a sample. Repeat your samples. Which extra information would make a guess trustworthy?

Learner choices: Choose a sampling rule, decide how many repetitions to collect, and propose an audit that might reveal a missing part of the population.

Prerequisite gates

Enter: O/C: select labeled cards and record a number.

Explore: F: calculate averages and compare the spread of repeated samples.

Explain: P: pair each equally likely sample with its contribution to the average.

Prove: F/A: use linearity of expectation for uniform draws.

Reading: Number cards plus two short selection rules; oral participation works.

Arithmetic: Means of two or four small integers; fractions.

Reasoning: Separate randomness of selection from representativeness of its sampling frame.

Hard stop: Correcting unknown real-world nonresponse needs extra information; no universal arithmetic adjustment is promised.

Materials and preparation

10 minutes estimated prep. Population cards with values 1,1,1,1,5,5,5,5, kept in two labeled sections. Prepare a full-population draw bag and a bag restricted to the four 1-cards; keep identities visible to the facilitator.

A flexible hour

0-10 explore guessing from covered cards; 10-15 define the target average and two selection rules; 15-35 collect independent samples with replacement; 35-40 exchange sampler and recorder roles; 40-55 reveal the population and explain contrasting records; 55-60 propose a better survey design.

Questions to explore

- Can one group be extremely consistent and still wrong?
- What is the difference between drawing randomly from a bag and including every population member in that bag?

Satisfying stop: Explain how a noisy estimate can be appropriately centered while a perfectly consistent estimate can be biased.

AP-04 / Facilitator reasoning

A sampling rule that always misses

The mathematical question

Why does collecting more observations fail to remove selection bias, and how does sampling design differ from sampling variability?

For uniform sampling of a finite population, the sample mean is unbiased. Sampling through a systematically selected subset can have persistent bias; increasing repetitions estimates that subset more precisely.

Worked instance

Problem: For two independent uniform draws with replacement from all eight cards, find the distribution of the sample mean. Compare draws restricted to the four 1-cards.

Solution: Each full-population draw is 1 or 5 with probability $1/2$. The two-draw mean is 1,3,5 with probabilities $1/4, 1/2, 1/4$, so its expected value is 3, the population mean. The restricted rule always gives mean 1, even after arbitrarily many draws; its bias is -2 . Sampling more from that same frame does not repair the omission.

Boundary: A single full-population sample can still give 1. Unbiasedness is a repeated-sampling property, not a promise that every individual estimate is right.

Hints to reveal in order

1. Write the target population above the record, separately from the draw bag.
2. After revealing all eight cards, compare the mean in each sampling frame.

Extensions

- Compare samples of one, two and four cards: uncertainty shrinks while the uniform target stays 3.
- Let sampling probabilities depend on card value and investigate how known inverse-probability weights can correct a specified design; unknown selection remains a genuine obstacle.

Connection and its limits

Bridge type: exact-special-case

Preserves: A concealed finite population and explicitly defined sampling mechanisms provide a genuine unknown-parameter inference problem.

Limits: The small population makes all calculations available; it does not certify any particular real survey.

Sources and inspection scope

[Allen Downey, Think Stats](#)

Chapter 8: estimation, bias and sampling distributions

Inspected: relevant exposition inspected; finite instance independently derived below

Prior use / what changes: No corresponding sampling-frame task is documented in the existing packets. Retain the selected population and sampling rule in future use logs.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-05 / Participation and launch

An interval-making machine

MSC primary 62 - Statistics | Secondary 60

The opening challenge

Our instrument reports $X = \theta + E$, where θ is a fixed hidden integer and E is a fresh uniform draw from $-2, -1, 0, 1, 2$. After each reading X , report $[X-1, X+1]$. Without knowing θ , can you predict how often this interval-making rule will cover it?

Learner choices: Choose an interval width, compare coverage against precision, and design a rule whose advertised coverage can be checked.

Prerequisite gates

Enter: O/C: move an interval marker to include three neighboring integer points.

Explore: F/A: interpret unknown integer θ plus a random measurement error.

Explain: P: list all five equiprobable errors and decide which intervals cover θ .

Prove: A/P: show the covering errors are the same for every integer θ .

Reading: Short instrument rules and interval endpoints; oral diagram explanation is possible.

Arithmetic: Add/subtract integers up to two; fifths.

Reasoning: Reason about a fixed unknown target and a random interval, not a moving target.

Hard stop: A probability distribution for θ after data requires an additional prior model; do not supply one silently.

Materials and preparation

10 minutes estimated prep. Integer number line, an opaque envelope containing a secret integer θ , five equally likely error cards $-2, -1, 0, 1, 2$, and transparent three-point interval strips.

A flexible hour

0-10 explore intervals on a number line; 10-15 establish the hidden target and error mechanism; 15-35 draw readings and place strips; 35-40 reveal θ and step to the five error positions; 40-55 prove coverage for any integer target or change width; 55-60 explain the guarantee in words.

Questions to explore

- Which errors make the interval miss, and why?
- If the observed interval is $[6, 8]$, what extra assumption would be needed to assign probabilities to possible hidden targets?

Satisfying stop: A proved 60% interval procedure and an explanation of what its percentage measures.

AP-05 / Facilitator reasoning

An interval-making machine

The mathematical question

What does a confidence procedure guarantee over repeated samples, and what does it leave unknown after one sample?

Coverage is $P(\theta \text{ lies in } I(X))$ for a sampling procedure I . It is a property indexed by the unknown parameter; it need not equal a posterior probability after observing a particular interval.

Worked instance

Problem: Compute coverage of $[X-1, X+1]$ and $[X-2, X+2]$. Is the first an exact 60% procedure for every integer θ ?

Solution: The first covers θ exactly when $-1 \leq E \leq 1$, which occurs for three of the five cards, so coverage is $3/5$ for every θ . The wider interval covers for all five errors and has 100% coverage under this bounded-error model. These probabilities describe repetitions of the random instrument with θ fixed.

Boundary: If the error distribution puts probability $1/2$ on each of -2 and 2 , the narrow procedure has zero coverage. Its claim depends on the stated error mechanism, even though its printed endpoints look unchanged.

Hints to reveal in order

1. Pretend θ is 0 temporarily; list all five readings.
2. Translate the whole picture. Does the list of successful error cards change?

Extensions

- Asymmetric intervals can have the same coverage: $[X-2, X]$ covers errors 0, 1, 2. Compare their practical consequences.
- Investigate confidence procedures for an unknown coin probability later; finite sample spaces permit exact checks but the coverage calculation is no longer a translation argument.

Connection and its limits

Bridge type: exact-special-case

Preserves: A finite shift experiment gives exactly calculable interval coverage without asymptotic approximation.

Limits: This deliberately simple instrument model is not the binomial proportion problem or a general method for constructing confidence intervals.

Sources and inspection scope

[NIST/SEMATECH e-Handbook of Statistical Methods](#)

Section 1.3.5.2, introductory interpretation of confidence intervals as repeated-sampling procedures; the five-error model here is independently derived, not the handbook's t-interval example

Inspected: introductory interpretation directly inspected; all five errors independently enumerated

Prior use / what changes: No confidence-coverage task is documented in the current collection. Do not label this a bootstrap activity; it uses a known error model.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-06 / Participation and launch

A root finder caught in a two-step loop

MSC primary 65 - Numerical analysis | Secondary 26, 37

The opening challenge

The robot finds a root by following the tangent line down to the x-axis, then repeating from there. Use $f(x)=x^3-2x+2$ and $f'(x)=3x^2-2$. Start at 0. Can you predict the robot's thousandth guess without doing a thousand calculations?

Learner choices: Choose starting points, compare exact and decimal records, and invent a warning rule that detects repeated states.

Prerequisite gates

Enter: V/A/D: coordinates, evaluating a cubic and the meaning of derivative as tangent slope.

Explore: A/F: calculate successive rational iterates or direct a calculator.

Explain: P: verify a repeating state and show why deterministic iteration then repeats forever.

Prove: D/X: a general convergence theorem needs quantitative derivative control near a simple root.

Reading: Symbolic formula and a short algorithm; shared recorder permitted.

Arithmetic: Signed integers and fractions; cubic substitution.

Reasoning: Distinguish the equation's root from an algorithm's current guess.

Hard stop: Do not remove tangency and call a number guessing game Newton's method; calculus is a legitimate entry gate here.

Materials and preparation

12 minutes estimated prep. Graph paper with $f(x)=x^3-2x+2$ drawn from -2 to 2 , ruler, and calculator. Prepare a table with x , $f(x)$, $f'(x)$, next x .

A flexible hour

0-10 explore the curve and tangent drawings; 10-15 calculate one complete Newton step together; 15-35 investigate starts 0, 1 and -2 ; 35-40 exchange graph and calculation roles; 40-55 contrast looping with a sign-changing bracket; 55-60 state what failed and what remains true.

Questions to explore

- Does a loop mean that the original equation has no root?
- What information does a bracket supply that a single tangent guess lacks?

Satisfying stop: A fully proved infinite loop and a guaranteed root bracket coexist.

AP-06 / Facilitator reasoning

A root finder caught in a two-step loop

The mathematical question

Why can a locally excellent numerical method fail completely from an unfortunate starting point?

Newton iteration $x_{\text{next}} = x - f(x)/f'(x)$ replaces a smooth equation by its tangent equation. Local convergence near a simple root does not imply convergence from every start.

Worked instance

Problem: Verify the loop from 0 and show that the cubic nevertheless has a real root.

Solution: $f(0)=2$ and $f'(0)=-2$, so Newton gives 1. At 1, $f(1)=1$ and $f'(1)=1$, so it gives 0. Thus even-numbered iterates are 0 and odd-numbered iterates 1. Meanwhile $f(-2)=-2$ and $f(-1)=3$; continuity guarantees a root between -2 and -1 . The equation is solvable while this algorithmic run fails.

Boundary: At $x=\sqrt{2/3}$, the derivative is zero, so this formula is undefined. A stopping rule based only on a small step is also not a universal accuracy certificate.

Hints to reveal in order

1. At 0, calculate the numerator and denominator separately.
2. Once both $0 \rightarrow 1$ and $1 \rightarrow 0$ are exact, every later iterate is determined.

Extensions

- Bisection $[-2, -1]$ to retain a guaranteed root bracket; after n halvings its width is $2^{-(n)}$.
- Explore Newton basins for other polynomials as a numerical-dynamics continuation; state precision and maximum iterations rather than treating a picture as proof.

Connection and its limits

Bridge type: exact-special-case

Preserves: The plotted tangent steps and rational calculations are the actual Newton algorithm on a specified cubic.

Limits: A plotted curve gives visual guidance; the exact loop is established algebraically. This is not a general convergence theorem.

Sources and inspection scope

[Yano, Penn, Konidaris and Patera, Math, Numerics, and Programming](#)

Section 29.2: Newton iteration and pathologies

Inspected: relevant exposition inspected; displayed instance independently derived

Prior use / what changes: The advisory atlas already uses this Newton counterexample. Record this as its developed facilitation card, not a newly discovered example.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-07 / Participation and launch

A cooling model that heats itself up

MSC primary 65 - Numerical analysis | Secondary 34, 80

The opening challenge

A model says the difference from room temperature falls at a rate equal to itself. Our computer approximates one step by “new difference = old difference – h times old difference.” Start at 8. Choose h from $1/2, 1, 3/2, 2, 3$. Which choices look like cooling, and which invent strange behavior?

Learner choices: Choose step sizes, decide what counts as a sensible approximation, and separate monotone decay from merely shrinking magnitude.

Prerequisite gates

Enter: F/A: signed multiplication and repeated updates.

Explore: V: plot a table at equally spaced times; compare signs and magnitudes.

Explain: A/P: classify powers of a constant multiplier.

Prove: D: relate the update to the differential equation; A suffices for recurrence stability.

Reading: A short recurrence; a partner can maintain the table.

Arithmetic: Halves, signed integers and powers.

Reasoning: Track three distinct objects: physical temperature difference, continuous model, and numerical approximation.

Hard stop: General ODE error bounds need calculus and regularity hypotheses. The recurrence analysis is a satisfying independent stop.

Materials and preparation

5 minutes estimated prep. Graph paper and two-color counters representing positive and negative differences from room temperature; calculator optional. No actual hot water is necessary.

A flexible hour

0-10 explore update cards; 10-15 identify the signed difference variable; 15-35 generate and plot sequences; 35-40 use body positions above/below a floor line for signs; 40-55 classify every positive h or compare equal elapsed time; 55-60 present a numerical failure.

Questions to explore

- Why does a negative value mean below room temperature rather than negative absolute temperature?
- Can a shrinking sequence still be a poor picture of monotone cooling?

Satisfying stop: A complete classification of one update rule and an explanation of why smaller steps can matter.

AP-07 / Facilitator reasoning

A cooling model that heats itself up

The mathematical question

How can discretizing a stable differential equation produce oscillation or explosive growth?

Forward Euler applied to $y' = -ay$ with $a > 0$ gives $y[n+1] = (1 - ah)y[n]$. The numerical magnitude decays exactly when $|1 - ah| < 1$, whereas the continuous solution $y(0)\exp(-at)$ always decays.

Worked instance

Problem: Compare $h=1/2$, $h=3/2$ and $h=3$ from $y_0=8$.

Solution: The multipliers are $1/2, -1/2, -2$. Sequences begin $8, 4, 2, 1$; $8, -4, 2, -1$; and $8, -16, 32, -64$. The first decays monotonically, the second decays while alternating, the third grows in magnitude. For $a=1$, stability is $0 < h < 2$. Only the first qualitatively matches positive monotone continuous decay.

Boundary: At $h=2$, values alternate $8, -8$ forever: bounded does not mean converging to equilibrium. At $h=1$, Euler reaches zero immediately, although the continuous solution remains positive at every finite time.

Hints to reveal in order

1. Rewrite the update as multiplication by $1-h$.
2. Classify whether the multiplier is between 0 and 1, between -1 and 0, exactly -1 , or below -1 .

Extensions

- For $a=2$, derive $0 < h < 1$ rather than copying the previous threshold.
- Compare equal elapsed times with $h=1/2$ and $h=1/4$; discuss convergence separately from stability.

Connection and its limits

Bridge type: exact-special-case

Preserves: The finite recurrence is precisely an Euler discretization of the specified scalar ODE. Comparing its multiplier with the continuous solution preserves a numerical-stability question.

Limits: No claim that every recurrence is an ODE or that stability alone guarantees small discretization error.

Sources and inspection scope

[Driscoll and Braun, Fundamentals of Numerical Computation](#)

Chapter 6 initial-value problems and chapter 11 diffusion located; scalar stability proof supplied here

Inspected: chapter scope inspected; scalar Euler stability proof supplied independently on this card

Prior use / what changes: No numerical time-stepping investigation is documented in the existing worksheets. This extends continuous mathematics beyond the existing bisection prototype.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-08 / Participation and launch

How much memory does the red-symbol machine need?

MSC primary 68 - Computer science | Secondary 03, 20

The opening challenge

Build a machine that reads R and B cards one at a time and says yes exactly when the number of R cards is a multiple of three, including zero. It may remember only which circle its token occupies. Can two circles do the job, or are three necessary?

Learner choices: Choose state labels, design transitions, invent tests that break a proposed machine, and decide how to convince another group that two states cannot suffice.

Prerequisite gates

Enter: O/C: follow one arrow per symbol and remember only the current circle.

Explore: M: count red symbols modulo three, or group counters in triples.

Explain: P: compare histories through the future strings that distinguish them.

Prove: P: contradiction using two histories assigned the same state.

Reading: Two symbol labels and arrow directions; spoken symbol streams work.

Arithmetic: Counts and remainders modulo three; no large computation.

Reasoning: The machine cannot secretly remember earlier symbols outside its state.

Hard stop: Pumping lemmas and arbitrary language proofs need quantified string reasoning; no need to introduce them before the concrete lower bound.

Materials and preparation

8 minutes estimated prep. Three floor circles or paper cards, R and B arrow cards, and a star marking the accepting state. Provide short example strings and blank cards for learner tests.

A flexible hour

0-10 play human machines on short strings; 10-15 enforce the memory rule; 15-35 build and attack two- and three-state designs; 35-40 walk the machine with spoken symbols; 40-55 develop a distinguishing-history proof; 55-60 demonstrate an accepting and rejecting example.

Questions to explore

- What should a B do to the state?
- If two different histories leave the same state, what happens when the same future cards follow both?

Satisfying stop: A working machine plus a convincing reason that fewer states cannot solve the unrestricted task.

AP-08 / Facilitator reasoning

How much memory does the red-symbol machine need?

The mathematical question

How can one prove that a finite-state machine needs at least a given number of states?

For strings over $\{R,B\}$, the language with a multiple of three R symbols has a three-state deterministic automaton. Prefixes with different R-count remainders are distinguishable by a common suffix, proving a three-state lower bound.

Worked instance

Problem: Construct a correct three-state machine and rule out every deterministic two-state machine.

Solution: States q_0, q_1, q_2 record remainder 0, 1, 2. R advances cyclically; B stays put; q_0 starts and accepts. Among empty, R, RR, a two-state machine puts some pair in the same state. Empty versus R are distinguished by empty suffix; empty versus RR likewise. R versus RR are distinguished by suffix R: RR rejects but RRR accepts. Identical states must react identically to identical suffixes, contradiction.

Boundary: If the task only allows input strings of length at most one, two states suffice. The lower bound depends on allowing every finite length.

Hints to reveal in order

1. Name states by red counts 0, 1, 2 after discarding complete triples.
2. Compare the three histories empty, R and RR. Can each pair be separated by an appropriate suffix?

Extensions

- Replace three by m and prove m states are necessary using remainder histories and suitable suffixes.
- Add a second condition, such as even B count, and investigate product states; prove necessity rather than assuming every product is minimal.

Connection and its limits

Bridge type: exact-special-case

Preserves: Floor circles are machine states, arrows are transitions, and ending at a marked circle is acceptance.

Limits: This finite-memory language is not an example of undecidability or proof of a complexity-class separation.

Sources and inspection scope

[Michael Sipser, MIT Theory of Computation](#)

Lecture 1: deterministic finite automata and acceptance

Inspected: relevant exposition inspected; displayed instance independently derived

Prior use / what changes: Weeks 3-4 use cyclic permutations and modular orbits. The new question concerns memory sufficiency and lower bounds, not another cycle table.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-09 / Participation and launch

Find the two dances hidden in coupled springs

MSC primary 70 - Mechanics of particles and systems | Secondary 15, 34

The opening challenge

Call the displacements from rest x_1 and x_2 . The model gives forces $F_1 = -2x_1 + x_2$ and $F_2 = x_1 - 2x_2$. Find starting displacement pairs for which the force pattern is a negative multiple of the starting pattern. Can you find two essentially different coordinated motions?

Learner choices: Choose displacement pairs, scale a successful pair, compare equal and opposite movements, and decide how an arbitrary position might be assembled from special patterns.

Prerequisite gates

Enter: F/V: signed displacements from two equilibrium marks and force arrows.

Explore: A: evaluate the two force formulas at chosen positions.

Explain: L: recognize proportional vectors and decompose a displacement into two modes.

Prove: D: differentiate cosine solutions twice and substitute into the ODE.

Reading: Two coordinate formulas; a diagram supports paired explanation.

Arithmetic: Signed sums, halves, and an optional square root.

Reasoning: Two masses move simultaneously; equal displacement differs from equal force.

Hard stop: The claim about temporal frequency needs the ODE and calculus. Before that, force-pattern eigenvectors are the honest mathematical result.

Materials and preparation

10 minutes estimated prep. Two movable paper mass markers on parallel number lines, three drawn springs, and arrows. A physical spring setup is optional and takes extra preparation; the paper core is sufficient.

A flexible hour

0-10 explore displacement markers and spring extension signs; 10-15 establish the force formulas; 15-35 test patterns and make a force table; 35-40 move in matched and opposite pairs; 40-55 decompose (2,0) or verify cosine motions; 55-60 share a mode and its scope.

Questions to explore

- Why does moving both masses equally leave the middle spring unchanged?
- Which pattern receives the stronger restoring force for the same displacement magnitude?

Satisfying stop: Two collective patterns and an exact decomposition of a non-collective starting shape.

AP-09 / Facilitator reasoning

Find the two dances hidden in coupled springs

The mathematical question

Why do coupled linear systems have collective modes, and how do these modes simplify motion?

Two unit masses joined to each other and fixed walls by three unit Hookean springs obey $x'' = -Kx$ with $K = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$. Eigenvectors $(1, 1)$ and $(1, -1)$ have eigenvalues 1 and 3, hence angular frequencies 1 and $\sqrt{3}$.

Worked instance

Problem: Find modes and express displacement $(2, 0)$ as their combination, with both initial velocities zero.

Solution: $K(1, 1) = (1, 1)$ and $K(1, -1) = 3(1, -1)$. Also $(2, 0) = (1, 1) + (1, -1)$. Therefore $x(t) = \cos(t)(1, 1) + \cos(\sqrt{3}t)(1, -1)$. Differentiating twice gives the prescribed restoring equation, and $x(0) = (2, 0)$, $x'(0) = 0$. Learners without calculus can stop at the exact force and decomposition statements.

Boundary: For an initial pattern $(1, 0)$, the force $(-2, 1)$ is not proportional to that pattern; the second mass cannot remain still under this model.

Hints to reveal in order

1. Try $(1, 1)$ and $(1, -1)$.
2. Add those two position patterns coordinate by coordinate.

Extensions

- Change the middle stiffness to $k > 0$; equal motion keeps eigenvalue 1, while opposite motion has eigenvalue $1 + 2k$.
- Generalize to longer chains and vibration spectra after matrix and eigenvalue prerequisites.

Connection and its limits

Bridge type: exact-special-case

Preserves: Two displacement coordinates and their restoring-force arrows realize the specified linear mechanical system exactly on paper.

Limits: Real springs and swinging demonstrations approximate the model; friction, unequal components and large deformations alter it.

Sources and inspection scope

[David Tong, Classical Dynamics](#)

Section 2.6: small oscillations and stability

Inspected: relevant exposition inspected; displayed instance independently derived

Prior use / what changes: No coupled-oscillation task is documented. This adds mechanics and linear algebra rather than relabeling a static balance puzzle.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-10 / Participation and launch

Which spring arrangement stretches farther?

MSC primary 74 - Mechanics of deformable solids | Secondary 15, 90

The opening challenge

You have two ideal springs. Spring A needs 2 force units per extension unit, and B needs 6. Predict how far a load of 6 stretches them when connected end-to-end, then when connected side-by-side between rigid bars. Defend which quantity is equal in each arrangement.

Learner choices: Choose how to draw the connections, select a test load, design a target stiffness, and decide whether a physical observation fits the ideal model.

Prerequisite gates

Enter: O/V: distinguish end-to-end and side-by-side arrangements on a diagram.

Explore: F/A: use $F=kx$ and add force or extension according to the connection.

Explain: P: identify the shared quantity before calculating.

Prove: A: eliminate internal forces or displacements from the two spring equations.

Reading: Labels k, F, x ; a facilitator can narrate loading directions.

Arithmetic: Fractions and reciprocals; small multiplication.

Reasoning: Avoid confusing stiffness with force, or total extension with each spring's extension.

Hard stop: Continuum elasticity requires strain, stress and calculus; these finite networks do not establish material failure or buckling laws.

Materials and preparation

10 minutes estimated prep. Reusable paper spring symbols, load arrows, and ruler scales. Optional identical safe elastic elements and small masses can test approximate behavior, with forces kept modest.

A flexible hour

0-10 explore connection diagrams or elastic pieces; 10-15 introduce each spring's law; 15-35 solve and compare arrangements; 35-40 trace force paths with partners; 40-55 build three-spring variants or examine measured deviations; 55-60 present a prediction with its assumptions.

Questions to explore

- Can adding a spring make the arrangement softer?
- Why is the load split in parallel but not in series?

Satisfying stop: A surprising, quantitatively checked reversal: adding a spring can make a device softer or stiffer.

AP-10 / Facilitator reasoning

Which spring arrangement stretches farther?

The mathematical question

How do local constitutive laws and connection constraints combine into an effective material response?

An ideal Hookean spring satisfies $F=kx$. Springs in parallel share extension and add forces; springs in series share force and add extensions. Thus $k_{\text{parallel}}=k_1+k_2$ and $1/k_{\text{series}}=1/k_1+1/k_2$.

Worked instance

Problem: For $k_A=2, k_B=6$ and load $F=6$, calculate total extension in both arrangements.

Solution: In series, each carries force 6, giving extensions 3 and 1, total 4. Effective stiffness is $6/4=3/2$. In parallel, both extend x ; total force is $2x+6x=8x$, so $x=3/4$ and stiffness is 8. The stiffer branch carries force $6 \times 3/4 = 9/2$, while the other carries $3/2$. Both totals check.

Boundary: Equal-force reasoning is wrong for parallel unequal springs: it would predict incompatible extensions $3/2$ and $1/2$ if each carried force 3. A rigid common bar requires equal extension.

Hints to reveal in order

1. Draw each spring as a separate object and label its two endpoint forces.
2. For series, add extensions. For parallel, add forces at the common moving bar.

Extensions

- Use two identical springs of stiffness k to get $k/2$ in series and $2k$ in parallel, explaining the factor without fractions first.
- Build series-parallel networks for a chosen target stiffness; non-series-parallel networks lead to linear systems and energy minimization.

Connection and its limits

Bridge type: exact-special-case

Preserves: The spring diagrams are finite elastic systems with specified force and displacement compatibility, not arbitrary weighted graphs.

Limits: The Hookean relation is an idealization. A rubber band's response must be measured rather than assumed linear.

Sources and inspection scope

[Allan Bower, Applied Mechanics of Solids](#)

Section 3.2: linear elastic behavior; spring-network equations derived explicitly here

Inspected: relevant exposition inspected; displayed instance independently derived

Prior use / what changes: No prior elastic-network task is documented; mathematical similarity to resistor networks will be cross-linked rather than counted as unrelated content.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-11 / Participation and launch

The narrow channel and the impossible flow map

MSC primary 76 - Fluid mechanics | Secondary 35, 65

The opening challenge

A tube carries incompressible water steadily with no leaks. At its wide section the area is 6 square units and average forward speed is 2 units per second. At the narrow section the area is 3. What average speed is possible? Invent a flow map that looks plausible but cannot stay steady.

Learner choices: Choose areas, propose speeds, decide where extra water would accumulate in a bad map, and add a branch or storage reservoir deliberately.

Prerequisite gates

Enter: O/C/V: compare cross-section sizes and count volume packets per time interval.

Explore: F: multiply area by speed and divide a fixed flux by area.

Explain: P: use a control-volume storage ledger to reject unequal in/out flow.

Prove: D/I: integrate the continuity equation or use the divergence theorem.

Reading: A few area and speed labels; arrows and oral rules suffice.

Arithmetic: Multiplication and division of small positive values; units.

Reasoning: Distinguish speed, volume rate, and water already stored between sections.

Hard stop: Pressure differences require momentum/energy laws and assumptions; turbulence and viscosity cannot be deduced from this ledger.

Materials and preparation

7 minutes estimated prep. Draw a wide-to-narrow tube with two cross-sections; use equal volume blocks and a one-second time card. Optional water transfer uses measuring cups, not a pressure demonstration.

A flexible hour

0-10 move equal-volume blocks between regions; 10-15 specify steady, incompressible and no-leak rules; 15-35 design and audit section labels; 35-40 act as equal-volume parcels crossing gates; 40-55 add branches or a changing reservoir; 55-60 explain one impossible flow map.

Questions to explore

- Why does a narrower channel require faster average flow under these assumptions?
- What would unequal inflow and outflow tell you if the region could store water?

Satisfying stop: Reject an impossible continuous flow map using a clear volume-accounting argument.

AP-11 / Facilitator reasoning

The narrow channel and the impossible flow map

The mathematical question

How does local conservation constrain a continuous flow, and how can a proposed flow field be rejected before solving its dynamics?

For steady incompressible flow through a stream tube with impermeable walls, volume flux $Q = \int (\mathbf{u} \cdot \mathbf{n} \, dA)$ is constant between cross-sections. If speed is uniform across a section, $Q = Av$; otherwise v denotes the section's mean normal speed.

Worked instance

Problem: Find narrow-section speed, then audit a proposed exit speed 3.

Solution: The incoming flux is $6 \times 2 = 12$ volume units per second. Conservation requires $3v = 12$, giving $v = 4$. If the proposed speed were 3, outgoing flux would be 9 and the intervening region would gain 3 volume units per second. That contradicts steady flow in a rigid full tube with no leaks and incompressible material.

Boundary: For a reservoir whose level rises, unequal inflow and outflow is allowed: storage changes. For compressible gas, conserve mass flux ρAv rather than volume flux Av .

Hints to reveal in order

1. Ask how much volume crosses each section in one second.
2. Inventory the water between the sections before and after that second.

Extensions

- Split flux 12 between two branches with areas 2 and 1; explore the infinitely many nonnegative flow allocations before additional physics chooses one.
- With calculus, derive the integral continuity law; with mechanics, add a justified Bernoulli relation and compare its stronger assumptions.

Connection and its limits

Bridge type: exact-special-case

Preserves: Area-and-mean-speed sections describe an ideal continuous incompressible stream tube. Conservation relates their fluxes directly.

Limits: Counters can represent fixed volume parcels but do not prove continuum dynamics. No pressure claim is inferred from continuity alone.

Sources and inspection scope

[David Tong, Fluid Mechanics](#)

Continuity equation and narrowing-pipe Bernoulli example, PDF pp. 23-24

Inspected: relevant exposition inspected; exact model calculation supplied here

Prior use / what changes: Week 10 route networks concern traversal; this question concerns conservation of flowing material and cannot be counted as another route puzzle.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-12 / Participation and launch

Where should the fastest path cross the shoreline?

MSC primary 78 - Optics, electromagnetic theory | Secondary 49, 26

The opening challenge

Travel from $A=(0,4)$ to $B=(7,-3)$, crossing the shoreline $y=0$ once. Above it speed is 3; below it speed is 4. Each segment is straight. Choose the crossing point $(x,0)$ to minimize total time. Must the fastest route be the shortest route?

Learner choices: Choose trial crossing points, organize numerical or geometric comparisons, and decide what extra reasoning certifies the best continuous choice.

Prerequisite gates

Enter: V/A: coordinates, Pythagoras and distance divided by speed.

Explore: F: compute and compare square-root travel times or measure scaled segments.

Explain: D: differentiate the time function and identify its zero derivative.

Prove: D/P: positive second derivative proves the stationary point is the unique global minimizer.

Reading: A diagram with speed labels and the time formula.

Arithmetic: Square roots and fractions; exact 3-4-5 triangles make the worked optimum accessible.

Reasoning: Distance and time are competing quantities; angles must be measured from a stated reference.

Hard stop: Without calculus, comparison gives evidence and an exact candidate, not a proof over all continuous crossing points.

Materials and preparation

10 minutes estimated prep. Coordinate paper, ruler, two colored half-planes separated by $y=0$, and a sliding crossing marker. A calculator is optional for exploratory comparisons.

A flexible hour

0-10 explore paths and speed labels; 10-15 define total time; 15-35 compare candidate crossings; 35-40 trace two routes with different walking tempos; 40-55 differentiate or verify the proposed optimum; 55-60 contrast an experimental minimum with a proof.

Questions to explore

- Which side should gain distance when its speed is greater?
- Does a zero derivative alone prove the crossing is globally best?

Satisfying stop: A concrete optimum whose geometry, time and proof all agree.

AP-12 / Facilitator reasoning

Where should the fastest path cross the shoreline?

The mathematical question

Why does a ray bend at a boundary between media, and how does minimizing travel time differ from minimizing distance?

In two homogeneous isotropic media with positive constant speeds, a path made of straight segments meeting a flat interface has travel time $T(x)$. Minimizing T yields the Snell relation $\sin(\theta_1)/v_1 = \sin(\theta_2)/v_2$, with angles measured from the normal.

Worked instance

Problem: Prove $x=3$ uniquely minimizes travel time over all real crossing positions.

Solution: $T'(x) = x/(3\sqrt{x^2+16}) - (7-x)/(4\sqrt{(7-x)^2+9})$. At $x=3$, both terms equal $1/5$. Also $T''(x) = 16/[3(x^2+16)^{3/2}] + 9/[4((7-x)^2+9)^{3/2}]$ is positive everywhere. Thus T is strictly convex and $x=3$ is its unique global minimizer, with time $5/3 + 5/4 = 35/12$. The direct straight route crosses at $x=4$, so shortest and fastest differ.

Boundary: If both speeds are equal, minimizing time is minimizing distance and the straight path is optimal. Changing the interface shape changes the parameterization.

Hints to reveal in order

1. Write $T(x) = \sqrt{x^2+16}/3 + \sqrt{(7-x)^2+9}/4$.
2. At $x=3$ both segment lengths are 5, making the derivative simple.

Extensions

- Vary the lower speed and predict the optimal crossing direction before calculating.
- Derive Snell's relation by identifying each derivative ratio as a sine; total internal reflection is a separate wave-boundary question, not established here.

Connection and its limits

Bridge type: exact-special-case

Preserves: The adjustable crossing point parameterizes actual geometric-optics travel paths in the two-speed idealization.

Limits: This variational calculation assumes the ray model; it does not derive Maxwell's equations or wave diffraction.

Sources and inspection scope

[David Tong, Electromagnetism](#)

Wave and boundary framework inspected; Fermat optimization for this geometry proved self-contained below

Inspected: relevant exposition inspected; exact model calculation supplied here

Prior use / what changes: Week 9 billiards uses reflections and unfolding. Refraction with unequal speeds adds a genuinely different continuous optimization mechanism.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-13 / Participation and launch

Add a wave and make the signal smaller

MSC primary 78 - Optics, electromagnetic theory | Secondary 42

The opening challenge

At one sensor, wave A has height $2 \sin(t)$. You may add wave B with height $\sin(t)$ or $-\sin(t)$. Add their heights at each time before comparing the resulting signal. Can adding a wave make the sensor's average squared reading smaller?

Learner choices: Choose relative phase, decide which sample times expose cancellation, and invent amplitude pairs that silence the sensor entirely.

Prerequisite gates

Enter: V/F: compare heights above and below an axis and add signed heights.

Explore: A: scale a sine-shaped template and shift it by half a period.

Explain: P: show pointwise reinforcement or cancellation.

Prove: Trigonometry and I: derive the time-average formula for an arbitrary phase.

Reading: Wave labels can be pictures; full formula is an optional symbolic extension.

Arithmetic: Signed sums and squares; fractions for averages.

Reasoning: Do not add intensities before adding coherent amplitudes; distinguish a single location from the whole field.

Hard stop: Without phase and coherence assumptions, the interference formula is not justified. General electromagnetic energy conservation needs more physics.

Materials and preparation

15 minutes estimated prep. Transparent sine-wave strips with amplitude 2 and amplitude 1, a shared horizontal time axis, and blank sum graphs. Preprint half-period alignment marks.

A flexible hour

0-10 overlay and slide wave strips; 10-15 define pointwise addition; 15-35 draw sums and compare squared amplitudes; 35-40 partners move in phase and opposite phase; 40-55 explore unequal amplitudes or intermediate phase; 55-60 explain why the word add can mislead.

Questions to explore

- Why is the sum of two nonzero waves sometimes zero at every time?
- What assumption would fail if the second wave's phase drifted unpredictably?

Satisfying stop: A surprising reduction verified by both a waveform drawing and an exact calculation.

AP-13 / Facilitator reasoning

Add a wave and make the signal smaller

The mathematical question

Why do coherent signals combine at the amplitude level, and how can adding a signal reduce local intensity?

For same-frequency coherent scalar waves at a fixed location, amplitudes add linearly. If $E(t) = A \sin(t) + B \sin(t+\phi)$, the normalized time-averaged intensity $2\langle E^2 \rangle$ is $A^2 + B^2 + 2AB \cos\phi$.

Worked instance

Problem: Compare normalized intensity before adding B, with B in phase, and with B opposite in phase.

Solution: Wave A alone has amplitude 2 and normalized intensity 4. In phase, $E = 3 \sin(t)$, giving 9. Opposite in phase, $E = \sin(t)$, giving 1. Adding the same amplitude-1 wave therefore either increases or decreases the sensor's intensity. With $A=B=1$ and opposite phase, the local field vanishes. These follow pointwise before any averaging.

Boundary: Independent drifting phases can average the cross term to zero, giving intensity $A^2 + B^2$ instead. Local cancellation does not mean energy vanished everywhere; source work and the spatial field require a larger model.

Hints to reveal in order

1. Start with the same phase and then turn one strip upside down.
2. For $C \sin(t)$, mean square is $C^2/2$ over a full period; use 2 times that average as normalized intensity.

Extensions

- Use a quarter-period shift and prove the normalized intensity is $A^2 + B^2$ for that coherent special phase.
- Generalize to rotating phasor vectors and Fourier superposition; unequal frequencies require a specified averaging interval.

Connection and its limits

Bridge type: exact-special-case

Preserves: Signed waveform strips and their pointwise sums are exact scalar coherent-wave calculations.

Limits: The scalar model fixes location, frequency and polarization. It does not describe every source, global energy flow, or incoherent light.

Sources and inspection scope

[David Tong, Electromagnetism](#)

Section 4.3: plane waves, polarization and superposition, PDF pp. 20-22

Inspected: relevant exposition inspected; exact model calculation supplied here

Prior use / what changes: No wave-superposition task is documented. Pattern repetition alone would not count as an equivalent investigation.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-14 / Participation and launch

Can the heat-engine sales pitch be true?

MSC primary 80 - Classical thermodynamics, heat transfer | Secondary 49

The opening challenge

Every engine must return to its starting internal state. It takes 12 heat units from a 600 K reservoir, sends Q_c heat units to a 300 K reservoir, and outputs W work units. Design the biggest work claim that passes both $W+Q_c=12$ and $Q_c/300-12/600\geq 0$.

Learner choices: Choose heat/work splits, test a rival sales pitch, and compare changing the hot temperature with changing the cold temperature.

Prerequisite gates

Enter: C/F: account for incoming heat, outgoing heat and work.

Explore: F/A: divide heat quantities by absolute temperature.

Explain: P: show why an energy-balanced proposal can still violate the entropy condition.

Prove: A: rearrange the two constraints; deriving the second law itself is a separate physical foundation.

Reading: Short proposal cards and a three-column ledger.

Arithmetic: Small subtraction and simple fractions; ratios use kelvin.

Reasoning: Distinguish a whole cycle from an intermediate state and internal changes from reservoir changes.

Hard stop: Entropy cannot be replaced with an undefined messiness metaphor. Advanced cycle analysis requires thermodynamic state variables and calculus.

Materials and preparation

8 minutes estimated prep. Two reservoir cards labeled 600 K and 300 K, twelve heat tokens, work tokens and proposal sheets. This is a paper model; no hot materials or actual engine are needed.

A flexible hour

0-10 explore token energy ledgers; 10-15 add the entropy constraint and absolute temperatures; 15-35 design, test and sort proposals; 35-40 physically transfer tokens between labeled stations; 40-55 change reservoir temperatures or cascade engines; 55-60 state the strongest valid bound.

Questions to explore

- Why is energy conservation insufficient to rule out converting all incoming heat to work?
- What happens to the maximum work when the two reservoir temperatures approach each other?

Satisfying stop: Reject a tempting perpetual-motion-style proposal with an exact second constraint.

AP-14 / Facilitator reasoning

Can the heat-engine sales pitch be true?

The mathematical question

How do conservation and entropy constrain possible transformations more strongly than either condition alone?

For a cyclic engine exchanging heat Q_h with a hot reservoir T_h and rejecting Q_c to a cold reservoir T_c , $W = Q_h - Q_c$. The reservoir entropy change is $Q_c/T_c - Q_h/T_h \geq 0$; hence $W \leq Q_h(1 - T_c/T_h)$, with equality for a reversible ideal engine.

Worked instance

Problem: Evaluate proposals $(W, Q_c) = (8, 4), (6, 6), (4, 8)$.

Solution: All conserve energy. Entropy changes are $4/300 - 12/600 = -1/150$, 0, and $1/150$. Thus the first violates the second-law condition, the second reaches the reversible bound, and the third passes with positive entropy production. The maximum permitted work is 6. Passing the two tests is necessary within this model, not a guarantee that a proposed machine exists.

Boundary: Using the corresponding Celsius temperatures in the efficiency ratio would be invalid: ratios require absolute temperature. A third reservoir, net internal-state change or external work input also changes the ledger.

Hints to reveal in order

1. First eliminate one variable using energy balance.
2. The entropy inequality gives a minimum rejected heat, which gives a maximum work.

Extensions

- With $T_h = 600$ K and $T_c = 400$ K, the same heat budget permits at most 4 work units.
- Cascade reversible engines through an intermediate temperature and show the combined reversible efficiency still equals $1 - T_c/T_h$.

Connection and its limits

Bridge type: exact-special-case

Preserves: Heat and work ledgers with two fixed absolute-temperature reservoirs are the exact algebraic constraints of an ideal thermodynamic cycle.

Limits: Passing these necessary tests is not a construction of a real engine. The model omits finite-rate losses and engineering details.

Sources and inspection scope

[David Tong, Statistical Physics](#)

Chapter 4: heat engines, entropy and Carnot efficiency

Inspected: relevant exposition inspected; exact model calculation supplied here

Prior use / what changes: No thermodynamic-cycle investigation is documented. This deliberately remains an algebraic thermodynamics entry rather than a generic sharing game.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-15 / Participation and launch

Does the order of two quantum questions matter?

MSC primary 81 - Quantum theory | Secondary 15, 60

The opening challenge

Start in state e_0 . A Z question measures in $\{e_0, e_1\}$; an X question measures in $\{+, -\}$. Use squared dot products for probabilities and replace the state by the outcome card. Compare X then Z with Z then X. Record the Z answer wherever it occurs in each order. How often is that answer e_1 ?

Learner choices: Choose measurement sequences, organize intermediate branches, and search for a sequence that changes a formerly certain answer.

Prerequisite gates

Enter: V/F/L: vectors $(1,0), (0,1)$ and diagonal unit vectors, dot products, squaring and probability trees.

Explore: L: distinguish basis coordinates from a physical state and apply the stated update.

Explain: P: sum probabilities over intermediate outcomes.

Prove: L: compute noncommuting projectors and their sequential measurement probabilities.

Reading: Symbolic rules; a facilitator may read the rule aloud and demonstrate one transition before learners construct the sequence trees.

Arithmetic: Halves, quarters and $1/\sqrt{2}$; calculator unnecessary with supplied dot products.

Reasoning: Separate unknown outcome from state update; discard unobserved intermediate labels only after summing their branches.

Hard stop: Do not remove the Born rule or projection update and still call the resulting coin game quantum measurement.

Materials and preparation

12 minutes estimated prep. Four state-arrow cards $e_0=(1,0)$, $e_1=(0,1)$, $+=(1,1)/\sqrt{2}$, $-=(1,-1)/\sqrt{2}$; branch-tree sheets and counters. Supply a dot-product table if needed.

A flexible hour

0-10 explore unit-vector projections; 10-15 establish probabilities and update separately; 15-35 build exact sequence trees; 35-40 exchange state keeper and probability recorder; 40-55 compare X-Z and Z-X or repeated measurements; 55-60 explain the order effect with a tree.

Questions to explore

- Why does asking Z twice immediately give the same second answer?
- Why can an unreported X result still change the later Z statistics?

Satisfying stop: An exact probability-tree comparison showing why a prior measurement changes a later question.

AP-15 / Facilitator reasoning

Does the order of two quantum questions matter?

The mathematical question

How can two projective measurements fail to commute, even when each has only two possible answers?

In a real two-component restriction of a two-dimensional quantum model, a normalized state ψ measured in an orthonormal basis yields outcome vector b with probability $|b \cdot \psi|^2$ and updates to b . Measurement order can change the recorded joint outcomes. General complex states require the complex inner product.

Worked instance

Problem: Compare the recorded Z outcome in $X \rightarrow Z$ and $Z \rightarrow X$; also check Z alone and $Z \rightarrow Z$.

Solution: In $X \rightarrow Z$, X gives $+$ or $-$ with probability $1/2$ each. Each branch then yields $Z=e1$ with probability $1/2$, totaling $1/4+1/4=1/2$. In $Z \rightarrow X$, the first Z returns $e0$ with certainty, so its recorded $e1$ probability is zero. Z alone and $Z \rightarrow Z$ also never yield $e1$. The claim concerns recorded outcome statistics: the final unconditional mixed state is the same for these particular two mutually unbiased dephasing orders, so no difference between those final mixed states is asserted.

Boundary: If the two measurements use the same basis, repeated measurement does not create the order effect. Ignoring the state update incorrectly predicts the original statistics throughout.

Hints to reveal in order

1. For $e0$ measured in X , both squared dot products equal $1/2$.
2. From either diagonal state, a Z measurement gives $e0$ and $e1$ with equal probability.

Extensions

- Using matrices $P0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $P+ = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, verify $P0P+$ differs from $P+P0$.
- Allow a continuously rotated real basis and derive $\sin^2\theta/\cos^2\theta$ probabilities; complex phases and entanglement are separate later branches.

Connection and its limits

Bridge type: exact-special-case

Preserves: Real normalized two-component states and projectors form a genuine restricted quantum model; the activity uses its exact probability/update rules.

Limits: Ordinary arrows are representations of amplitudes, not hidden particle directions. General states have complex amplitudes, and this card does not establish an interpretation of measurement.

Sources and inspection scope

[David Tong, Quantum Mechanics](#)

Sections 3.1-3.4: states, measurement, operators and commutation

Inspected: relevant exposition inspected; exact model calculation supplied here

Prior use / what changes: No finite quantum-measurement family is documented. Classical polarizer experiments may motivate it but are not substitutes for these stated quantum rules.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-16 / Participation and launch

Count the tiny magnet before running it

MSC primary 82 - Statistical mechanics, structure of matter | Secondary 05, 60

The opening challenge

Make every red/blue arrangement of four counters in a row. Count its agreeing neighboring pairs. Its raffle weight is 2 raised to that count: 1, 2, 4 or 8. Which total weight belongs to perfectly aligned arrangements, and which belongs to the more numerous partly aligned arrangements?

Learner choices: Choose an enumeration method, predict the largest weight class, and compare uniform sampling with the interaction-weighted model.

Prerequisite gates

Enter: O/C: flip four counters and count agreeing neighbors.

Explore: M: organize configurations and attach powers of two as weights.

Explain: F/P: distinguish counts from weighted probabilities.

Prove: P: encode a configuration by its first spin and three agree/disagree choices.

Reading: Color symbols and a short weight rule; oral explanation works.

Arithmetic: Counts to 16, weights to 8, totals to 54; probability fractions optional.

Reasoning: Configurations with equal energy share weights, but groups contain unequal numbers of configurations.

Hard stop: Infinite-volume phase transitions and criticality require limits and additional analysis; no claim that this toy has a critical temperature.

Materials and preparation

8 minutes estimated prep. Four reversible red/blue counters per group, a strip showing exactly three neighbor links, and a table for agreement count 0, 1, 2, 3. Do not wrap the chain into a ring.

A flexible hour

0-10 play with spin patterns; 10-15 establish open ends and the weight rule; 15-35 catalog all sixteen configurations; 35-40 act as four neighbors agreeing or disagreeing; 40-55 calculate class probabilities or change the weight factor; 55-60 explain energy-versus-multiplicity competition.

Questions to explore

- Can a group of individually less-favored arrangements outweigh the two best-favored arrangements?
- Why does changing the first spin not change the agreement pattern?

Satisfying stop: A complete sixteen-state ensemble whose most numerous and most favored classes differ.

AP-16 / Facilitator reasoning

Count the tiny magnet before running it

The mathematical question

How can energetic preference compete with the number of available microscopic arrangements?

A four-spin open Ising chain has spins ± 1 and three adjacent interactions. Assign each configuration weight 2^a , where a is its number of agreeing adjacent pairs. This is a canonical Ising distribution at a specified positive inverse-temperature coupling, up to a common factor.

Worked instance

Problem: Find class multiplicities, total weight and probability of complete alignment.

Solution: For $a=0,1,2,3$ the multiplicities are 2,6,6,2. Their total weights are 2,12,24,16, summing to 54. Complete alignment therefore has probability $16/54=8/27$. The $a=2$ class has greater total weight, $24/54$, despite each member having weight 4 rather than 8. Under uniform sampling alignment has probability $2/16=1/8$.

Boundary: Closing the row into a ring adds an interaction. The three independent edge-choice count no longer applies; a ring must have an even number of disagreements.

Hints to reveal in order

1. Choose the first color, then choose agree or disagree independently at each of the three links.
2. For a fixed agreement count a , there are 2 times “three choose a ” configurations.

Extensions

- Replace factor 2 by $b>0$ and derive total weight $2(1+b)^3$; alignment probability becomes $b^3/(1+b)^3$.
- Compare longer open chains with rings, then distinguish finite partition functions from infinite-system phase transitions.

Connection and its limits

Bridge type: exact-special-case

Preserves: Four two-sided counters are literal spin configurations; adjacent-pair weights define a genuine finite statistical-mechanics ensemble.

Limits: Sixteen states cannot exhibit a true thermodynamic-limit phase-transition singularity. The weights are a stipulated model, not measured magnetic behavior.

Sources and inspection scope

[David Tong. Statistical Physics](#)

Chapters 1 and 5: canonical ensembles and Ising models; four-spin enumeration supplied here

Inspected: canonical-ensemble exposition and Ising chapter scope inspected; finite chain independently enumerated

Prior use / what changes: No statistical-ensemble activity is documented. Coin colors alone are not a new family; the interaction weights and multiplicity question define this one.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-17 / Participation and launch

Two clocks, one reunion, different elapsed times

MSC primary 83 - Relativity and gravitational theory | Secondary 53, 26

The opening challenge

Both clocks leave event $(t,x)=(0,0)$ and reunite at $(10,0)$. Clock A stays at $x=0$. Clock B travels to $(5,3)$, then returns to $(10,0)$. For each straight segment add $\sqrt{(\Delta t)^2 - (\Delta x)^2}$ to that clock. Which clock records more time, and can a different legal turn point change the gap?

Learner choices: Choose turn positions with speed below 1, compare symmetric routes, and decide which features of the drawing affect each clock.

Prerequisite gates

Enter: V/A: plot time and position, interpret speed and use the given interval rule.

Explore: F: calculate squares and square roots using 3-4-5 triangles.

Explain: P: distinguish coordinate duration from each clock's proper duration.

Prove: A/P: compare piecewise constant-speed routes; D/I for arbitrary smooth paths.

Reading: Symbolic interval rule and axis labels.

Arithmetic: Squares of 3,4,5 and addition; units chosen so $c=1$.

Reasoning: The vertical axis is time, not a second space direction. A geometrically longer drawing need not mean more clock time.

Hard stop: General curved spacetime requires a metric and differential geometry; rubber sheets are not a substitute.

Materials and preparation

8 minutes estimated prep. Graph paper with vertical t-axis and horizontal x-axis, rulers, colored worldline segments, and two clock counters.

A flexible hour

0-10 explore legal and too-fast segments; 10-15 introduce the interval rule; 15-35 compute several reunion paths; 35-40 move markers along equal coordinate-time slices; 40-55 prove the symmetric comparison or vary reunion duration; 55-60 explain the result without Euclidean-length language.

Questions to explore

- Why must $|\Delta x|$ be smaller than Δt for a massive clock segment?
- What changes if B turns around at $x=0$ instead?

Satisfying stop: An exact 10-versus-8 comparison and a clear statement of the model's causal constraint.

AP-17 / Facilitator reasoning

Two clocks, one reunion, different elapsed times

The mathematical question

Why is elapsed proper time path dependent in spacetime, even between the same departure and reunion events?

In flat 1+1 spacetime with $c=1$, an inertial timelike segment has proper duration $\sqrt{(\Delta t)^2 - (\Delta x)^2}$. Proper durations add along a piecewise inertial path. This Lorentzian rule differs from Euclidean path length.

Worked instance

Problem: Calculate both proper times and prove every symmetric nonzero excursion with $|d| < 5$ gives B less than 10.

Solution: A records $\sqrt{(100)}=10$. Each B leg records $\sqrt{(25-9)}=4$, for total 8. For a general symmetric excursion to d , total is $2\sqrt{(25-d^2)}$, which is less than 10 whenever $0 < |d| < 5$. Equality occurs at $d=0$. No numerical approximation is required for the worked route.

Boundary: At $|d|=5$ the legs are lightlike and proper duration is zero; that is not a possible worldline for the massive clock assumed here. At $|d| > 5$ the square root is not real because the proposed segment is spacelike.

Hints to reveal in order

1. Compute each leg separately; both B legs have $\Delta t=5$ and $|\Delta x|=3$.
2. Compare $\sqrt{(25-d^2)}$ with 5 for a symmetric excursion to distance d .

Extensions

- Fix the excursion distance and change turnaround coordinate time; investigate why constant-speed segments maximize each leg's proper time under its endpoints.
- With calculus, evaluate proper-time integrals on smooth subluminal curves and examine how an actual turnaround approximates the ideal diagram.

Connection and its limits

Bridge type: exact-special-case

Preserves: Coordinate events and piecewise inertial worldlines are exact special-relativistic objects under the stated metric.

Limits: The diagram is not a physical scale model; an instantaneous turnaround is an idealization. The activity does not solve Einstein's gravitational field equations.

Sources and inspection scope

[David Tong, General Relativity](#)

Section 1.2.1, particle in Minkowski spacetime; proper time equation (1.16)

Inspected: author-hosted proper-time exposition inspected; finite piecewise inertial calculation independently checked

Prior use / what changes: No spacetime activity is documented. Ordinary shortest-path investigations do not cover the Lorentzian sign change or proper time.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-18 / Participation and launch

Weigh an unseen center from its orbiters

MSC primary 85 - Astronomy and astrophysics | Secondary 70, 62

The opening challenge

An unseen object is orbited by three tiny test particles. Assume their orbits are circular, all motion is visible in the page, gravity is inverse-square, and the central object dominates. With $G=1$, infer its mass from radius/speed pairs (1,6),(4,3),(9,2). Then decide whether the new card (4,4) fits the same story.

Learner choices: Choose which observation to use first, predict an unmeasured speed, and propose reasons why inconsistent data might arise rather than forcing agreement.

Prerequisite gates

Enter: V/F/A: radius, speed, square and the supplied circular-balance formula.

Explore: A: infer one common mass from several orbiters and test consistency.

Explain: P: show a model predicts a specific speed-radius relation.

Prove: D/V: derive centripetal acceleration and the Newtonian circular formula.

Reading: A small data table and clearly labeled assumptions.

Arithmetic: Small squares, products and square roots.

Reasoning: Distinguish fitting a parameter under a model from proving the model is true.

Hard stop: Noncircular orbits, projected velocities and extended mass distributions require new equations; do not silently apply the point-mass formula.

Materials and preparation

8 minutes estimated prep. Orbit-circle drawings at radii 1,4,9, colored orbit markers, and cards giving speeds 6,3,2 in units with $G=1$. Provide an alternative deliberately inconsistent data card.

A flexible hour

0-10 inspect orbit diagrams and what a measurement means; 10-15 state the force-balance model; 15-35 infer masses and predict missing data; 35-40 trace fast inner and slow outer orbits; 40-55 audit inconsistent or projected data; 55-60 distinguish conditional inference from discovery of a law.

Questions to explore

- Why does a slower outer particle not necessarily imply a smaller central mass?
- What information would you seek before blaming an inconsistent observation on arithmetic?

Satisfying stop: An unseen parameter inferred three ways and an inconsistency recognized honestly.

AP-18 / Facilitator reasoning

Weigh an unseen center from its orbiters

The mathematical question

How can observable orbital motion reveal an unobserved mass, and how can inconsistent estimates expose a failed model?

For a negligible-mass test particle in a circular Newtonian orbit around a dominant point mass, $v^2/r = GM/r^2$, hence $M = v^2 r / G$. The inference relies on circularity, the true three-dimensional speed and radius, and the assumed force law.

Worked instance

Problem: Find the common central mass and test the extra data card.

Solution: The first three estimates are $6^2 \times 1 = 36$, $3^2 \times 4 = 36$, $2^2 \times 9 = 36$. The new card gives $4^2 \times 4 = 64$ and is incompatible with the same exact point-mass circular model. The predicted speed at $r=16$ is $\sqrt{(36/16)} = 3/2$. The inconsistency identifies a failed joint set of assumptions or data, not which component failed.

Boundary: A measured projected speed can underestimate true speed. At a turning point of an eccentric orbit, $v^2 r$ need not equal GM , despite the same inverse-square force.

Hints to reveal in order

1. Compute $v^2 r$ for each observation.
2. For a fixed inferred mass, solve $v = \sqrt{(M/r)}$ to make a prediction.

Extensions

- Use small intervals for observed radii and speeds to obtain a mass interval and test overlap between orbiters.
- Investigate flat galaxy rotation curves: under spherical symmetry, enclosed mass scales as $v^2 r / G$, but this is a changed distributed-mass model, not the original point mass.

Connection and its limits

Bridge type: exact-special-case

Preserves: Radius/speed data cards are exact observations of a specified circular-orbit model; learners perform a genuine inverse parameter inference.

Limits: The supplied observations are invented and exact. Real astronomy requires projection, uncertainty and distributed-mass modeling.

Sources and inspection scope

[MIT, Astrophysics II lecture notes](#)

PDF pp. 15-16: dynamical mass inference and virial assumptions; circular special case derived from force balance here

Inspected: virial and dynamical-inference exposition inspected; circular-orbit formula independently derived from stated force balance

Prior use / what changes: No astronomical inverse problem is documented. A drawing of ellipses alone would not preserve this inference mechanism.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-19 / Participation and launch

Can travel times reveal the water depth?

MSC primary 86 - Geophysics | Secondary 35, 65

The opening challenge

A long, small wave crosses two 6-unit segments. In this ideal model its speed in depth H is $\sqrt{H}$. The first segment has depth 4. The total travel time is 5. Infer the second depth. Then hide both depths: does the same total time still determine them?

Learner choices: Choose candidate depths, decide which extra measurement would resolve ambiguity, and compare forward prediction with inverse reconstruction.

Prerequisite gates

Enter: V/F/A: distance, speed, time and square roots.

Explore: A: infer one unknown depth from a measured total time.

Explain: P: distinguish one unknown from two and show nonuniqueness by two witnesses.

Prove: D: derive wave speed from the linear shallow-water equations; A suffices for the finite inverse calculation.

Reading: A labeled cross-section and three numerical data cards.

Arithmetic: Small divisions, squares and fractions.

Reasoning: Wave propagation is not the same as the journey of a particular water parcel; keep model and observation separate.

Hard stop: Continuous bathymetric inversion and real seismic/tidal data require more wave physics and uncertainty modeling.

Materials and preparation

7 minutes estimated prep. A two-segment channel sketch, two movable depth labels, travel-time strips and a calculator optional. Use units with $g=1$ explicitly; no real water is needed.

A flexible hour

0-10 explore speed-versus-depth cards; 10-15 state long-wave and piecewise-depth assumptions; 15-35 infer and check the hidden depth; 35-40 move wave markers using travel-time strips; 40-55 construct two indistinguishable depth pairs; 55-60 explain what the data determine.

Questions to explore

- Why can deeper water produce faster wave travel in this model?
- How would measuring the arrival time at the interface change the inverse problem?

Satisfying stop: A hidden depth found and a clear counterexample to overinterpreting one measurement.

AP-19 / Facilitator reasoning

Can travel times reveal the water depth?

The mathematical question

How can wave travel times infer hidden geometry, and which assumptions make that inverse problem identifiable?

Small-amplitude long gravity waves in nonrotating shallow water of constant depth H have speed $c=\sqrt{gH}$. For one-dimensional travel through specified constant-depth segments, total ray travel time is the sum of segment length divided by local speed.

Worked instance

Problem: Recover the second depth, then exhibit nonuniqueness when both depths are unknown.

Solution: The first time is 3, leaving 2 for the second segment. Thus its speed is $6/2=3$ and its depth is 9. If both depths are unknown, $(H_1, H_2)=(4, 9)$ and $(9, 4)$ both give total time 5. A genuinely different pair is $(144/25, 144/25)$: each speed is $12/5$ and each time $5/2$, again total 5.

Boundary: A total time of 3 or less is impossible with first depth 4 and a finite positive second depth: the first segment alone takes 3 and the second adds a positive time.

Hints to reveal in order

1. The known first segment takes $6/\sqrt{4}$ time units.
2. Subtract that time before solving for the remaining speed; square only after finding speed.

Extensions

- Measure one intermediate arrival time and show how both depths become individually recoverable.
- From $\eta_t + H u_x = 0$ and $u_t + g\eta_x = 0$ derive $\eta_{tt} = gH\eta_{xx}$, then verify traveling waves $f(x-ct)$ with $c^2 = gH$.

Connection and its limits

Bridge type: exact-special-case

Preserves: Travel-time equations are an exact inverse problem inside the stated piecewise homogeneous long-wave approximation.

Limits: Interface reflections, rapidly varying-depth corrections, rotation and finite-amplitude effects are omitted. A real tsunami is not established by this two-segment calculation.

Sources and inspection scope

[Geoffrey Vallis, Atmospheric and Oceanic Fluid Dynamics](#)

Section 7.1, equations (7.18)–(7.23), PDF pp. 75–76: shallow-water limit and parcel motion

Inspected: relevant exposition or named scope inspected; finite calculation self-contained below

Prior use / what changes: No geophysical inverse-data family is documented. Ordinary path-length problems omit the depth-dependent wave model and identifiability issue.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-20 / Participation and launch

A price certificate for the best factory plan

MSC primary 90 - Operations research, mathematical programming | Secondary 15, 05

The opening challenge

Make products without reusing ingredients. A sells for 3 and uses two red plus one blue; B sells for 4 and uses one red plus two blue. You have five red and four blue. Find the most revenue, then price the raw ingredients so your total ingredient budget proves nobody can earn more.

Learner choices: Choose product mixes, organize feasible cases, propose resource prices, and improve a loose upper bound until it matches a plan.

Prerequisite gates

Enter: O/C: make two kinds of bundles from limited colored counters.

Explore: C: enumerate integer plans and total revenue.

Explain: F/P: assign fractional resource prices and show a universal bound.

Prove: A: multiply resource inequalities by nonnegative prices and add them.

Reading: Two recipe cards and values; oral description and physical bundles work.

Arithmetic: Small counts initially; thirds for the sharp certificate.

Reasoning: Distinguish finding a good plan from ruling out every better plan.

Hard stop: General linear-programming duality, existence and algorithms require algebraic/convex prerequisites; this finite certificate is complete without them.

Materials and preparation

8 minutes estimated prep. Five red counters, four blue counters, product recipe cards $A=(2 \text{ red}, 1 \text{ blue})$, $B=(1 \text{ red}, 2 \text{ blue})$, and revenue labels 3 and 4. Include blank resource-price tags.

A flexible hour

0-10 build and undo bundles; 10-15 establish resource and sale rules; 15-35 search for good production plans; 35-40 exchange builder and auditor roles; 40-55 invent a price certificate with fractions or a scaled integer version; 55-60 present plan and matching bound.

Questions to explore

- Why does giving every raw ingredient price 1 fail to cover product B's sale price?
- What makes a resource-price argument apply even to plans we never listed?

Satisfying stop: A best plan accompanied by a short universal certificate, not only a search record.

AP-20 / Facilitator reasoning

A price certificate for the best factory plan

The mathematical question

How can a proposed optimum be certified without examining every feasible allocation?

A nonnegative resource-price vector whose priced ingredients cover every product's sale value gives an upper bound on the revenue of every feasible production plan. Equality between such a bound and a feasible plan proves optimality by weak duality.

Worked instance

Problem: Prove the maximum revenue is 10 for five red and four blue counters.

Solution: Two A and one B use five red and four blue and earn 10. Prices $r=2/3, b=5/3$ make A's ingredients worth 3 and B's worth 4. All available ingredients are worth $5(2/3)+4(5/3)=10$. Since no product sells above its ingredient value, every feasible plan earns at most 10. Equality proves optimality, even if fractional production were allowed.

Boundary: With four red and four blue, fractional $A=B=4/3$ earns $28/3$, whereas the best integer plan is two B, earning 8. A fractional bound need not be attained by whole products.

Hints to reveal in order

1. Find prices r, b with $2r+b$ at least 3 and $r+2b$ at least 4.
2. Try making both equalities hold; alternatively multiply all money values by 3.

Extensions

- Scale all prices by three so the certificate uses integers 2 and 5 against product values 9 and 12.
- Change resource supplies and identify when the same price certificate remains sharp; this introduces sensitivity and complementary slackness.

Connection and its limits

Bridge type: exact-special-case

Preserves: Products, resource inequalities and price certificates constitute a genuine finite linear/integer optimization problem and its dual bound.

Limits: One integral optimum does not imply all linear programs have integral optima; general duality requires separate hypotheses.

Sources and inspection scope

[Stephen Boyd and Lieven Vandenbergh, Convex Optimization](#)

Chapter 5: weak duality; finite resource certificate verified algebraically here

Inspected: relevant exposition or named scope inspected; finite calculation self-contained below

Prior use / what changes: Existing tiling packets use upper bounds for packing. This connects the bounding habit to resource prices and linear optimization with a distinct feasible set.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-21 / Participation and launch

Spend the battery now, or save it?

MSC primary 90 - Operations research, mathematical programming | Secondary 68, 93

The opening challenge

Start with two energy tokens and no recharge. On each day you may accept that day's offer once if you can pay its energy cost, or skip it. Tomorrow's offers are already visible. Find the best plan, then make a table that gives the best decision from every day and remaining-energy state.

Learner choices: Choose a plan, create a counterexample to a greedy rule, and decide how to organize reusable information rather than rechecking every complete story.

Prerequisite gates

Enter: O/C: spend energy tokens once, collect score counters and follow three dates.

Explore: C: list legal plans and compare their scores.

Explain: P: justify why optimal remaining decisions depend only on day and energy.

Prove: P: backward induction over the number of remaining days.

Reading: Three short offer cards; pictures can replace text.

Arithmetic: Small whole-number subtraction and addition.

Reasoning: An immediate attractive action may remove a more valuable future choice; the record must keep remaining energy.

Hard stop: Continuous optimal control and stochastic decision theory are continuations, not claims proved by the three-day table.

Materials and preparation

8 minutes estimated prep. Two energy tokens, three ordered offer cards: day 1 costs 2 and pays 5; day 2 costs 1 and pays 4; day 3 costs 1 and pays 4. Prepare a 3×3 day/energy grid.

A flexible hour

0-10 act out possible spending plans; 10-15 agree on one-time offers and no recharge; 15-35 enumerate or build a decision tree; 35-40 swap planner and battery-keeper roles; 40-55 compress the tree into a backward table; 55-60 explain why a tempting first choice fails.

Questions to explore

- Why can the highest-value single offer be part of a worse overall plan?
- Which details of the earlier days matter after the remaining energy is known?

Satisfying stop: A greedy counterexample and a complete optimal-decision table.

AP-21 / Facilitator reasoning

Spend the battery now, or save it?

The mathematical question

Why can looking backward through a state space beat choosing the best immediate reward?

For a finite deterministic horizon with state (day, remaining energy), the optimal value obeys Bellman's recurrence: compare the reward plus optimal value of the resulting state for every legal action. The state must retain all information relevant to future feasibility and reward.

Worked instance

Problem: Find the optimal score and the backward values for energy 0,1,2.

Solution: On day 3, values are 0,4,4. Before day 2, values are 0,4,8: with two tokens take both remaining offers. Before day 1, values are 0,4,8; with two tokens, taking day 1 gives 5 while skipping gives 8. Thus skip day 1 and take days 2 and 3. Enumerating feasible subsets gives $\{\}, \{1\}, \{2\}, \{3\}, \{2,3\}$, with scores 0,5,4,4,8, confirming the table.

Boundary: If energy recharges by one unit overnight, the state transitions change and taking day 1 can become optimal. The old table must not be reused unchanged.

Hints to reveal in order

1. Begin on the last day, when there is no later consequence.
2. At each state compare skip with take plus the best value already recorded for the next day.

Extensions

- Add a fourth dated offer and update the same state table.
- Introduce a random future offer with a stated distribution and compare a precommitted plan with a policy reacting to the observed offer.

Connection and its limits

Bridge type: exact-special-case

Preserves: A two-unit battery, dated offers and accept/skip actions form an exact finite-horizon resource-allocation problem.

Limits: Future offers are known here. Unknown or random future offers require an explicit information model and possibly different policies.

Sources and inspection scope

[Jeffrey Chasnov, Mathematical Biology](#)

Chapter 7.3: dynamic programming located; this finite resource recurrence is self-contained

Inspected: dynamic-programming chapter located in authored contents; this finite decision recurrence has a complete independent proof

Prior use / what changes: Subtraction games use backward winning-state analysis. Here the payoff is accumulated reward under resource constraints, with optimization rather than alternating adversarial turns.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-22 / Participation and launch

How should you hide an unequal choice?

MSC primary 91 - Game theory, economics, finance, and other social and behavioral sciences | Secondary 60, 90

The opening challenge

Choose your action simultaneously and reveal together. The row player wants a high score; the column player wants a low score. Everyone knows the payoff table. Can the row player guarantee a positive average even against an opponent who knows the row player's randomization rule?

Learner choices: Choose mixtures, test predictable play, build a private randomizer, and seek a strategy whose guarantee does not depend on the opponent's choice.

Prerequisite gates

Enter: O/C: choose cards secretly and score the four outcomes.

Explore: F: compare long-run averages and design a three-card randomizer.

Explain: A: calculate payoff against each opposing pure choice.

Prove: F/P: show one strategy guarantees a lower bound and the other caps every strategy at the same value.

Reading: A two-by-two score table; colors can replace U,D,L,R.

Arithmetic: Thirds and multiplication by two; no money or betting required.

Reasoning: Keep the chosen distribution public but the realized action private; players have opposite objectives.

Hard stop: General finite-game equilibrium existence uses more machinery; the exact two-by-two certificate suffices here.

Materials and preparation

10 minutes estimated prep. Row cards U,D and column cards L,R, score counters, and two three-card draw packs. Row payoff is 2 for U/L, 1 for D/R, and 0 otherwise; column minimizes that score.

A flexible hour

0-10 play several rounds; 10-15 clarify simultaneous secrecy and opposite goals; 15-35 compare pure play and mixtures; 35-40 exchange roles; 40-55 derive matching lower and upper bounds; 55-60 explain why the mixture is unequal.

Questions to explore

- Why is choosing U and D equally often not necessarily best?
- What could the opponent do if it saw your realized action before choosing?

Satisfying stop: Two three-card randomizers with a complete minimax certificate.

AP-22 / Facilitator reasoning

How should you hide an unequal choice?

The mathematical question

How can randomization protect a player against an opponent who knows the strategy but not its realized action?

A two-player zero-sum matrix game allows each player to choose a distribution independently and privately. For payoff matrix $[[2,0],[0,1]]$, maximizing the guaranteed expected row payoff gives a mixed minimax equilibrium with value $2/3$.

Worked instance

Problem: Find both optimal mixtures and certify the game value.

Solution: Set $2p=1-p$, giving $p=1/3$ and guarantee $2/3$. If column chooses L with $q=1/3$, pure U earns $2q=2/3$ and pure D earns $1-q=2/3$. Therefore every row mixture earns exactly $2/3$ against that column strategy, proving no larger guarantee is possible. The two matching bounds certify the equilibrium.

Boundary: If column sees the row action first, it chooses R after U and L after D, forcing payoff zero. The guarantee depends on simultaneous hidden realization.

Hints to reveal in order

1. If U is chosen with probability p , payoffs against L and R are $2p$ and $1-p$.
2. Make those two threats equal, then find a column mixture giving the same payoff to U and D.

Extensions

- Change the diagonal rewards to positive a, b and derive row U probability $b/(a+b)$ and value $ab/(a+b)$.
- Contrast expected guarantee with every-round guarantee: the latter remains zero in this game.

Connection and its limits

Bridge type: exact-special-case

Preserves: Simultaneous hidden action cards and score transfers realize the specified strategic-form game exactly.

Limits: Observed short-run winnings need not equal expectation. Real people need not play equilibrium, and visible random draws change the information structure.

Sources and inspection scope

[Giacomo Bonanno, Game Theory](#)

Sections 5.2-5.3: mixed strategies, indifference and best replies

Inspected: mixed-strategy exposition inspected; this matrix solved independently

Prior use / what changes: Weeks 7-8 cover perfect-information combinatorial games. Simultaneous mixed strategies are a different branch and use a different notion of solution.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-23 / Participation and launch

A stable traffic pattern that costs more

MSC primary 91 - Game theory, economics, finance, and other social and behavioral sciences | Secondary 90

The opening challenge

Three travelers choose A or B. Everyone on A pays the number of travelers on A; everyone on B pays 2. A configuration is stable if no single traveler can switch and strictly lower their own cost. Find every stable occupancy and compare total costs.

Learner choices: Choose configurations, act as a player searching for improvement, propose coordinated changes, and distinguish fairness from total efficiency.

Prerequisite gates

Enter: O/C: place one token per player on a route and count users.

Explore: C: compute each player's cost and total cost.

Explain: P: test every kind of unilateral switch, including the cost change caused by joining.

Prove: P: enumerate the four possible route occupancies.

Reading: Two route cost labels; oral rules sufficient.

Arithmetic: Counts to three and totals to nine.

Reasoning: A switch changes the destination route's crowding. Equal cost is not a strictly profitable deviation.

Hard stop: General equilibrium existence and price-of-anarchy bounds require additional game-theoretic definitions and proofs.

Materials and preparation

5 minutes estimated prep. Three distinct player counters, route A labeled "cost equals number using A," and route B labeled "cost always 2." Provide a table for A-occupancies 0,1,2,3.

A flexible hour

0-10 explore costs by moving counters; 10-15 define unilateral and strictly lower; 15-35 classify the four occupancies; 35-40 become three travelers and rehearse switches; 40-55 compare stable states with a planner's optimum; 55-60 explain one surprising stable pattern.

Questions to explore

- Why can a B traveler not simply use the cost A had before they joined it?
- Can a stable pattern still have a cheaper alternative arrangement?

Satisfying stop: A complete table separating stable behavior from globally efficient behavior.

AP-23 / Facilitator reasoning

A stable traffic pattern that costs more

The mathematical question

Why can individually stable choices fail to minimize total cost?

A pure Nash equilibrium has no strictly profitable unilateral deviation. In a finite congestion game, a player's route cost depends on how many use it; equilibrium and social optimum are distinct conditions.

Worked instance

Problem: Find Nash occupancies and the social optimum.

Solution: Let n be the number on A. Total costs for $n=0,1,2,3$ are 6,5,6,9. At $n=0$ a traveler improves from 2 to 1 by joining A. At $n=3$ a traveler improves from 3 to 2 by switching to B. At $n=1$, a B traveler joining A would pay 2, unchanged, and the A traveler would worsen from 1 to 2: stable. At $n=2$, an A traveler moving to B stays at 2, while the B traveler joining A would pay 3: also stable. The social optimum is $n=1$ with total 5; $n=2$ is a costlier equilibrium.

Boundary: If ties also trigger moves, stability is a different notion. Two A travelers may be indifferent to a move even though that move improves total cost.

Hints to reveal in order

1. Record each current cost and then recompute the destination cost after one token moves.
2. Compare total costs separately from individual improvement tests.

Extensions

- Change B's constant cost to $3/2$ and recompute equilibria and total optimum with fractions.
- Explore a small toll or coordination rule, stating whose objective it serves; no policy is universally best without an objective.

Connection and its limits

Bridge type: exact-special-case

Preserves: Three route-choice counters are literal players in an explicitly specified congestion game.

Limits: This is a mathematical cost model, not a prediction of actual children's traffic choices or a recommendation for a real transport policy.

Sources and inspection scope

[Giacomo Bonanno, Game Theory](#)

Section 1.6: Nash equilibrium; finite congestion instance checked directly

Inspected: named source scope inspected; displayed finite result proved directly here

Prior use / what changes: No congestion-game task is documented. The existing route packets ask how to traverse edges, not how route users create strategic costs.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-24 / Participation and launch

A neutral population can lose a color

MSC primary 92 - Biology and other natural sciences | Secondary 60

The opening challenge

Make a new generation of exactly two counters. For each offspring, draw a parent color uniformly from the current two, record that color, and replace the parent before drawing again. After both offspring are recorded, they become the new parents. Can a neutral rule eventually erase a color?

Learner choices: Choose how to record generations, compare several independent populations, and design a changed rule that would prevent or reverse loss.

Prerequisite gates

Enter: O/C: draw, replace, and record a parent color for each offspring.

Explore: F: compare probabilities for two independent draws and repeat generations.

Explain: P: distinguish an unchanged expectation from an unchanged realized population.

Prove: F/A: derive transition probabilities and eventual absorption for the two-individual case.

Reading: Color-only generation rows; spoken rules suffice.

Arithmetic: Halves and quarters; optional powers for survival probability.

Reasoning: The new generation replaces the old only after both offspring have been drawn; no within-generation updating.

Hard stop: Selection, diploidy, mutation and real demographic inference require changed models and biological assumptions.

Materials and preparation

5 minutes estimated prep. Two parent counters, one red and one blue, an opaque draw cup, and a row of generation boxes. Use colored pencils to record offspring while replacing every parent draw.

A flexible hour

0-10 practice drawing with replacement; 10-15 establish simultaneous generation replacement; 15-35 run independent lineages; 35-40 exchange sampler and recorder roles; 40-55 enumerate one-generation outcomes and explain absorption; 55-60 compare expected counts with observed endpoints.

Questions to explore

- Why can both colors be equally successful in expectation while one disappears in a run?
- Once a color is gone, can this particular model recover it?

Satisfying stop: An exact explanation of random loss under a neutral rule.

AP-24 / Facilitator reasoning

A neutral population can lose a color

The mathematical question

Why can random inheritance drive fixation even when the expected allele fraction remains unchanged?

A neutral finite Wright-Fisher generation of size N samples N offspring independently with replacement from the current generation's allele frequencies. Conditional expected next frequency equals current frequency, while absorbing all-one-color states remain possible.

Worked instance

Problem: Starting from one red and one blue, find one-step probabilities and eventual fixation probabilities.

Solution: RR, RB, BR, BB each have probability $1/4$. Red counts 2, 1, 0 therefore have probabilities $1/4, 1/2, 1/4$ and expected count 1. The mixed state survives k generations with probability $(1/2)^k$, tending to zero.

Symmetry gives eventual all-red and all-blue probabilities $1/2$ each. Once all red or all blue, every parent draw has that color, so these states are absorbing.

Boundary: Drawing without replacement would preserve one red and one blue forever for $N=2$. That small rule change removes the drift in this example.

Hints to reveal in order

1. From one red and one blue, list RR, RB, BR, BB as distinct ordered offspring outcomes.
2. Only the two mixed outcomes keep both colors alive for another generation.

Extensions

- Increase population size and compare time to loss; derive the binomial one-step distribution with fraction p .
- Add a specified small mutation chance after each draw and observe that formerly absorbing states no longer remain absorbing.

Connection and its limits

Bridge type: exact-special-case

Preserves: Parent-color draws with replacement implement a genuine finite neutral allele-frequency model.

Limits: This is a highly simplified haploid population with fixed size, no mutation and no selection. It is not a model of individual human traits.

Sources and inspection scope

[Jeffrey Chasnov, Mathematical Biology](#)

Section 5.5: random genetic drift; $N=2$ transition calculation provided below

Inspected: named source scope inspected; displayed finite result proved directly here

Prior use / what changes: Year-1 probability examples do not document this generational sampling process. Record population size and rule version to distinguish future variants.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-25 / Participation and launch

What can never change in the reaction box?

MSC primary 92 - Biology and other natural sciences | Secondary 15, 80

The opening challenge

A legal forward reaction trades exactly one A and two B for one C. A reverse reaction trades one C for one A and two B. Start with $(A,B,C)=(3,4,0)$. Find every reachable collection. What totals stay fixed even though the number of counters changes?

Learner choices: Choose reaction sequences, invent a conserved counting method, propose impossible target states, and decide whether a certificate rules them out.

Prerequisite gates

Enter: O/C: exchange one A and two B for one C, or reverse it.

Explore: C: record reachable count triples without creating negative counts.

Explain: M/P: discover weighted totals unchanged by each exchange.

Prove: A/L: solve $w \cdot (-1, -2, 1) = 0$ and use invariance under sequences of moves.

Reading: Three symbols or colors with a pictorial recipe; oral rules work.

Arithmetic: Counts and doubling; optional linear equations.

Reasoning: The total number of tokens is not conserved; different quantities may be conserved simultaneously.

Hard stop: Mass-action differential equations add rates and continuous concentration assumptions; do not infer them from token reachability alone.

Materials and preparation

7 minutes estimated prep. A counters, B counters and distinct C counters, a forward/backward recipe mat, and count-triple records. Begin with 3 A and 4 B and no C.

A flexible hour

0-10 practice both reactions freely; 10-15 establish nonnegative whole counts; 15-35 map reachable states and test targets; 35-40 partners act as molecule trades; 40-55 derive weighted invariants or add a second reaction; 55-60 explain an impossibility.

Questions to explore

- Why does simply counting all tokens fail as an invariant?
- Are invariant equations alone always enough to prove reachability in an arbitrary reaction network?

Satisfying stop: A complete reaction-state list and two visible conservation laws.

AP-25 / Facilitator reasoning

What can never change in the reaction box?

The mathematical question

How do stoichiometric constraints produce conserved quantities independently of reaction rates?

For reaction $A+2B \leftrightarrow C$, a forward firing changes counts by $(-1,-2,+1)$ and a backward firing reverses it. Linear quantities $w \cdot n$ are conserved exactly when w is orthogonal to that change vector; rates determine timing but not these conservation laws.

Worked instance

Problem: List every reachable state and decide whether $(0,0,3)$ can be reached.

Solution: $A+C=3$ and $B+2C=4$ stay fixed. Writing $C=c$ gives $A=3-c$ and $B=4-2c$. Nonnegative integer counts force $c=0,1,2$, yielding $(3,4,0), (2,2,1), (1,0,2)$. Each is reached by zero, one or two forward reactions. Target $(0,0,3)$ violates $B+2C=4$ since its value is 6, so it is impossible. Total molecule count falls by two per forward firing, demonstrating why the unweighted total fails.

Boundary: For a larger network, nonnegative invariant-compatible counts need not be reachable; modular obstructions or missing reactants needed to enable a move may prevent reachability. Here sufficiency was proved by an explicit sequence.

Hints to reveal in order

1. Imagine each C as a sealed packet containing the A and B used to make it.
2. Count $A+C$ and $B+2C$ separately.

Extensions

- Add $A \leftrightarrow B$ and determine which conserved totals survive both reaction vectors.
- Assign positive forward and backward rates and compare dynamical equilibrium with stoichiometric reachability; conservation alone cannot locate the equilibrium.

Connection and its limits

Bridge type: exact-special-case

Preserves: Colored molecule counters and legal reaction firings are the exact discrete stoichiometric state system for the stated reversible reaction.

Limits: Tokens are hypothetical species; legal reachability does not determine real reaction speed, energy, equilibrium or chemical feasibility.

Sources and inspection scope

[Jeffrey Chasnov, Mathematical Biology](#)

Section 6.1: law of mass action; conservation proof is explicit for this network

Inspected: named source scope inspected; displayed finite result proved directly here

Prior use / what changes: Weeks 2 and 4 use invariants for toggles and remainders. This is a distinct chemical-stoichiometry instance with changing molecule count and several simultaneous conservation laws.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-26 / Participation and launch

The controller is correcting yesterday's error

MSC primary 93 - Systems theory; control | Secondary 37

The opening challenge

The plant adds your correction to its current position once each tick. Your sensor reports the previous tick's position. Begin with $\text{previous} = \text{current} = 1$. First always negate the reported number. Does the position settle at zero? Then try correcting only half the reported error.

Learner choices: Choose full or half correction, test a different initial state pair, and decide what information is necessary to predict the future.

Prerequisite gates

Enter: F/C: signed positions and one-step memory represented by two cards.

Explore: F: apply full or half correction and plot a sequence.

Explain: P: identify a repeating pair of states, not merely a repeated single reading.

Prove: A: manipulate the two-step recurrence to prove periodicity or decay.

Reading: Current, previous and correction labels; spoken role instructions support participation.

Arithmetic: Signed addition, halves and quarters.

Reasoning: Separate the current position from the sensor's delayed report; update both cards in the correct order.

Hard stop: General delay-system stability and frequency-domain design require further control theory; the exact recurrence suffices.

Materials and preparation

7 minutes estimated prep. A signed number line, two state cards labeled previous/current, correction arrows, and a two-person controller/plant station. Start both state cards at 1.

A flexible hour

0-10 move positions with chosen corrections; 10-15 establish the delayed-report order; 15-35 compare full and half feedback; 35-40 exchange controller and plant roles; 40-55 prove the repeating pair or four-step shrink rule; 55-60 explain why strong correction need not be better.

Questions to explore

- Why is seeing a zero position once not enough to show the system has settled?
- What changes when the controller sees the current state instead?

Satisfying stop: A proved delay-induced cycle and a weaker correction that converges.

AP-26 / Facilitator reasoning

The controller is correcting yesterday's error

The mathematical question

Why can delayed feedback destabilize a correction rule that works perfectly with current information?

The discrete plant is $x[n+1]=x[n]+u[n]$. Current feedback $u[n]=-x[n]$ sets the next state to zero. One-step delayed feedback $u[n]=-x[n-1]$ instead creates a second-order recurrence whose behavior depends on the stored previous state.

Worked instance

Problem: Analyze full and half delayed correction from $x[-1]=x[0]=1$.

Solution: Full correction gives $x[1..6]=0,-1,-1,0,1,1$, returning the pair to $(1,1)$: a six-step cycle. Current full correction would instead reach zero in one tick and stay there. Half delayed correction gives $1,1/2,0,-1/4,-1/4,-1/8,0,1/16,\dots$ from $x[0]$. Algebra from $x[n+1]=x[n]-x[n-1]/2$ yields $x[n+4]=-x[n]/4$, so every four-step subsequence decays to zero.

Boundary: A single repeated position does not prove a cycle: its preceding state might differ. The complete state is the ordered previous/current pair.

Hints to reveal in order

1. Keep a two-column record of (previous,current).
2. A deterministic repeated pair makes every future pair repeat; for half correction, compare entries four steps apart.

Extensions

- Test other gains k in $x[n+1]=x[n]-k x[n-1]$ and study characteristic roots with algebra.
- Add bounded disturbances and compare perfect nominal convergence with robustness; a new disturbance rule must be specified.

Connection and its limits

Bridge type: exact-special-case

Preserves: Position, correction input and a one-step-old sensor report are exactly the state, control and observation of a discrete feedback system.

Limits: This is not Euler's approximation to a cooling ODE. Its clocked update and communication delay are the model itself.

Sources and inspection scope

[Åström and Murray, Feedback Systems](#)

Chapter 7: state feedback; delay example is a self-contained discrete system

Inspected: relevant source framework inspected; finite example explicitly proved below

Prior use / what changes: AP-07 also studies recurrence stability, but there the update approximates an ODE. Here delayed information changes the exact closed-loop state dimension; record the shared proof mechanism.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-27 / Participation and launch

Can one sensor see two hidden tanks?

MSC primary 93 - Systems theory; control | Secondary 15, 62

The opening challenge

A sensor reports only the sum of two hidden amounts. First both amounts halve each tick. Can any number of total readings reveal the original split? Now let the first halve and the second quarter each tick. Readings are 8 initially and 3 one tick later. Find the original amounts.

Learner choices: Choose indistinguishable hidden states, decide what extra sensor or changed dynamics would help, and test the reconstruction with a partner's secret split.

Prerequisite gates

Enter: O/C/F: hide two amounts and read only their total.

Explore: F: halve or quarter the amounts each step.

Explain: P: produce two distinct states with the same entire reading history.

Prove: A/L: solve two linear equations or find an unobservable difference vector.

Reading: Two update arrows and sensor labels; diagram-first explanation possible.

Arithmetic: Halves and quarters, signed linear combinations.

Reasoning: The visible total is not the full state; avoid assuming an unknown split is equal.

Hard stop: Kalman filtering requires a noise model and probabilistic estimation; it is not established by this exact reconstruction.

Materials and preparation

8 minutes estimated prep. Two covered amount cards representing tanks, one total-readout card, and an update table. Use counters for integer starting values and fraction strips after decay.

A flexible hour

0-10 hide and guess two-counter splits; 10-15 fix the sensor and update rules; 15-35 compare histories under identical decay; 35-40 exchange hidden-state and observer roles; 40-55 reconstruct under distinct decay or inspect near-equal rates; 55-60 state an observability conclusion.

Questions to explore

- Why does collecting the same kind of information many times sometimes add nothing?
- How can changing the dynamics make one sensor sufficient?

Satisfying stop: A proof of permanent ambiguity and a contrasting exact reconstruction.

AP-27 / Facilitator reasoning

Can one sensor see two hidden tanks?

The mathematical question

When do output observations determine a system's hidden initial state, and when can infinitely many observations remain insufficient?

For $x[n+1]=Ax[n]$ and $y[n]=Cx[n]$, two states are observationally indistinguishable if $C A^n$ times their difference vanishes for every observed n . Distinct decay rates can make two total-output readings identify two initial components.

Worked instance

Problem: Give indistinguishable states in the first model and solve the second model's two readings.

Solution: Under identical halving, states (2,6) and (4,4) both produce totals 8,4,2,1,..., so no number of those readings distinguishes them. Under rates 1/2 and 1/4, equations $a+b=8$ and $2a+b=12$ give $a=b=4$. In general $a=4y_1-y_0$ and $b=2y_0-4y_1$. Substitution recovers both measured totals.

Boundary: If the second decay factor is changed back to 1/2, the two equations become dependent. If rates are merely very close, exact uniqueness can coexist with high sensitivity to measurement error.

Hints to reveal in order

1. For identical decay, express every future total as the original total times a power of 1/2.
2. For different decay, write $a+b=8$ and $a/2+b/4=3$.

Extensions

- Add a sensor reading the difference; one simultaneous sum and difference then identifies both components without waiting.
- Build the two-row observability matrix and investigate its rank and conditioning as the decay rates approach equality.

Connection and its limits

Bridge type: exact-special-case

Preserves: Hidden component amounts, linear updates and a sum-only sensor are a literal finite-dimensional linear observation system.

Limits: Exact noiseless identification does not imply stable inference from noisy measurements. Physical tank drainage need not follow these fractional updates.

Sources and inspection scope

[Åström and Murray, Feedback Systems](#)

Chapter 8: observability and state estimation; two-state examples derived below

Inspected: observability chapter scope inspected; both finite linear systems solved and checked directly

Prior use / what changes: Week 6 asks optimal queries about codes. Here the measurements arise from evolving hidden state and fixed sensor design; this is a substantive systems-theory connection.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-28 / Participation and launch

Four messages that survive one damaged bit

MSC primary 94 - Information and communication theory, circuits | Secondary 05, 68

The opening challenge

Send one of four messages using a five-bit strip. Your partner may secretly flip at most one bit. Design a decoding rule that always recovers the original. Then investigate whether four bits could ever give the same guarantee, even with a different code.

Learner choices: Choose a message, choose a damaging position as adversary, organize all received patterns, and seek either a shorter construction or an impossibility proof.

Prerequisite gates

Enter: O/C: copy black/white patterns and flip at most one tile.

Explore: C/P: count disagreements and compare candidate codewords.

Explain: P: show a received word cannot be within one flip of two codewords distance three apart.

Prove: M/P: count disjoint radius-one balls to rule out four-bit codes.

Reading: Color patterns only; rule can be spoken.

Arithmetic: Counts to five and powers of two up to 32.

Reasoning: The error position is unknown; the guarantee must cover every single flip, not just errors tried experimentally.

Hard stop: Asymptotic capacity, optimal code families and decoding complexity are later questions; the finite code is complete.

Materials and preparation

10 minutes estimated prep. Five-position black/white strips, four message symbols and codewords 00000,11100,10011,01111. Give an independent partner the role of flipping zero or one bit.

A flexible hour

0-10 practice damage and recovery; 10-15 state the at-most-one-flip rule; 15-35 test and audit the four codewords; 35-40 exchange sender/noise/decoder roles; 40-55 count neighborhoods to attack four-bit coding; 55-60 show a recovery and its universal justification.

Questions to explore

- Why must two message patterns differ in at least three places?
- How many distinct received patterns can one four-bit message produce with at most one flip?

Satisfying stop: A working optimal-length code with both a correction proof and a lower bound.

AP-28 / Facilitator reasoning

Four messages that survive one damaged bit

The mathematical question

How much redundancy is required to correct an adversarial error, and how can geometry in a discrete space certify a limit?

Binary words of minimum Hamming distance at least three have disjoint radius-one Hamming balls and uniquely correct any one-bit error by nearest-codeword decoding. Four messages require at least five bits for that guarantee.

Worked instance

Problem: Decode received word 10111 and prove four-bit coding is impossible for four messages.

Solution: 10111 differs from 10011 only at the third bit, so it decodes to that codeword. The six pairwise distances of the supplied code are 3,3,4,4,3,3, hence the guarantee holds. For four-bit words, each radius-one ball contains 5 patterns. Four messages would require 20 distinct received patterns, but only $2^4=16$ exist. Any shorter code could be padded with fixed zeros to length four without decreasing pairwise distance, so ruling out length four rules out every shorter length. Therefore four bits cannot suffice; the displayed five-bit code attains the minimum.

Boundary: Two flipped bits can mislead nearest decoding. From 00000, flipping the first two gives 11000, which lies one bit from 11100.

Hints to reveal in order

1. If a received word is one flip from both originals, the originals are at most two flips apart.
2. A four-bit message has one unchanged pattern and four one-flip patterns; those five-pattern sets must be disjoint.

Extensions

- Separate detection from correction: distance two detects any single flip but need not identify the original.
- Compare a threefold repetition code with the five-bit construction, then explore Hamming codes after linear algebra over two-element arithmetic.

Connection and its limits

Bridge type: exact-special-case

Preserves: Bit strips, adversarial single flips and nearest-codeword decoding are exact finite error-correcting communication.

Limits: This finite worst-case guarantee is distinct from Shannon's asymptotic probabilistic channel theorem and from secret encryption.

Sources and inspection scope

[Claude Shannon, A Mathematical Theory of Communication](#)

Part II: noisy communication framework; finite distance and sphere-packing proofs given here

Inspected: relevant source framework inspected; finite example explicitly proved below

Prior use / what changes: Week 6 uses optimal code queries. This adds channel damage, Hamming distance and sphere-packing rather than duplicating the existing guessing questions.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-29 / Participation and launch

Give the common message a shorter name

MSC primary 94 - Information and communication theory, circuits | Secondary 05, 60

The opening challenge

Our eight messages contain A four times, B twice, and C and D once each. Give each symbol a binary codeword so a receiver can decode a concatenation without separators. Restrict to prefix codes. Can you use fewer than the 16 bits required by two bits per symbol?

Learner choices: Choose tree shape and label placement, invent difficult concatenations, and compare total length with worst-case length.

Prerequisite gates

Enter: O/C: follow left/right branches and associate leaves with pictures.

Explore: M: count total bits for a specified collection of messages.

Explain: P: explain immediate greedy decoding in a prefix tree and why a prefix violation can make greedy stopping fail. A non-prefix code may still be uniquely decodable by another method.

Prove: P: classify full binary trees with four leaves and use a label-swap argument.

Reading: Four picture symbols and binary digits; a spoken “left/right” alternative works.

Arithmetic: Weighted counts up to 16; division by eight optional.

Reasoning: A codeword ends at a leaf. Preserve boundaries without secretly inserting separators for free.

Hard stop: Entropy bounds require logarithms and probability; optimality among unrestricted uniquely decodable codes needs an additional theorem.

Materials and preparation

8 minutes estimated prep. Eight message tokens: four A, two B, one C, one D; binary-tree mats and movable leaf labels. Provide code strips but no free spaces or punctuation between words.

A flexible hour

0-10 send and decode short strings; 10-15 establish no-free-separator and prefix rules; 15-35 construct trees and count bits; 35-40 partners walk left/right paths; 40-55 prove the best four-leaf tree choice; 55-60 show a savings calculation and decoding demonstration.

Questions to explore

- Why should the most frequent symbol get a shallow leaf?
- Could the shortest average code make one rare message longer than before?

Satisfying stop: An unambiguous code that saves two bits, with a finite optimality proof.

AP-29 / Facilitator reasoning

Give the common message a shorter name

The mathematical question

How can unequal frequencies be converted into shorter unambiguous communication?

A binary prefix code represents messages by leaves of a binary tree, preventing a codeword from prefixing another. For positive message weights, an optimal tree has no unary internal node, and shorter leaves should receive larger weights by an exchange argument.

Worked instance

Problem: Optimize weighted total prefix-code length for weights 4,2,1,1.

Solution: Code A=0,B=10,C=110,D=111 has total $4 \times 1 + 2 \times 2 + 1 \times 3 + 1 \times 3 = 14$ bits, average $7/4$. An optimal four-leaf binary tree can be taken full. Its root splits leaf counts either 2+2, giving lengths 2,2,2,2 and total 16, or 1+3, giving 1,2,3,3. Swapping a larger weight into a shorter depth never increases cost, so the displayed assignment minimizes the second shape at 14. These exhaust full four-leaf shapes up to symmetry.

Boundary: Codes A=0 and B=01 are not prefix-free; without the rest of a specified code and a unique-decoding proof, greedily stopping at 0 can misread B.

Hints to reveal in order

1. Try A=0, then put all remaining words beneath the 1 branch.
2. An optimal positive-weight tree needs no node with only one child; remove such a step to shorten its descendants.

Extensions

- Change the frequencies to all equal; the balanced two-bit code now has total 8 and beats lengths 1,2,3,3 totaling 9.
- Use the two-smallest-weight merge idea to develop Huffman coding, proving why sibling placement permits recursive optimization.

Connection and its limits

Bridge type: exact-special-case

Preserves: Messages, binary strings and weighted bit totals are exact finite lossless source coding.

Limits: The known message frequencies and prefix condition are part of the model. This does not prove general entropy bounds or all properties of non-prefix uniquely decodable codes.

Sources and inspection scope

[Claude Shannon, A Mathematical Theory of Communication](#)

Part I, PDF p. 18: code 0,10,110,111 and weighted average length

Inspected: relevant source framework inspected; finite example explicitly proved below

Prior use / what changes: Week 6 query trees share a branching representation. Here path length is weighted communication cost, and unknown separators create the central decoding constraint.

Status: reviewed plan v1; exact instance checked; not classroom-piloted

AP-30 / Participation and launch

The sensor changes the circuit it measures

MSC primary 94 - Information and communication theory, circuits | Secondary 15, 78

The opening challenge

First find node X's voltage with no meter branch. Then attach a meter modeled as a $2\ \Omega$ resistor between X and 0 V and recalculate. Can the act of measuring lower the voltage being measured? Use current conservation rather than treating the original voltage as fixed.

Learner choices: Choose which node equation to write, test different meter resistances, and decide how a meter could disturb the circuit less.

Prerequisite gates

Enter: V/F/A: interpret two labeled nodes and voltage differences along resistors.

Explore: A: solve one linear node equation.

Explain: P: distinguish a current split from equal currents and check the new measurement branch.

Prove: A/L: derive node potentials from Kirchhoff equations and verify energy balance.

Reading: Circuit sketch and Ohm's law; partner calculation allowed.

Arithmetic: Fractions and small linear equations; optional squares for power.

Reasoning: A measuring device is part of the modeled system unless its loading is idealized away.

Hard stop: Capacitors, inductors and signal-frequency behavior add state and dynamics; they are not covered by a steady DC calculation.

Materials and preparation

8 minutes estimated prep. Paper circuit with ideal 6 V source, a $2\ \Omega$ resistor from 6 V to node X, and a $1\ \Omega$ resistor from X to 0 V. Supply an optional meter branch of $2\ \Omega$ from X to 0 V.

A flexible hour

0-10 explore circuit connections and voltage differences; 10-15 introduce current=drop/resistance; 15-35 solve before and after attaching the meter; 35-40 trace and split current counters at the node; 40-55 vary meter resistance or check energy; 55-60 explain loading in a complete sentence.

Questions to explore

- Why is the incoming current no longer equal to the current through the original bottom resistor alone?
- What should happen as the meter's resistance becomes extremely large?

Satisfying stop: A quantitatively checked case in which measurement changes the object measured.

AP-30 / Facilitator reasoning

The sensor changes the circuit it measures

The mathematical question

How do component laws and conservation determine a circuit, and why can attaching a measurement device alter the measured voltage?

In an ideal DC resistor network, each edge current is voltage drop divided by resistance and currents balance at every internal node. A finite-resistance meter connected across a component becomes an extra parallel branch, changing the network solution.

Worked instance

Problem: Compute X and all currents before and after adding the $2\ \Omega$ meter.

Solution: Without the meter, $(6-X)/2=X$ gives $X=2\text{ V}$ and current 2 A through both original resistors. With the meter, $(6-X)/2=X+X/2$ gives $X=3/2\text{ V}$. Incoming current is $9/4\text{ A}$; outgoing currents are $3/2\text{ A}$ through $1\ \Omega$ and $3/4\text{ A}$ through the meter, totaling $9/4\text{ A}$. The displayed voltage falls by $1/2\text{ V}$ while the source supplies more current.

Boundary: An ideal infinite-resistance voltmeter draws zero current and does not change this circuit. A zero-resistance branch would short X to 0 V and requires separate handling rather than division by zero.

Hints to reveal in order

1. Without the meter, equate $(6-X)/2$ to $X/1$.
2. With the meter, add $X/2$ to the outgoing side of that equation.

Extensions

- For meter resistance $R>0$, solve $X=6R/(3R+2)$ and show it approaches 2 V as R grows.
- Verify source power equals the sum of resistor dissipation before and after loading; later add capacitance to study measurement transients.

Connection and its limits

Bridge type: exact-special-case

Preserves: Node voltages, Ohmic edge laws and current conservation form a literal ideal resistive circuit model; the meter is modeled as a specified resistor.

Limits: Real meters, sources and components have tolerances and additional effects. Paper calculations avoid claiming that all physical devices obey these ideal values.

Sources and inspection scope

[David Tong, Electromagnetism](#)

Section 4.1.3, equation (4.4), printed pp. 74-77 / PDF pp. 8-11: Ohm's law and Joule heating; the finite Kirchhoff equations are explicit below

Inspected: relevant source framework inspected; finite example explicitly proved below

Prior use / what changes: AP-10 spring networks share linear constitutive and compatibility reasoning. Keep that cross-link explicit; the new circuit question is sensor loading, not a renamed spring-combination exercise.

Status: reviewed plan v1; exact instance checked; not classroom-piloted